Weight-Space Programming and Gradient Optimization on Nonconvex Algorithmic Tasks

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Abstract

The universal approximation theorem guarantees arbitrarily accurate approximation of continuous functions on compact domains with suitable activations and sufficient width. How should an optimizer find the weights, and which architecture should it use? We compare weight-space programming (WSP) with gradient optimization on ten algorithmic tasks. The fixed-network study covers 29 nonconvex task–architecture pairs and three random starts, using 3–50 coefficients in rectified linear, residual and recurrent networks. Astra, a pretrained language model, returns numerical weights in one reply or revises them using training and development feedback. Both examples-only WSP procedures match every extrapolation reference in the same 30 of 87 settings, including addition and running sums. Full-batch gradient descent using backpropagation achieves this in zero settings, as does descent followed by limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) refinement. Summing each task's mean raw extrapolation root mean squared error (RMSE) ranks the two iterative WSP conditions first among 14 evaluated pipelines. The examples-only sum falls from 1.06×10^{12} to 2.03×10^{9}, driven mainly by improved Fibonacci predictions. The median scaled extrapolation error instead increases, and stored square-network coefficients show how central validation can favor a worse outer continuation. Thirty further requests let Astra choose the graph and coefficients from input/output examples, with extrapolation as the primary objective. Twenty-one proposals are exact on every outer input. Square uses a single product node with zero stored coefficients, while sine attains RMSE 7.79×10^{-16}. Two self-power proposals are invalid. We report coefficient storage, live tensor memory and arithmetic costs under the declared operation library, and reproduce prompts, returned graphs and public construction explanations.

1 Introduction

Universal approximation leaves open how to find the weights of a particular small network. Consider a network given twelve examples of addition. How closely do the returned weights fit those sums, and do the same weights still add correctly when both numbers are larger? We evaluate each solver on both questions. The unit of comparison is a *task and architecture*: fixing addition as the target still leaves a choice between a linear map with two coefficients and a network with a trainable hidden layer. The former has a convex least-squares training loss. The latter can have several weight vectors that fit equally well and a much worse fit halfway between them.

Weight-space programming (WSP) uses a specification or examples to assign numerical values directly to a network's weights. Our solver, GPT-6 Astra (Astra), receives the architecture, a random initial weight vector and labeled training examples. In *one-shot WSP*, it returns numerical weights for every trainable coefficient in a single reply. In *iterative WSP*, we start with those very weights, evaluate them on training and development examples, and invite up to three revisions. We keep a re-

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vision only if it improves the development loss, using the training loss to break a tie. Each procedure is run with examples only and with the generating formula supplied alongside the same examples.

Backpropagation (BP) computes a derivative; its use does not specify how to update the weights. Our first numerical baseline uses that derivative for full-batch gradient descent. Our second begins at the same gradient-descent endpoint and uses a limited-memory approximation to local curvature to refine it. We report both checkpoints. Additional comparisons use adaptive gradient steps and trust-region least squares. We evaluate the weights on the original training data and on disjoint tests inside and outside the original input range. For integer sequences both tests concern unseen later indices.

The benchmark covers four elementary functions, two arithmetic operations and four sequence or sorting tasks. For every included fixed-network task, architecture and training set we exhibit two weight vectors whose midpoint has higher loss than the average endpoint loss. This strict violation of the convexity inequality certifies that the empirical objective is nonconvex in its trainable parameters. It permits the comparison to include linear recurrent networks whose repeated trainable matrices make their losses nonconvex. We measure how often each method fits the observations more closely, how often that advantage carries beyond the observed range, and how the outcome changes with the number of parameters. The question concerns the quality of the returned weights at stated budgets. The tasks have known mathematical answers and allow direct checks of exactness.

We also ask Astra to choose the architecture from examples. Thirty further requests specify the observations, deployment domain and supported neural operations. Astra chooses the graph, its size and coefficients, with extrapolation accuracy as the primary objective and inference cost as the secondary objective. We evaluate every returned proposal and report its coefficient storage, activation memory and arithmetic cost.

2 Related Work

Comparing optimizers. DeepOBS pairs tasks with specified networks and evaluates optimization methods on those problems [18]. Schmidt et al. show experimentally how optimizer performance varies with task and tuning budget [17]. We use small networks and few observations so that a language model can receive the entire parameter array, numerical solvers can use individual output derivatives, and some training optima can be computed independently. Our data also permit separate measurements of interpolation and extrapolation. We compare final predictions rather than speed at equal computation.

Constructing weights. Algorithms have long been translated into neural weights, including recurrent computations [19, 25] and sorting in transformers [12]. Recent demonstrations used Astra to write game-policy weights and modify a pretrained language model [21]. Other WSP studies tested bounded arithmetic and sorting through repeated proposals [20] or compiled numerical formulas into neural networks [24]. Our primary comparison fixes every graph, initial array, set of training labels and scoring objective before collecting its WSP responses. We report both the objective attained on those labels and accuracy on undisclosed test inputs.

Capacity and extrapolation. Studies of arithmetic networks and algorithmic reasoning show how computation outside the training range depends on network structure [23, 14, 27, 26]. Universal approximation gives existence of a network that approximates a continuous function on a compact domain to a specified positive tolerance [3, 11]. It leaves open the weights an optimizer will find and the size required to round an answer exactly. Vapnik–Chervonenkis dimension concerns the ability to realize every binary labeling of a point set [1]. We instead use lower bounds for the particular task and graph, including a bound on the number of linear pieces of a shallow square network. Floating-point arithmetic introduces additional expressivity and rounding questions [16, 9, 15, 8].

Selecting the architecture. Neural architecture search selects network structures using measured performance, as in the recurrent controller of Zoph and Le [29]. Our additional experiment asks Astra to infer a rule from examples and return both a graph and its numerical coefficients in one response. We measure extrapolation and execution cost for each returned graph.

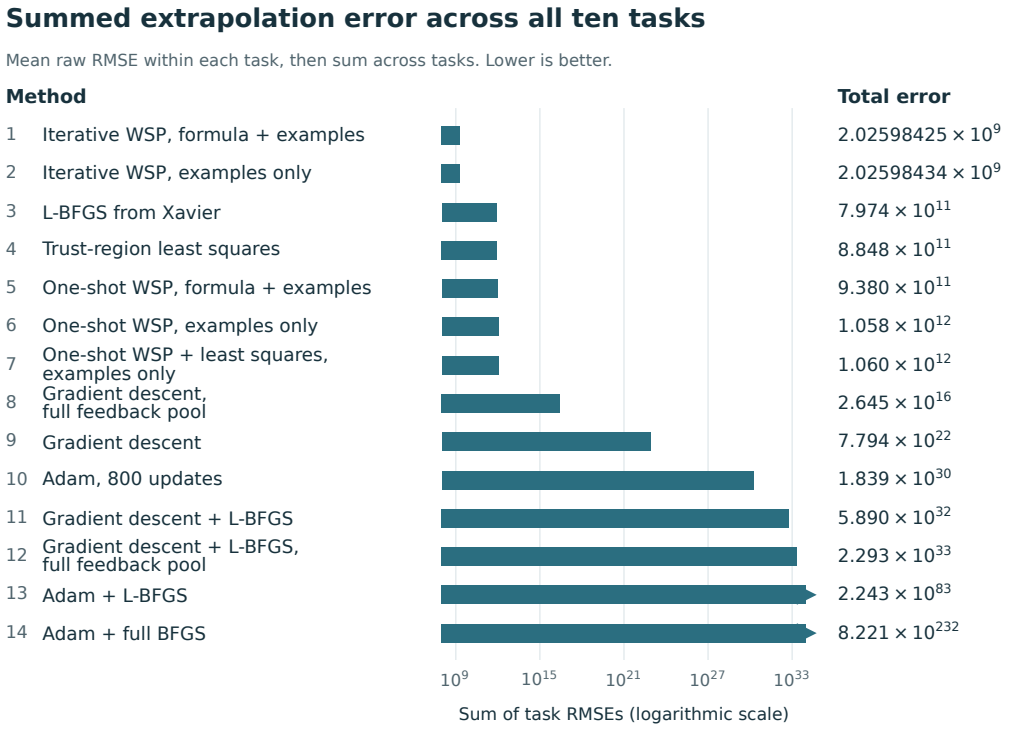


Figure 1: **Iterative WSP has the lowest summed raw extrapolation error in this benchmark.** Each method is evaluated on the same 87 task, architecture and start combinations. For each task, we average raw root mean squared error (RMSE) across its tested sizes and three starts, then sum those ten task values. Each task contributes one mean; output scales are not normalized. Ranks use unrounded totals. The two iterative WSP totals agree to six significant figures. The full feedback-pool controls use additional labels, and WSP followed by least squares receives an additional numerical stage. Adam means adaptive moment estimation; BFGS means Broyden–Fletcher–Goldfarb–Shanno. Fibonacci supplies 99.8% of the examples-only one-shot total; positive self-power supplies 78.1% of its iterative counterpart. The horizontal axis is logarithmic. Arrows mark the two totals beyond 10^{34}, with their actual values printed at the right. Figure 12 shows the entire range.

3 Tasks, Architectures, and Optimization Methods

An *architecture* specifies the operations and which scalar coefficients may change. A *rectified linear unit* (ReLU) applies $R(u) = \max(0, u)$ to an input u. A multilayer perceptron (MLP) applies a sequence of affine maps (weighted sums plus biases) and activations. A recurrent neural network (RNN) carries a state vector from one step to the next and reuses the same weights at each step. Table 1 lists every primary graph. The counts include all trainable biases and shared weights, with each stored scalar counted once.

Graph definitions. The scalar targets are x^2, $\sin x$ in radians, x^x on positive inputs, and the hyperbolic tangent $\tanh x$. For each hidden unit j let u_j, v_j, c_j denote its input slope, offset and output weight, and let w be the width. With a direct linear path of slope a and offset b, the scalar prediction is $ax + b + \sum_{j=1}^{w} c_j R(u_j x + v_j)$. Each unit introduces at most one change in slope, called a hinge. The addition and subtraction networks take two inputs, have no direct path and learn two input weights, a bias and an output weight per hidden unit, plus a final output bias.

For running sum the scalar input at tick t is x_t. With a state of width d, a learned transition matrix A, input column b and readout column v, the graph computes $h_t = Ah_{t-1} + bx_t$ from $h_0 = 0$ and predicts $v^\top h_t$ at every tick. Here $\top$ denotes transpose. For Fibonacci and triangular numbers

Table 1: Architectures in the primary comparison. Sizes and parameter counts correspond left to right. A width counts hidden units or recurrent state coordinates. For sorting, size counts consecutive residual blocks, each acting on the same four wires.

Target	Network	Sizes	Trainable parameters
Square, sine, positive self-power, hyperbolic tangent	One hidden ReLU layer with a direct input path	$1, 4, 16$	$5, 14, 50$
Addition, subtraction	Two-input ReLU MLP	$1, 2, 4$	$5, 9, 17$
Running sum	Linear RNN with input	$1, 2$	$3, 8$
Sort four numbers	Four-wire residual ReLU network	$4, 5, 6$	$32, 40, 48$
Fibonacci numbers	Autonomous linear RNN	$1, 2, 3$	$3, 8, 15$
Triangular numbers	Autonomous linear RNN	$2, 3, 4$	$8, 15, 24$

the learned column b is instead the initial state h_0, there are no inputs after initialization, and each tick applies $h_{t+1} = Ah_t$. At integer index $n \geq 0$, Fibonacci numbers start at $0, 1$ and each later term is the sum of its two predecessors. Triangular numbers equal $n(n+1)/2$. For sorting, the state h initially holds the four inputs. Each block updates it to $h + vR(u^\top h)$ with its own learned four-component columns u, v. The final four values should give the original inputs in ascending order.

How we checked nonconvexity. For each of the 87 fixed-network settings, we exhibit two weight vectors whose midpoint has loss strictly greater than the average endpoint loss. Each witness uses two trainable coefficients that multiply along an active path. Swapping those coefficients preserves the endpoint prediction while changing the midpoint prediction. Exact rational evaluation on the actual training observations certifies the violation. Appendix A.9 supplies the construction and its scope. The additional architecture-selection experiment can choose a different graph and is reported separately.

Initialization and observations. We used three random starts per task and architecture. Under Xavier-uniform initialization a layer with p inputs and q outputs draws its weights uniformly between $-\sqrt{6/(p+q)}$ and $\sqrt{6/(p+q)}$. Designated biases start at zero. All primary methods start with the same initial arrays and 12 central labeled inputs for targets with sampled inputs. A running-sum example is a length-six stream and a sorting example is a four-item list. The two integer sequences instead use the nine fixed indices 0 through 8. A new seed changes both the random training data and the weights where inputs are sampled. Test inputs are fixed across sizes and seeds. The three sizes (two for running sum) share a single WSP request, and the three integer-sequence starts share the same training indices.

Objective and methods. For observed scalar prediction $\hat{y}$ with label y, the scaled residual is $(\hat{y} - y)/\max(1, |y|)$. The common training objective is half the mean squared scaled residual, over every output coordinate. Both WSP prompts explicitly prioritize this objective and request extrapolation among similarly good fits. An *endpoint* is the weight array a method returns using only the specified training information. The original four endpoints are the two WSP conditions and two numerical solvers.

- **One-shot WSP.** In the *Examples only* condition, Astra sees the graph, initial weights, intended input range and examples. In *Formula and examples*, it also sees the target rule. It returns one array per size in a single reply without tools or scoring feedback.
- **Iterative WSP.** We retain both original information conditions. At each active size, Astra receives the selected weights, exact-match counts, errors and up to three worst labeled examples from each of the training and development sets. The development pool contains 256 new central inputs per sampled task and start, or indices 9–16 for the integer sequences. A revision replaces the current weights only when its development loss is strictly lower or, at equal development loss, its training loss is lower. An invalid proposal counts as a rejection. Each size stops when every training and development example is exact, after two consecutive rejections, or after three revision attempts. This is a finite stopping rule, not a guarantee of convergence. Requests for different cases may run concurrently. Each revision uses a fresh Astra session with the same task and selected weights; the prompt

forbids tools, and we inspect the recorded events for tool use. Test inputs and scores never enter a revision. Appendix A.1 describes the audit and controls.

- **Gradient descent and gradient descent followed by L-BFGS.** Full-batch gradient descent uses the derivative computed by backpropagation and a training-loss line search for up to 800 updates. From that endpoint, the limited-memory Broyden–Fletcher–Goldfarb–Shanno method (L-BFGS) approximates curvature with ten recent pairs of weight and gradient changes, for up to 1,000 iterations [13]. We score the two endpoints separately.
- **Adam followed by BFGS.** Adaptive moment estimation (Adam) scales each update using running averages of the gradient and its square [10]. From Xavier we run 800 full-batch steps and then up to 1,000 iterations of the Broyden–Fletcher–Goldfarb–Shanno method (BFGS), which estimates local curvature from changes in gradients. The solver returns the lower training loss of its two stages.
- **Trust-region least squares.** Starting at the identical Xavier weights, this solver uses the matrix of first derivatives of individual residuals (the Jacobian) to select local steps. It allows up to 250 residual evaluations followed by up to ten local polishing steps accepted on training loss.

Appendix A.3 also reports L-BFGS directly from Xavier and from the 800-step Adam endpoint. The primary numerical solvers fit the original training labels. A separate control fits both gradient baselines to the entire labeled training and development pool.

The archive also scores the initial Xavier array. Astra brings pretrained knowledge of familiar formulas and operates under a different computational budget. Examples-only prompts disclose the deployment range and a task-specific graph. A match on visible information therefore controls the experiment more narrowly than a match on prior knowledge or compute.

Evaluation. We evaluate root mean squared error (RMSE) in target units over all scalar outputs. An exact prediction equals its reference in the standard 64-bit floating-point format (binary64), treating both signs of zero as equal. For running sums and sorting, one example counts as exact only if every coordinate matches. For the integer sequences we count matching indices on a single trajectory. Metrics are computed per endpoint before aggregation across starts. The central test uses fresh values in $[-3, 3]$, and the outer test uses $[-10, -3]$ and $[3, 10]$ with both signs for signed tasks. Positive self-power uses training inputs in $[0.02, 3]$, central and development inputs drawn uniformly from $[0, 3]$ (positive almost surely), and outer inputs in $[3, 10]$. Each scalar evaluation has 2,048 samples per available outer side. Each arithmetic, sum or sorting test has 2,048 examples per split. The integer sequences use unseen indices 17–32 for early transfer and 33–64 for late transfer. The original one-shot study used 9–32 for early transfer. We preserve those scores separately because indices 9–16 are now used for feedback. Running sums in the primary comparison keep length six as magnitudes change. The supplement separately tests length 32. We report raw errors within tasks. For the aggregate ranking, we average raw outer RMSE over the sizes and starts of each task and then sum those ten task means. The supplementary plots also retain scaled errors and paired comparisons.

4 Results

Total error across all tasks. Figure 1 ranks all 14 evaluated optimizer pipelines by summed raw extrapolation RMSE. Each task contributes its mean error across sizes and starts. Iterative WSP has the smallest sum in both information conditions, approximately 2.03×10^9. The examples-only one-shot sum is 1.06×10^{12}. More than 99.8% of that one-shot total comes from Fibonacci, while positive self-power contributes 78.1% of the iterative total. The aggregate therefore emphasizes errors on targets with large values. Figure 12 shows the full range including divergent recurrent fits.

Central and extrapolation behavior. The median scaled error answers a different aggregation question. With examples alone, iteration reduces the held-out central or early median from 0.0214 to 0.000765, while the extrapolation median rises from 0.057 to 0.235. The scaled error divides each output residual by the larger of one and the target magnitude before computing RMSE. Figure 13 retains that view in the appendix. Table 2 gives the eight primary endpoints.

What changes during revision? Figure 2 separates training fit, development feedback and untouched extrapolation across three revision opportunities. Development loss cannot rise under the selection rule, but neither training nor extrapolation error has that guarantee. With formulas and examples, median scaled outer RMSE falls from 0.0731 to 0.00590 after the first opportunity, then rises to 0.227 after the second. The final table alone hides this transient improvement. Test scores are retrospective and never select weights; stopped runs retain their last state. Appendix A.2 separates proposed and retained weights and gives raw errors by task.

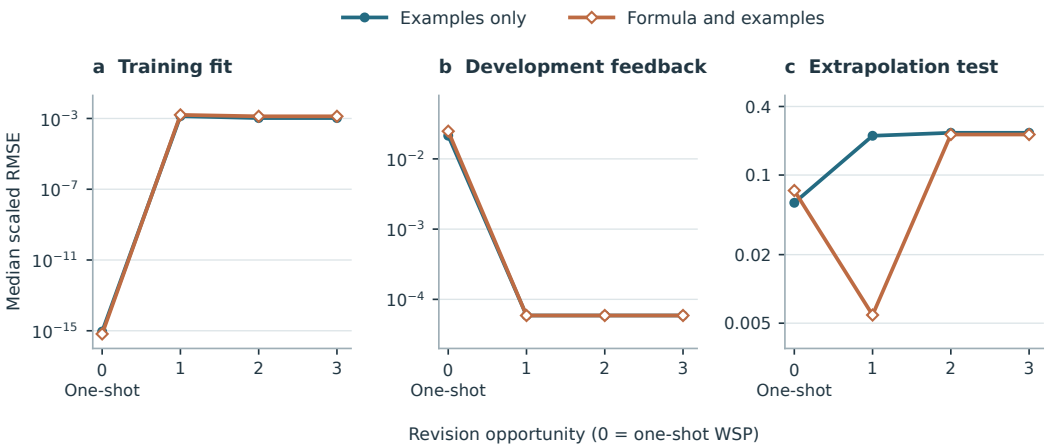


Figure 2: **Development gains do not guarantee extrapolation gains.** Each point is the median scaled root mean squared error (RMSE) of selected weights across the same 87 settings in one information condition. Round 0 is one-shot WSP; rounds 1–3 are revision opportunities, with stopped runs retaining their last weights. Development data supply feedback and select revisions. Extrapolation scores were computed only after selection was frozen and never supplied feedback or selected weights. All vertical axes are logarithmic with different ranges. Thirty settings per condition stay exact on all three splits; their zeros remain in the medians. Lines join recorded summaries, not continuous optimization steps.

Table 2: Final prediction errors on the same 87 settings. Each error column gives the median scaled root mean squared error (RMSE) across settings. For each prediction, the error is divided by the larger of one and the absolute target value. The final column counts settings in which every extrapolation output was exactly correct. Bold marks the best value in each column, including ties.

	Median scaled RMSE			Fully exact
Method	Training	Central / early	Extrapolation	outer sets (of 87)
Weight-space programming (WSP)				
One-shot, examples only	9.03×10^{-16}	0.0214	**0.0571**	**30**
Iterative, examples only	0.00106	7.65×10^{-4}	0.235	**30**
One-shot, formula and examples	$\mathbf{6.62\times10^{-16}}$	0.0236	0.0731	**30**
Iterative, formula and examples	0.00132	$\mathbf{2.73\times10^{-4}}$	0.227	**30**
Gradient methods				
Gradient descent	0.072	0.254	0.961	0
Gradient descent then L-BFGS	0.0331	0.168	0.775	0
Adam then full BFGS	0.0327	0.169	0.818	0
Trust-region least squares	0.0254	0.116	0.536	0

Does feedback improve extrapolation? With examples only, iterative WSP returned lower outer RMSE than one-shot WSP in 28 of 87 settings, the same RMSE in 35, and higher RMSE in 24. With the formula supplied, the counts were 27, 35 and 25. The number of settings exact on every outer reference changed from 30 to 30 with examples only and from 30 to 30 with the formula.

In both conditions, the selected development loss fell strictly in 52 settings and tied in 35. In the examples-only condition the loss on the original training examples rose in 46, fell in six and tied in 35. The prompt requests exact unseen-domain outputs among similarly fitting solutions. The revision selector gives priority to central development loss. Figure 3 shows the computed functions for three fixed scalar cases. Across all six width-16 square networks, iteration moves every change of slope into $[-3, 3]$, leaving affine predictions on the outer intervals. At input 10, one examples-only run changes its prediction from approximately 100 to 46.7. Appendix A.5 reproduces the instructions, and Appendix A.1 gives the coefficient-level diagnosis.

How does WSP compare with backpropagation-based updates? One-shot examples-only WSP gave lower outer RMSE than plain gradient descent in 75 of 87 settings and lower RMSE than gradient descent followed by L-BFGS in 75. For iterative examples-only WSP these counts were 78 and 77, respectively. Neither training-only gradient procedure returned an entirely exact outer set. Refinement nevertheless mattered: at width two on subtraction, median RMSE fell from 0.831 after plain gradient descent to 1.44×10^{-13} after L-BFGS. The two examples-only WSP procedures both matched all 2,048 inputs at each start. The original examples-only reply had lower outer RMSE than Adam followed by full BFGS in 73 settings and than trust-region least squares in 76. Its training loss was lower than Adam followed by BFGS in 79 settings. The full-information gradient control in Appendix A.1 checks what changes if the numerical solvers also have access to the development labels.



Figure 3: **Improved central fit can coexist with a worse extrapolation rule.** Each panel uses a 16-unit scalar network with 50 coefficients, the first recorded random start (164), and examples-only WSP. Shading marks the training and development interval; dots are the twelve training pairs. Curves are evaluations of the frozen networks, not averages over runs. These illustrative regressions show how similar central predictions conceal different outer continuations; they do not estimate how often this happens. Tables 3 and 4 summarize errors across all three starts.

How large were the errors? Table 3 in the appendix gives the size of the errors and the exact-match counts at a specified network size for every task. Figures 4–8 plot those same values for all eight methods. We chose sizes from the architecture and representability arguments: the largest tested scalar width, the two-unit arithmetic construction, and the smallest known sufficient real-arithmetic size for the sequence and sorting tasks. Table 4 gives all 29 pairs on the outer test. Tables 5 and 6 report the central or early test and the training examples. No test score was used to pick an optimizer checkpoint.

Did the optimizer reach the global training minimum? For the four scalar tasks with one hidden unit we can compute the global optimum on the supplied labels in real arithmetic. After sorting the training inputs there are finitely many intervals for the change of slope. For each interval we fit the two affine pieces subject to continuity, also checking the boundaries, and take the smallest feasible loss. Appendix N gives the rational arithmetic procedure for these same training data. This yields an independent benchmark in 12 settings. At an absolute tolerance of 10^{-10} above the

certified optimum, one-shot examples-only WSP reached it in 1 of 12 settings, one-shot formula-and-examples WSP in 2, Adam followed by BFGS in 1, and trust-region least squares from Xavier in 5. Iterative WSP reached it in 0 and 0 settings in the two information conditions, respectively; gradient descent reached it in 0 and gradient descent followed by L-BFGS in 1. This reverses the ordering of examples-only WSP and least squares seen in the outer-error comparisons.

What if least squares starts from the WSP weights? We took each of the 87 examples-only WSP arrays and applied the same trust-region solver and training-loss stopping rules used from Xavier. On the 12 certified fits this initialization brought the solver within tolerance of the global minimum in 12 of 12. Across the full benchmark the refinement lowered training loss by more than 10^{-10} in 22 settings. Among those 22, mean squared scaled outer error (with the same factor of one half as training) increased by more than 10^{-10} in 13 and decreased by more than 10^{-10} in nine. The 13 deteriorations also increased raw outer RMSE. The number of settings with every outer prediction exact stayed at 30. This two-stage experiment gives the refiner an additional model call through its starting weights. Zero extra numeric solver steps were available to the single-response WSP methods.

5 Choosing the Architecture from Examples

The fixed-network comparison asks each method to choose weights in a supplied graph. Here we ask Astra to choose the graph as well. Thirty independent requests cover ten tasks, with three requests per task using the original example sets. Each request supplies the input/output pairs, input shape and deployment domain. The prompt makes extrapolation the primary objective and asks for economical coefficient storage, inference memory and arithmetic among comparably accurate solutions.

The model can compose affine maps, rectified linear or hyperbolic tangent activations, multiplicative gates, recurrence, convolution and attention. It chooses every node, connection and coefficient within a maximum of 64 nodes and 4,096 stored coefficients. A convolution applies shared weights to local input windows. Attention uses input-dependent weights to combine values. These operations support multilayer, recurrent, convolutional and attention-based architectures. Direct multiplication and hyperbolic tangent can implement individual targets in this task set, so operator choice is part of the result.

We request one graph and a short construction explanation in each final answer. These are one-shot proposals; the calls use no tools or external numerical solver. The collector freezes all responses before a separate evaluator scores the unseen inputs. We count every request, including malformed graphs or nonfinite outputs. A floating-point operation (FLOP) is one scalar arithmetic operation. Comparisons and calls to exponential or hyperbolic tangent are counted separately. Memory includes stored 64-bit coefficients and peak live tensor storage for one input under the declared schedule. The memory count excludes operator-internal temporary arrays and Python or allocator overhead; it is not a measured process-memory footprint. Appendix A.6 gives the full operator grammar, per-trial outcomes and archived explanations. Across 30 independent architecture proposals from input/output pairs, 28 were executable and 21 matched every original extrapolation output. All three proposals were exact for square, hyperbolic tangent, addition, subtraction, running sum, Fibonacci and triangular numbers. The sine proposals had median outer RMSE 7.79×10^{-16}; sorting errors were about 5.2×10^{-16}. Positive self-power had one executable proposal with RMSE 1.68×10^{4} and two invalid proposals. Astra chose feedforward arithmetic or activation graphs and recurrent networks; no proposal chose convolution or attention. Square used one allowed multiplication, zero stored coefficients and 16 bytes of live activation storage. Its graph specification still occupied 109 serialized bytes. Operator choice explains this reduction relative to a ReLU-only network. The trials establish observed implementations and resource costs, with no global architecture optimum claimed.

6 Discussion and Limitations

The benchmark asks two questions of the same weights. The training objective measures how well a method optimized the observations it received. Error on new input values or sequence indices measures whether those weights continued the rule. On an unseen input no optimizer can infer a

unique continuation from central labels alone. For example, for any real number a, the functions 0 and $aR(x-3)$ agree on the entire central interval $[-3,3]$. At $x=10$ their predictions differ by $7a$. Knowledge of the task family, the architecture and Astra's pretraining can all affect which continuation a solver chooses.

The fixed-network comparison covers ten deterministic numerical tasks, with up to 50 trainable coefficients. The architecture-selection experiment additionally changes the allowed graph and cost. Continuous tasks have three sampled training sets. Integer tasks have one training set and three initializations. Multiple sizes in the same WSP response share context. We give counts over these fixed cases and show each size. Population claims about optimizers in vision, language or general large-scale training require other workloads, models and budgets. The iterative method uses its development set for both feedback and selection. Three attempts cap the cost, and a rejected proposal can still perform better on unseen inputs. The original numerical baselines see fewer labels; we therefore also report an intervention giving the two gradient procedures the entire feedback pool. Neither design matches computation or pretraining. The examples-only condition still discloses the architecture and intended range, and there is no common unit equating an Astra inference to a gradient or Jacobian evaluation. Each hosted-model response is archived because the provider model name alone cannot guarantee its reproduction.

Which sizes are known to suffice? For a one-state running sum, setting all three weights to one returns every sum in the same addition order as the reference. Autonomous Fibonacci and triangular generators require minimum real-arithmetic state widths two and three, respectively, for their infinite sequences (Appendix I and the realization argument in Appendix E). For arithmetic, write $t = x_1 + x_2$ or $x_1 - x_2$. Even a single hidden unit can return $R(t+20)-20=t$ throughout the bounded real domain. The large intermediate offset can change its binary64 rounding. Two units give $R(t)-R(-t)=t$ without that offset. For the shallow scalar architecture the best possible uniform real error on square over $[-10,10]$ is $50/(w+1)^2$ at width w (Appendix R). A different deep graph returned by Astra provably rounds every square correctly on the outer shell; an engineered extension covers the full binary64 interval (Appendices Y and Z). Those architectures and proofs give constructive bounds. They do not establish a minimal deep network for the smooth functions.

The supplementary edit theorem fixes a single output coordinate and counts how many initialized coefficients must change to implement a recurrence (Appendix C). That question has a different unit of cost from the primary comparison, where each solver can change every weight. The supplement also studies a register machine, an abstract computer that updates integer registers according to conditional instructions. Its proof concerns unbounded exact integers; the reported binary64 runs test bounded executions (Appendix V). Neither construction is counted as a win in the optimizer comparison.

Interpreting aggregate error and model size. The raw-RMSE sum gives each task one averaged contribution, but the target scale determines that contribution's magnitude. Large Fibonacci errors dominate many totals, especially at state widths that cannot exactly represent the recurrence. A rank under this measure differs from a rank under median scaled error. The architecture experiment measures the outputs and resource costs of 30 proposals. Its declared operations include a direct product and a hyperbolic tangent activation, so exactness on square and hyperbolic tangent can follow from one primitive. Zero stored coefficients attain the nonnegative coefficient-count lower bound. A smallest graph, smallest total storage or least arithmetic for every task would require additional proofs. Serialized graph storage is reported even when there are no stored coefficients.

7 Conclusion

Holding the task and graph fixed makes it possible to measure three different properties of an optimizer's final weights: how well they fit the displayed examples, how close they come to the right answer on new inputs, and whether they reproduce every reference exactly. Across the fixed-network benchmark, iterative WSP has the smallest summed raw extrapolation RMSE among the 14 evaluated pipelines. Its lower total coexists with regressions on individual tasks and an unchanged count of 30 fully exact settings. Allowing Astra to choose the graph yields 21 entirely exact extrapolation sets in 30 further requests, with costs determined by the permitted neural operations. Reporting the returned weights, input/output error and execution cost together exposes these tradeoffs.

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A Final weights at each network size

Figures 4–8 show the exact counts and errors from Table 3 as bars at one network size per task. Each task has its own error scale in the units of its output. Tables 4–6 then give all 29 task and architecture pairs on the extrapolation, central or early, and training sets. Even on addition and

Table 3: Outer-test results at one prespecified size per task. Continuous inputs have training magnitudes up to 3 and outer values in $[-10, -3]$ or $[3, 10]$, with positive self-power restricted to $[3, 10]$. Sequences test indices 33–64. Entries are three-start medians. Exact counts whole examples out of N. Root mean squared error (RMSE) is in output units. Bold marks the highest positive exact count and lowest RMSE across both panels, including all ties before rounding. Size counts hidden units, state coordinates or residual blocks. Slots count trainable coefficients. Adam is adaptive moment estimation. L-BFGS is limited-memory Broyden–Fletcher–Goldfarb–Shanno (BFGS).

A. Weight-space programming				One-shot Examples only		Iterative Examples only		One-shot Formula and examples		Iterative Formula and examples	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	16	50	4096	0	**1.83**	0	23.9	0	1.9	0	23.8
Sine	16	50	4096	0	0.1725	0	3.85	0	**0.1724**	0	3.85
Positive self-power	16	50	2048	0	1.79×10^{9}	0	1.58×10^{9}	0	$\mathbf{1.37 \times 10^{9}}$	0	1.58×10^{9}
Hyperbolic tangent	16	50	4096	0	$\mathbf{4.45 \times 10^{-4}}$	0	0.0769	0	6.8×10^{-4}	0	0.0781
Addition	2	9	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Subtraction	2	9	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Running sum	1	3	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Sort four	5	40	2048	**1096**	5.23×10^{-16}	1091	$\mathbf{4.94 \times 10^{-16}}$	**1096**	5.23×10^{-16}	**1096**	5.23×10^{-16}
Fibonacci	2	8	32	**32**	**0**	**32**	**0**	**32**	**0**	**32**	**0**
Triangular	3	15	32	**32**	**0**	**32**	**0**	**32**	**0**	**32**	**0**

B. Gradient methods				Gradient descent Full-batch		Gradient descent then L-BFGS		Adam then full BFGS		Trust-region least squares	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	16	50	4096	0	34.8	0	33.5	0	29.5	0	31.9
Sine	16	50	4096	0	2.33	0	2.52	0	2.46	0	2.49
Positive self-power	16	50	2048	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}
Hyperbolic tangent	16	50	4096	0	0.649	0	0.343	0	0.52	0	0.394
Addition	2	9	2048	0	4.9	0	4.2	1	2.1	324	4.55
Subtraction	2	9	2048	0	0.831	14	1.44×10^{-13}	50	3.64	769	1.22×10^{-15}
Running sum	1	3	2048	0	7.02×10^{-14}	0	7.02×10^{-14}	0	2.27×10^{-14}	65	1.73×10^{-15}
Sort four	5	40	2048	26	4.44	8	6.09	12	5.95	4	5.11
Fibonacci	2	8	32	0	2.44×10^{10}	0	0.0715	0	0.269	0	0.0102
Triangular	3	15	32	0	8.3×10^{5}	0	7.51×10^{-8}	0	2.12×10^{-8}	0	5.78×10^{-10}

subtraction, whose generating rule is linear, training a two-layer network introduces nonconvexity in the weights. On these two tasks the width-two and width-four one-shot WSP networks in both information conditions reproduced every outer output in all 12 settings. At width two on subtraction, the least-squares optimizer had median outer RMSE 1.22×10^{-15} and matched 769 of 2,048 inputs exactly at the median. At the same width on addition its median outer RMSE was 4.55. Training loss at or below 10^{-10} occurred in 17 of the 18 arithmetic least-squares fits. A threshold on training loss alone would miss both the near-exact subtraction and the large error on addition. The archived width-two one-shot WSP coefficients implement the target with a pair of opposite hidden weights in all six replies for each arithmetic task. Appendix A.10 decodes a literal reply and tests the symmetries of the scalar networks.

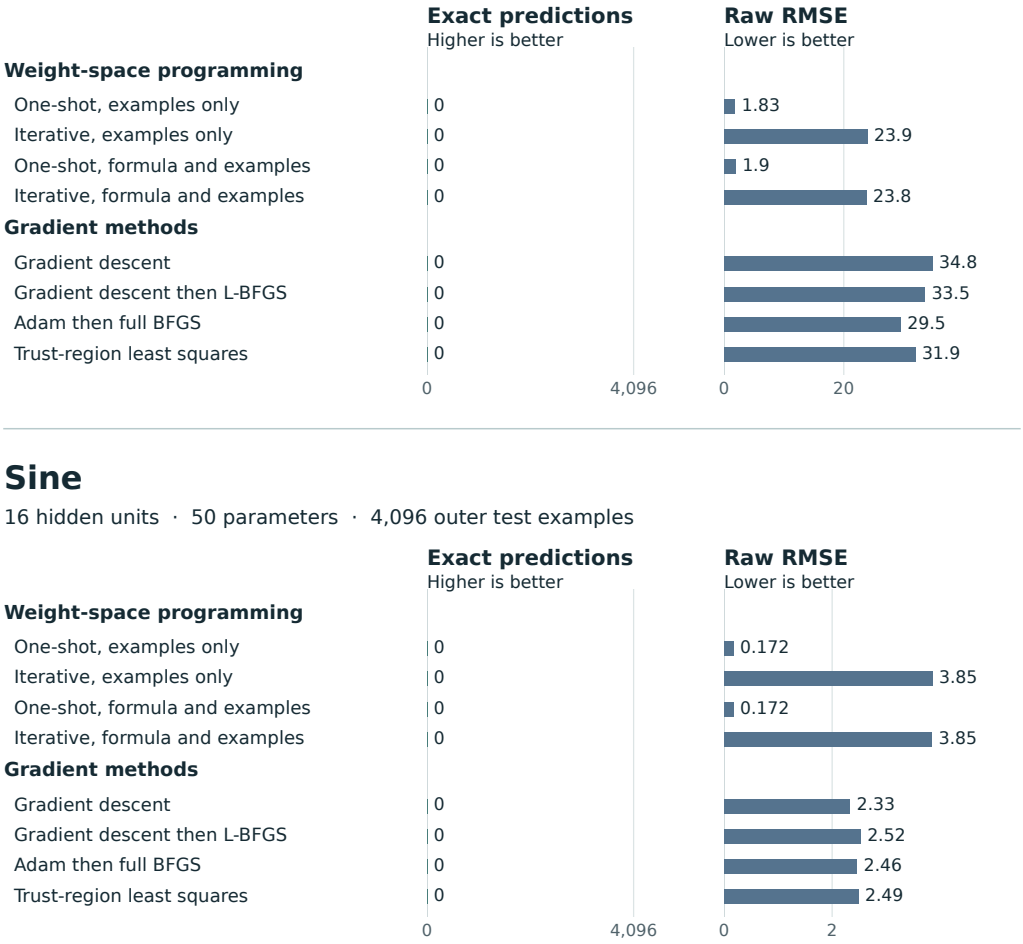


Figure 4: Square and sine on the extrapolation test at the sizes in Table 3. Bars show medians over three starts. Exact predictions count whole examples that match the reference. Raw root mean squared error (RMSE) is in the target's units. Each task has its own linear error axis starting at zero.

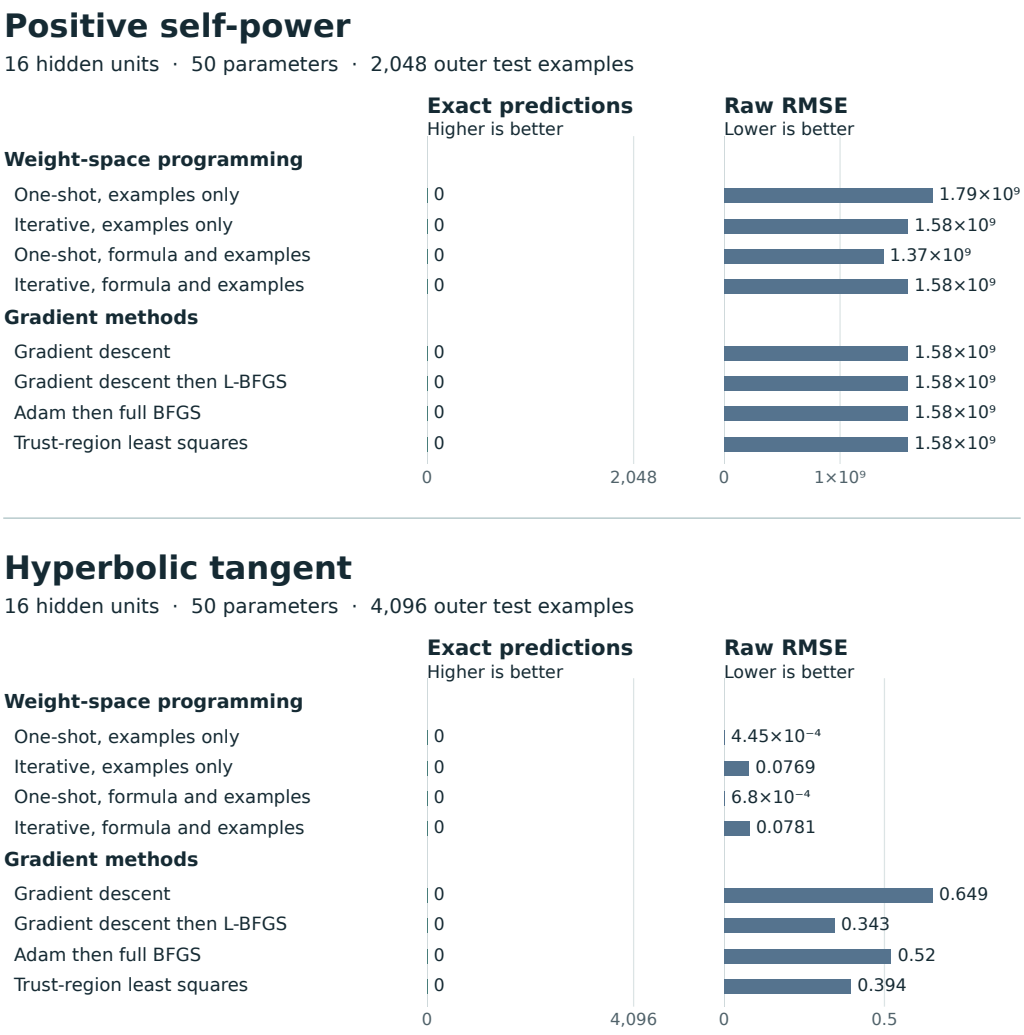


Figure 5: Positive self-power and hyperbolic tangent on the extrapolation test at the sizes in Table 3. Bars show medians over three starts. Exact predictions count whole examples that match the reference. Raw root mean squared error (RMSE) is in the target's units. Each task has its own linear error axis starting at zero.

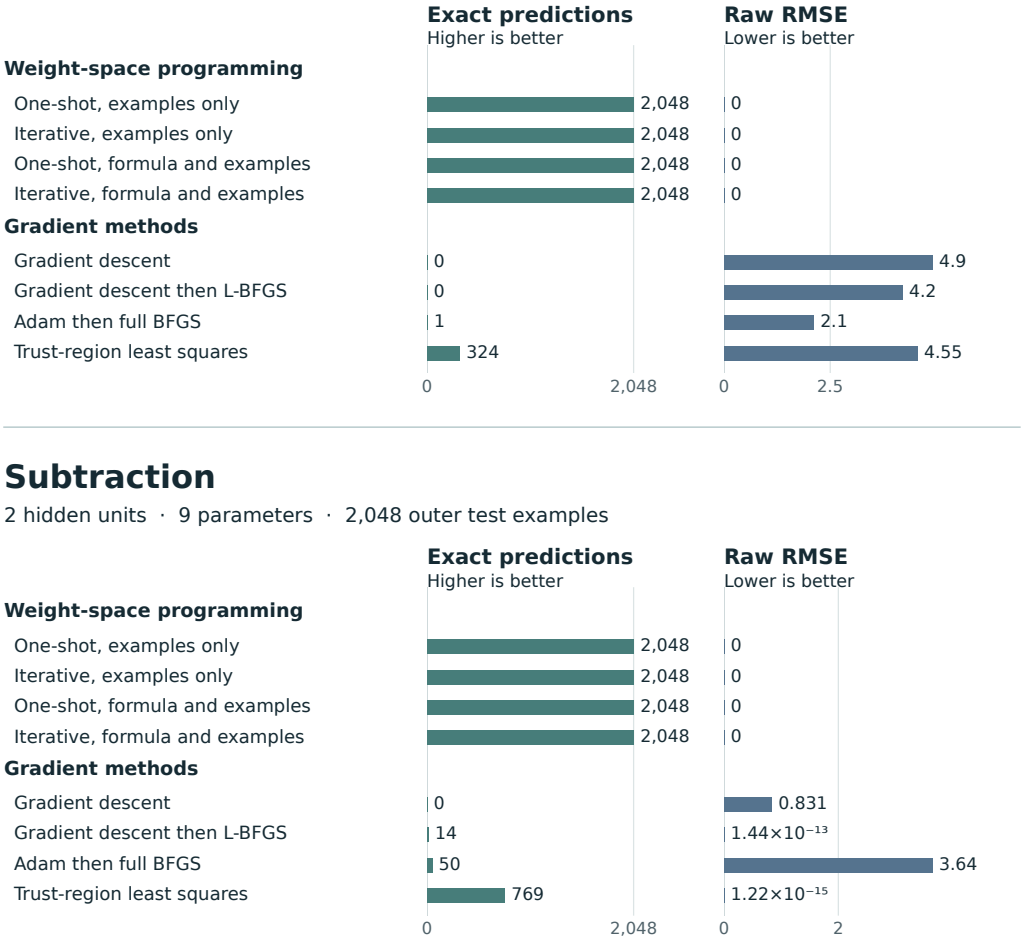


Figure 6: Addition and subtraction on the extrapolation test at the sizes in Table 3. Bars show medians over three starts. Exact predictions count whole examples that match the reference. Raw root mean squared error (RMSE) is in the target's units. Each task has its own linear error axis starting at zero.

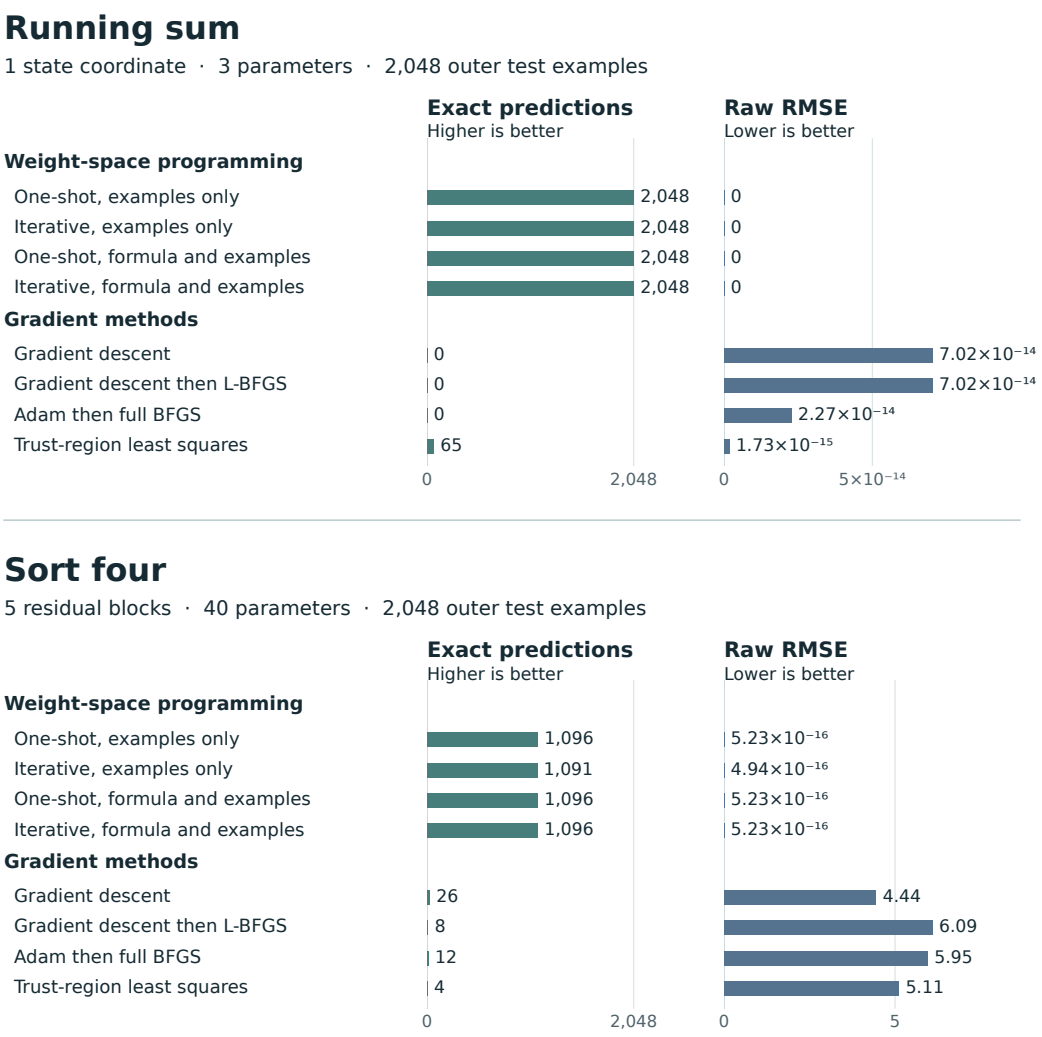


Figure 7: Running sum and sort four on the extrapolation test at the sizes in Table 3. Bars show medians over three starts. Exact predictions count whole examples that match the reference. Raw root mean squared error (RMSE) is in the target's units. Each task has its own linear error axis starting at zero.

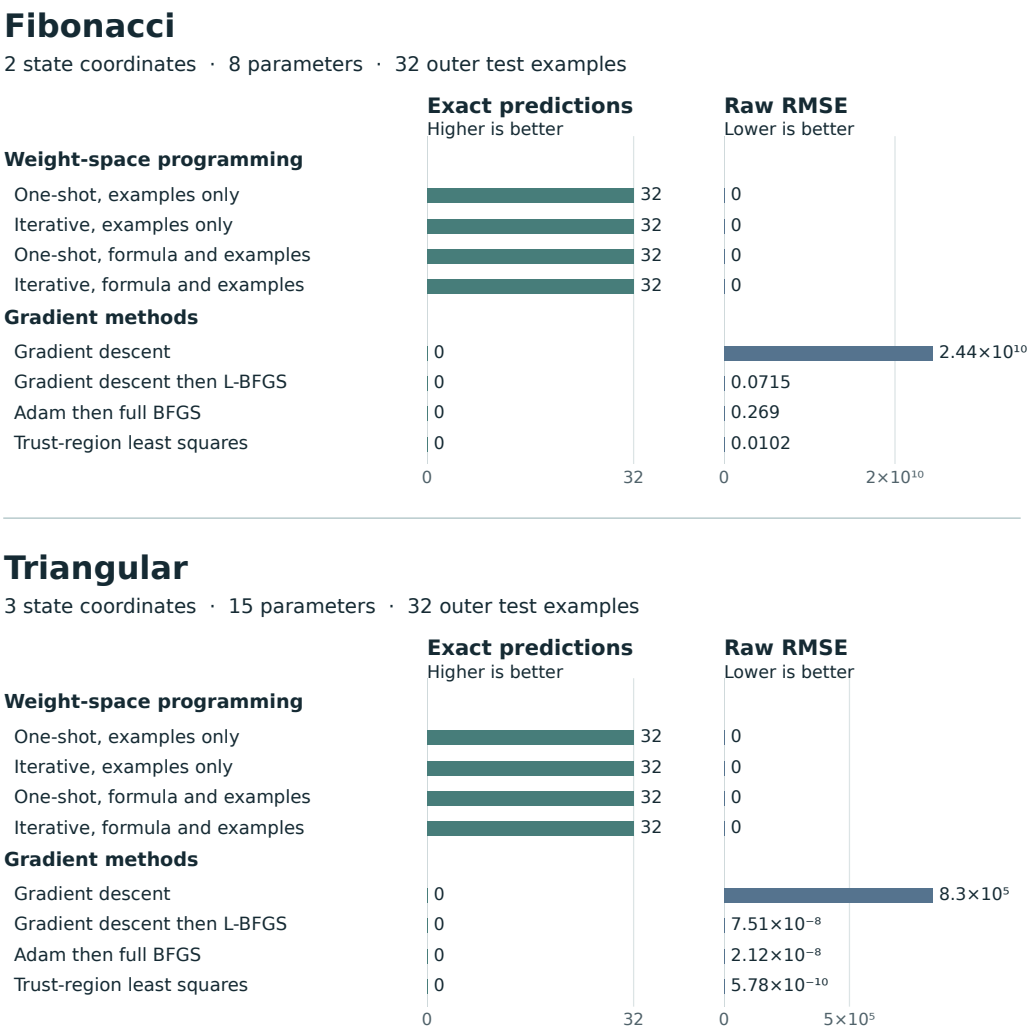


Figure 8: Fibonacci and triangular on the extrapolation test at the sizes in Table 3. Bars show medians over three starts. Exact predictions count whole examples that match the reference. Raw root mean squared error (RMSE) is in the target's units. Each task has its own linear error axis starting at zero.

Table 4: Outer extrapolation for weight-space programming (WSP) and gradient methods. Continuous inputs have training magnitudes up to 3 and outer values in $[-10, -3]$ or $[3, 10]$, with positive self-power restricted to $[3, 10]$. Sequences test indices 33–64. Entries are three-start medians. Exact counts whole examples out of N. Root mean squared error (RMSE) is in output units. Bold marks the highest positive exact count and lowest RMSE across both panels, including all ties before rounding. Size counts hidden units, state coordinates or residual blocks. Slots count trainable coefficients. Adam is adaptive moment estimation. L-BFGS is limited-memory Broyden–Fletcher–Goldfarb–Shanno (BFGS).

A. Weight-space programming				One-shot Examples only		Iterative Examples only		One-shot Formula and examples		Iterative Formula and examples	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	4096	0	**39.38**	0	40.5	0	39.39	0	40.5
	4	14	4096	0	**28.4**	0	29.6	0	28.42	0	29.6
	16	50	4096	0	**1.83**	0	23.9	0	1.9	0	23.8
Sine	1	5	4096	0	3.58	0	3.03	0	3.58	0	3.03
	4	14	4096	0	2.6	0	3.08	0	2.6	0	3.08
	16	50	4096	0	0.1725	0	3.85	0	**0.1724**	0	3.85
Positive self-power	1	5	2048	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}	0	$\mathbf{1.58\times10^{9}}$
	4	14	2048	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}	0	$\mathbf{1.58\times10^{9}}$
	16	50	2048	0	1.79×10^{9}	0	1.58×10^{9}	0	$\mathbf{1.37\times10^{9}}$	0	1.58×10^{9}
Hyperbolic tangent	1	5	4096	0	2.1	0	1.72	0	2.1	0	**1.71**
	4	14	4096	0	0.298	0	0.235	0	0.298	0	**0.227**
	16	50	4096	0	$\mathbf{4.45\times10^{-4}}$	0	0.0769	0	6.8×10^{-4}	0	0.0781
Addition	1	5	2048	1081	$\mathbf{1.21\times10^{-15}}$	**1287**	3.89	1153	5.19	1282	3.82
	2	9	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
	4	17	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Subtraction	1	5	2048	1097	$\mathbf{1.2\times10^{-15}}$	**1288**	3.76	1097	$\mathbf{1.2\times10^{-15}}$	**1288**	3.76
	2	9	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
	4	17	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Running sum	1	3	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
	2	8	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Sort four	4	32	2048	**715**	2.99	200	2.27	**715**	2.65	95	**2.09**
	5	40	2048	**1096**	5.23×10^{-16}	1091	$\mathbf{4.94\times10^{-16}}$	**1096**	5.23×10^{-16}	**1096**	5.23×10^{-16}
	6	48	2048	**1096**	5.23×10^{-16}	1091	$\mathbf{4.94\times10^{-16}}$	**1096**	5.23×10^{-16}	**1096**	4.98×10^{-16}
Fibonacci	1	3	32	0	3.17×10^{12}	0	$\mathbf{1.33\times10^{9}}$	0	3.17×10^{12}	0	1.33×10^{9}
	2	8	32	**32**	**0**	**32**	**0**	**32**	**0**	**32**	**0**
	3	15	32	**32**	**0**	**32**	**0**	**32**	**0**	**32**	**0**
Triangular	2	8	32	0	1.54×10^{9}	0	4.05×10^{3}	0	1.42×10^{9}	0	$\mathbf{4.01\times10^{3}}$
	3	15	32	**32**	**0**	**32**	**0**	**32**	**0**	**32**	**0**
	4	24	32	**32**	**0**	**32**	**0**	**32**	**0**	**32**	**0**

B. Gradient methods				Gradient descent Full-batch		Gradient descent then L-BFGS		Adam then full BFGS		Trust-region least squares	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	4096	0	51.8	0	51.8	0	51.8	0	40
	4	14	4096	0	39.1	0	39.1	0	37.3	0	32.9
	16	50	4096	0	34.8	0	33.5	0	29.5	0	31.9
Sine	1	5	4096	0	2.57	0	2.57	0	**2.18**	0	3.42
	4	14	4096	0	2.15	0	2.45	0	2.26	0	**2.08**
	16	50	4096	0	2.33	0	2.52	0	2.46	0	2.49
Positive self-power	1	5	2048	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}
	4	14	2048	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}
	16	50	2048	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}	0	1.58×10^{9}
Hyperbolic tangent	1	5	4096	0	2.23	0	2.23	0	1.77	0	2.26
	4	14	4096	0	0.916	0	0.618	0	0.741	0	0.701
	16	50	4096	0	0.649	0	0.343	0	0.52	0	0.394
Addition	1	5	2048	0	9.68	0	9.68	0	9.63	398	5.03
	2	9	2048	0	4.9	0	4.2	1	2.1	324	4.55
	4	17	2048	0	0.266	2	0.266	32	0.815	344	0.0921
Subtraction	1	5	2048	0	4.92	30	4.77	223	4.46	443	4.33
	2	9	2048	0	0.831	14	1.44×10^{-13}	50	3.64	769	1.22×10^{-15}
	4	17	2048	0	0.754	2	0.00435	0	2.53×10^{-13}	366	2.91×10^{-15}
Running sum	1	3	2048	0	7.02×10^{-14}	0	7.02×10^{-14}	0	2.27×10^{-14}	65	1.73×10^{-15}
	2	8	2048	0	1.01×10^{-13}	0	1.01×10^{-13}	23	1.95×10^{-15}	17	2.05×10^{-15}
Sort four	4	32	2048	29	5	46	5.86	14	6.17	32	5.73
	5	40	2048	26	4.44	8	6.09	12	5.95	4	5.11
	6	48	2048	0	5.71	0	5.52	0	6.7	0	4.41
Fibonacci	1	3	32	0	3.17×10^{12}	0	3.17×10^{12}	0	3.17×10^{12}	0	2.39×10^{12}
	2	8	32	0	2.44×10^{10}	0	0.0715	0	0.269	0	0.0102
	3	15	32	0	4.94×10^{11}	0	0.0279	0	0.212	1	0.00238
Triangular	2	8	32	0	1.47×10^{9}	0	1.47×10^{9}	0	1.47×10^{9}	0	1.47×10^{9}
	3	15	32	0	8.3×10^{5}	0	7.51×10^{-8}	0	2.12×10^{-8}	0	5.78×10^{-10}
	4	24	32	0	1.59×10^{7}	0	6.88×10^{-7}	0	2.61×10^{-7}	0	3.13×10^{-5}

Table 5: Weight-space programming (WSP) and gradient methods on central and early-index generalization. Fresh continuous inputs have magnitudes up to 3. Number sequences use held-out indices 17–32. The central test excludes all revision feedback inputs. Entries are three-start medians. Exact counts whole examples out of N. Root mean squared error (RMSE) is in output units. Bold marks the highest positive exact count and lowest RMSE across both panels, including all ties before rounding. Size counts hidden units, state coordinates or residual blocks. Slots count trainable coefficients. Adam is adaptive moment estimation. L-BFGS is limited-memory Broyden–Fletcher–Goldfarb–Shanno (BFGS).

A. Weight-space programming				One-shot Examples only		Iterative Examples only		One-shot Formula and examples		Iterative Formula and examples	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	2048	0	**1.245**	0	1.3	0	1.246	0	1.3
	4	14	2048	0	0.278	0	0.189	0	0.276	0	**0.188**
	16	50	2048	0	0.383	0	0.0146	0	0.358	0	**0.0141**
Sine	1	5	2048	0	0.42	0	**0.338**	0	0.42	0	0.339
	4	14	2048	0	0.0766	0	0.02821	0	0.0977	0	**0.02818**
	16	50	2048	0	0.0326	0	0.00273	0	0.0326	0	**0.00268**
Positive self-power	1	5	2048	0	3.03	0	2.03	0	3.03	0	**1.92**
	4	14	2048	0	1.23	0	0.393	0	1.23	0	**0.383**
	16	50	2048	0	0.355	0	0.0366	0	1.16	0	**0.0333**
Hyperbolic tangent	1	5	2048	0	0.229	0	**0.1877**	0	0.229	0	0.1881
	4	14	2048	0	0.0236	0	**0.0118**	0	0.0236	0	0.0119
	16	50	2048	0	0.0122	0	**0.00114**	0	0.0112	0	0.00116
Addition	1	5	2048	333	$\mathbf{1.02\times10^{-15}}$	1046	0.00454	**1262**	0.431	1022	1.53×10^{-4}
	2	9	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
	4	17	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Subtraction	1	5	2048	315	1.03×10^{-15}	**1004**	$\mathbf{3.32\times10^{-16}}$	315	1.03×10^{-15}	**1004**	3.38×10^{-16}
	2	9	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
	4	17	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Running sum	1	3	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
	2	8	2048	**2048**	**0**	**2048**	**0**	**2048**	**0**	**2048**	**0**
Sort four	4	32	2048	**802**	0.791	207	0.562	**802**	0.655	100	**0.501**
	5	40	2048	**1335**	$\mathbf{1.15\times10^{-16}}$	1286	1.16×10^{-16}	**1335**	$\mathbf{1.15\times10^{-16}}$	**1335**	$\mathbf{1.15\times10^{-16}}$
	6	48	2048	**1335**	$\mathbf{1.15\times10^{-16}}$	1286	1.16×10^{-16}	**1335**	$\mathbf{1.15\times10^{-16}}$	1334	1.16×10^{-16}
Fibonacci	1	3	16	0	3.15×10^{5}	0	**137**	0	3.15×10^{5}	0	137
	2	8	16	**16**	**0**	**16**	**0**	**16**	**0**	**16**	**0**
	3	15	16	**16**	**0**	**16**	**0**	**16**	**0**	**16**	**0**
Triangular	2	8	16	0	3.72×10^{4}	0	52	0	3.65×10^{4}	0	**50.8**
	3	15	16	**16**	**0**	**16**	**0**	**16**	**0**	**16**	**0**
	4	24	16	**16**	**0**	**16**	**0**	**16**	**0**	**16**	**0**

B. Gradient methods				Gradient descent Full-batch		Gradient descent then L-BFGS		Adam then full BFGS		Trust-region least squares	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	2048	0	3.27	0	3.27	0	3.27	0	1.246
	4	14	2048	0	1.24	0	1.24	0	1.05	0	0.665
	16	50	2048	0	0.765	0	0.575	0	0.193	0	0.576
Sine	1	5	2048	0	0.445	0	0.445	0	0.386	0	0.443
	4	14	2048	0	0.242	0	0.109	0	0.169	0	0.171
	16	50	2048	0	0.132	0	0.0671	0	0.106	0	0.0852
Positive self-power	1	5	2048	0	5.59	0	5.59	0	5.59	0	5.59
	4	14	2048	0	5.59	0	5.59	0	3.62	0	5.59
	16	50	2048	0	3.81	0	3.81	0	4.45	0	3.62
Hyperbolic tangent	1	5	2048	0	0.237	0	0.237	0	0.204	0	0.237
	4	14	2048	0	0.114	0	0.133	0	0.0923	0	0.0423
	16	50	2048	0	0.0421	0	0.0368	0	0.0322	0	0.0273
Addition	1	5	2048	0	2.43	0	2.43	0	2.44	727	0.344
	2	9	2048	0	0.248	1	0.0323	1	0.335	492	0.448
	4	17	2048	0	0.0161	9	0.0156	10	0.109	550	0.0807
Subtraction	1	5	2048	0	0.304	185	0.24	286	0.152	585	0.106
	2	9	2048	0	0.167	65	2.85×10^{-14}	453	0.0888	860	3.29×10^{-16}
	4	17	2048	0	0.152	6	0.00129	3	6.14×10^{-14}	557	6.73×10^{-16}
Running sum	1	3	2048	0	1.79×10^{-14}	0	1.79×10^{-14}	0	5.79×10^{-15}	86	4.23×10^{-16}
	2	8	2048	0	2.56×10^{-14}	0	2.56×10^{-14}	21	5.09×10^{-16}	21	5.19×10^{-16}
Sort four	4	32	2048	21	1.32	43	1.58	9	1.47	27	1.5
	5	40	2048	25	1.11	9	1.65	16	1.53	3	1.27
	6	48	2048	0	1.44	0	1.43	0	1.64	0	1.09
Fibonacci	1	3	16	0	6.93×10^{5}	0	6.93×10^{5}	0	6.93×10^{5}	0	6.93×10^{5}
	2	8	16	0	3.34×10^{3}	0	9.4×10^{-9}	0	3.57×10^{-8}	0	1.26×10^{-9}
	3	15	16	0	6.04×10^{4}	0	3.78×10^{-9}	0	2.78×10^{-8}	4	1.83×10^{-10}
Triangular	2	8	16	0	3.68×10^{4}	0	3.68×10^{4}	0	3.68×10^{4}	0	3.68×10^{4}
	3	15	16	0	1.59×10^{3}	0	1.28×10^{-9}	0	4.96×10^{-10}	0	1.21×10^{-11}
	4	24	16	0	6.23×10^{3}	0	1.69×10^{-8}	0	5.63×10^{-9}	0	8.18×10^{-11}

Table 6: Weight-space programming (WSP) and gradient methods on observed training examples. Observed data contain 12 examples per sampled task or the nine sequence indices 0–8. Entries are three-start medians. Exact counts whole examples out of N. Root mean squared error (RMSE) is in output units. Bold marks the highest positive exact count and lowest RMSE across both panels, including all ties before rounding. Size counts hidden units, state coordinates or residual blocks. Slots count trainable coefficients. Adam is adaptive moment estimation. L-BFGS is limited-memory Broyden–Fletcher–Goldfarb–Shanno (BFGS).

A. Weight-space programming				One-shot Examples only		Iterative Examples only		One-shot Formula and examples		Iterative Formula and examples	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	12	0	**0.7916**	0	1.2	0	0.7924	0	1.2
	4	14	12	0	**0.00612**	0	0.183	0	0.00664	0	0.185
	16	50	12	7	$\mathbf{2.9\times10^{-16}}$	0	0.014	**9**	4.62×10^{-16}	0	0.0144
Sine	1	5	12	0	**0.236**	0	0.326	0	0.236	0	0.313
	4	14	12	0	0.00334	0	0.0306	0	**0.00237**	0	0.0303
	16	50	12	0	$\mathbf{6.7\times10^{-9}}$	0	0.00298	0	9.39×10^{-9}	0	0.00278
Positive self-power	1	5	12	0	0.603	0	**0.597**	0	0.603	0	0.61
	4	14	12	0	**0.00863**	0	0.219	0	0.00863	0	0.201
	16	50	12	0	2.32×10^{-8}	0	0.0147	0	$\mathbf{7.05\times10^{-9}}$	0	0.0213
Hyperbolic tangent	1	5	12	0	**0.154**	0	0.201	0	0.154	0	0.187
	4	14	12	0	0.00151	0	0.0111	0	**0.00151**	0	0.0136
	16	50	12	0	2.33×10^{-7}	0	0.00106	0	$\mathbf{1.48\times10^{-8}}$	0	0.00132
Addition	1	5	12	2	1.02×10^{-15}	4	3.27×10^{-16}	**8**	$\mathbf{2.31\times10^{-16}}$	4	3.07×10^{-16}
	2	9	12	**12**	**0**	**12**	**0**	**12**	**0**	**12**	**0**
	4	17	12	**12**	**0**	**12**	**0**	**12**	**0**	**12**	**0**
Subtraction	1	5	12	2	7.28×10^{-16}	**6**	$\mathbf{2.79\times10^{-16}}$	2	7.28×10^{-16}	**6**	$\mathbf{2.79\times10^{-16}}$
	2	9	12	**12**	**0**	**12**	**0**	**12**	**0**	**12**	**0**
	4	17	12	**12**	**0**	**12**	**0**	**12**	**0**	**12**	**0**
Running sum	1	3	12	**12**	**0**	**12**	**0**	**12**	**0**	**12**	**0**
	2	8	12	**12**	**0**	**12**	**0**	**12**	**0**	**12**	**0**
Sort four	4	32	12	**3**	0.248	1	0.488	**3**	0.362	1	0.472
	5	40	12	**6**	$\mathbf{1.24\times10^{-16}}$	**6**	$\mathbf{1.24\times10^{-16}}$	**6**	$\mathbf{1.24\times10^{-16}}$	**6**	$\mathbf{1.24\times10^{-16}}$
	6	48	12	**6**	$\mathbf{1.24\times10^{-16}}$	**6**	$\mathbf{1.24\times10^{-16}}$	**6**	$\mathbf{1.24\times10^{-16}}$	**6**	1.4×10^{-16}
Fibonacci	1	3	9	0	0.296	0	0.19	0	0.297	0	**0.19**
	2	8	9	**9**	**0**	**9**	**0**	**9**	**0**	**9**	**0**
	3	15	9	**9**	**0**	**9**	**0**	**9**	**0**	**9**	**0**
Triangular	2	8	9	0	0.535	0	3.46	0	**0.485**	0	3.5
	3	15	9	**9**	**0**	**9**	**0**	**9**	**0**	**9**	**0**
	4	24	9	**9**	**0**	**9**	**0**	**9**	**0**	**9**	**0**

B. Gradient methods				Gradient descent Full-batch		Gradient descent then L-BFGS		Adam then full BFGS		Trust-region least squares	
Target	Size	Slots	N	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	12	0	2.8	0	2.8	0	2.8	0	1.07
	4	14	12	0	0.792	0	0.792	0	0.434	0	0.405
	16	50	12	0	0.296	0	0.29	0	0.00464	0	0.225
Sine	1	5	12	0	0.399	0	0.399	0	0.399	0	0.238
	4	14	12	0	0.0679	0	0.0327	0	0.0339	0	0.0339
	16	50	12	0	0.0272	0	0.00294	0	0.0017	0	0.0111
Positive self-power	1	5	12	0	3.03	0	3.03	0	3.03	0	3.03
	4	14	12	0	3.03	0	3.03	0	1.48	0	2.67
	16	50	12	0	0.912	0	0.912	0	2.07	0	1.48
Hyperbolic tangent	1	5	12	0	0.211	0	0.211	0	0.21	0	0.195
	4	14	12	0	0.0327	0	0.0279	0	0.0135	0	0.0146
	16	50	12	0	0.0097	0	0.00527	0	0.00312	0	0.00709
Addition	1	5	12	0	1.72	0	1.72	0	1.72	5	5.48×10^{-16}
	2	9	12	0	0.427	0	1.09×10^{-13}	0	3.15×10^{-14}	6	2.72×10^{-16}
	4	17	12	0	0.00363	0	2.08×10^{-14}	0	4.93×10^{-15}	7	2×10^{-16}
Subtraction	1	5	12	0	0.0851	2	1.15×10^{-15}	1	8.33×10^{-16}	4	5.91×10^{-16}
	2	9	12	0	0.106	0	5.02×10^{-15}	3	8.86×10^{-16}	7	3.17×10^{-16}
	4	17	12	0	0.00799	0	3.14×10^{-14}	0	6.16×10^{-14}	5	3.45×10^{-16}
Running sum	1	3	12	0	1.25×10^{-14}	0	1.25×10^{-14}	0	5.07×10^{-15}	1	3.14×10^{-16}
	2	8	12	0	2.03×10^{-14}	0	2.03×10^{-14}	0	4.84×10^{-16}	0	5.09×10^{-16}
Sort four	4	32	12	0	1.07	0	**0.243**	0	0.371	0	0.29
	5	40	12	0	0.279	0	0.238	0	0.133	0	0.303
	6	48	12	0	0.722	0	0.167	0	0.145	0	0.213
Fibonacci	1	3	9	0	7.53	0	6.78	0	5.51	0	8.91
	2	8	9	0	0.00781	1	6.71×10^{-15}	0	4.92×10^{-14}	4	1.33×10^{-15}
	3	15	9	0	0.0644	0	6.71×10^{-15}	0	3.75×10^{-14}	6	1.06×10^{-16}
Triangular	2	8	9	0	0.491	0	0.491	0	0.491	0	0.491
	3	15	9	0	0.18	0	2.71×10^{-14}	1	7.8×10^{-15}	2	3.3×10^{-15}
	4	24	9	0	0.214	0	2.44×10^{-13}	1	2.68×10^{-14}	6	1.2×10^{-15}

A.1 Validation feedback and additional gradient controls

What the model saw. Each revision request reproduced the original specification and included only the active sizes, their incumbent weights, and diagnostics. For each size and split the model received the scaled loss, raw RMSE, whole-example exact count, and at most three non-exact inputs with the largest absolute scaled error on any coordinate. For each of those inputs it saw the target and current prediction. The 256 central development inputs were fixed within task and start and shared across sizes and prompt conditions. Integer indices 9–16 instead extended the original nine labels. Original one-shot replies, new prompts, replies, decisions and final vectors are retained. Appendix X is a different experiment that allows a circuit description to change.

Stopping and selection. The collector made 161 revision requests after the original 60 model calls. Their recorded median latency was 51.7 seconds per request, with 198,446 output tokens in total; requests for different tasks ran concurrently. These requests produced 332 size-specific proposals, of which 217 were accepted; one width-16 square proposal had the wrong number of coefficients. None of the 161 requests used tools. The 174 size-by-condition trajectories share requests within an original task, start and prompt condition. Of those trajectories, 60 were exact before any revision. At the end, 60 had stopped on exact training and development outputs, 33 after two consecutive rejected proposals, and 81 at the three-attempt cap. After all selections were frozen, a separate evaluator reconstructed the feedback from training and development data, checked every decision against the saved replies and then scored the selected weights. Table 7 follows all 87 settings per condition through each revision opportunity. The cap limits what we can conclude about additional rounds.

What if gradient descent can use all feedback labels? We separately fit both gradient procedures from the same Xavier start using the original training observations pooled with every feedback label, keeping both optimizer budgets and the test sets fixed. This gives the gradient solver 268 labeled examples per sampled task or 17 integer indices; Astra saw diagnostics and selected examples from that pool. The baseline therefore has access to more individual labels than Astra. With examples only, iterative WSP had lower outer RMSE than pooled-data gradient descent in 78 settings and than its L-BFGS refinement in 80 (out of 87). The pooled procedures matched every outer reference in 0 and 0 settings, respectively. Machine-readable files include training, feedback, independently held-out central or early, and outer scores for both pooled endpoints.

Independent test draw. After the weight selection was frozen, we generated a second set of central and outer inputs using a new random seed. On the 69 settings with sampled inputs, the iterative-versus-one-shot outer RMSE comparison was 22 lower, 23 tied and 24 higher with examples only, and 20 lower, 23 tied and 26 higher with the formula supplied. The 18 integer settings have fixed target indices and are not repeated as independent draws. The companion data include the scores from both draws.

Table 7: What changed with each opportunity to revise? Each information condition keeps all 87 settings throughout. Attempted and accepted count individual size-specific proposals. After a size stops, its last retained weights carry forward. The last four columns are retrospective extrapolation scores; these scores were never shown to Astra. Exact counts settings in which every test prediction matched. Lower, equal and higher compare the retained outer RMSE with that setting's one-shot outer RMSE.

Information	Round	Attempted	Accepted	Exact	Lower	Equal	Higher
Examples only	0	–	–	30	0	87	0
	1	57	51	30	24	36	27
	2	57	32	30	28	35	24
	3	52	22	30	28	35	24
Formula and examples	0	–	–	30	0	87	0
	1	57	51	30	27	36	24
	2	57	34	30	27	35	25
	3	52	27	30	27	35	25

Why did iterative WSP lose extrapolation accuracy? The prompts explicitly asked for prediction beyond the observations. In the original eight-task prompt, Astra was told, *Among similarly fitting solutions, aim for exact outputs on unseen inputs across the stated domain.* The arithmetic prompt gave the equivalent instruction for *equally good training fits.* Each revision repeated the original prompt and added, *Seek a rule that also generalizes to examples not shown.* These instructions made extrapolation a secondary aim. The revision selector accepted weights with lower development loss, using training loss only to break a tie. That selection criterion rewards accuracy on the central interval and permits a worse continuation outside it.

The stored coefficients show the resulting tradeoff on square. In all six width-16 square networks, across both information conditions and three starts, the original reply put three or four changes of slope outside $[-3, 3]$. After iteration, every change of slope lay inside $[-3, 3]$. Each revised network therefore used a single affine continuation on either extrapolation interval. For the examples-only run at seed 164, the rightmost change of slope moved from 7.4 to 2.419 and the right-tail slope fell from 17.4 to 5.3915. At input 10, the reference is 100, the original network's real-arithmetic prediction is approximately 100, and the revised prediction is 46.6765. The selected development RMSE fell from 0.396 to 0.0145, while outer RMSE rose from 1.22 to 24.0. This coefficient-level analysis identifies the mechanism of that failure. The aggregate effect of the wording and selection rule would require a controlled ablation.

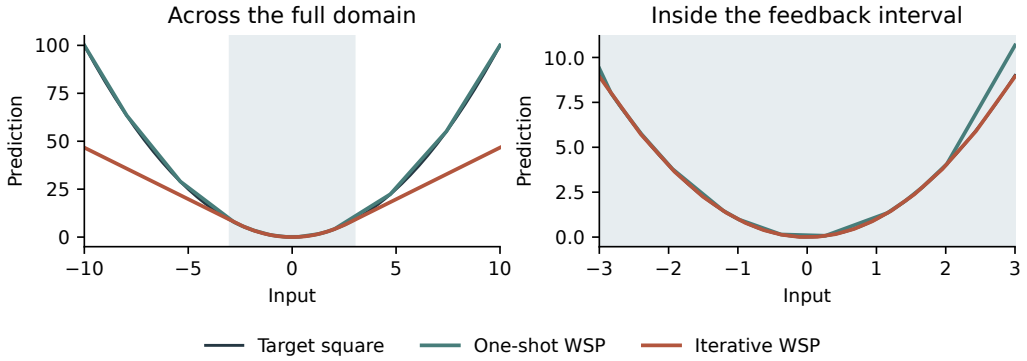


Figure 9: The actual width-16 square networks from the examples-only condition at random start 164. Training and development inputs lie in the central interval. The stored one-shot weights retain changes of slope outside that interval. Iteration concentrates all changes of slope inside it and extends an affine function into each outer interval. The curves explain this specific extrapolation regression.

A.2 Recorded iterative WSP learning curves

The plots reconstruct all 174 trajectories from their original interaction records: 87 task, size and start settings in each information condition. Each selected training and development score agrees with the separately archived evaluator, and each final state agrees with the published endpoint. A rejected proposal leaves the selected weights unchanged. After early stopping, later opportunities display that retained state; they are not additional model calls. No new fitting or test-guided selection was performed for these plots. The data archive includes every selected state, all proposal decisions, raw and scaled errors, and exact counts.

Figure 2 uses the same scaled error as the main endpoint summary: divide each output error by the larger of one and the target magnitude before computing RMSE. It aggregates by the median over settings, not by first averaging within each task. No uncertainty band is drawn; these are descriptive summaries of the fixed cases. The median training error need not decrease because acceptance prioritizes development loss. The examples-only median development error falls from 0.0214 to 5.89×10^{-5}, while the median training error rises from 9.03×10^{-16} to 1.06×10^{-3}. Formula-and-examples medians change from 0.0248 to 5.89×10^{-5} on development and from 6.62×10^{-16} to 1.32×10^{-3} on training. The maximum of three attempted rounds does not establish convergence.

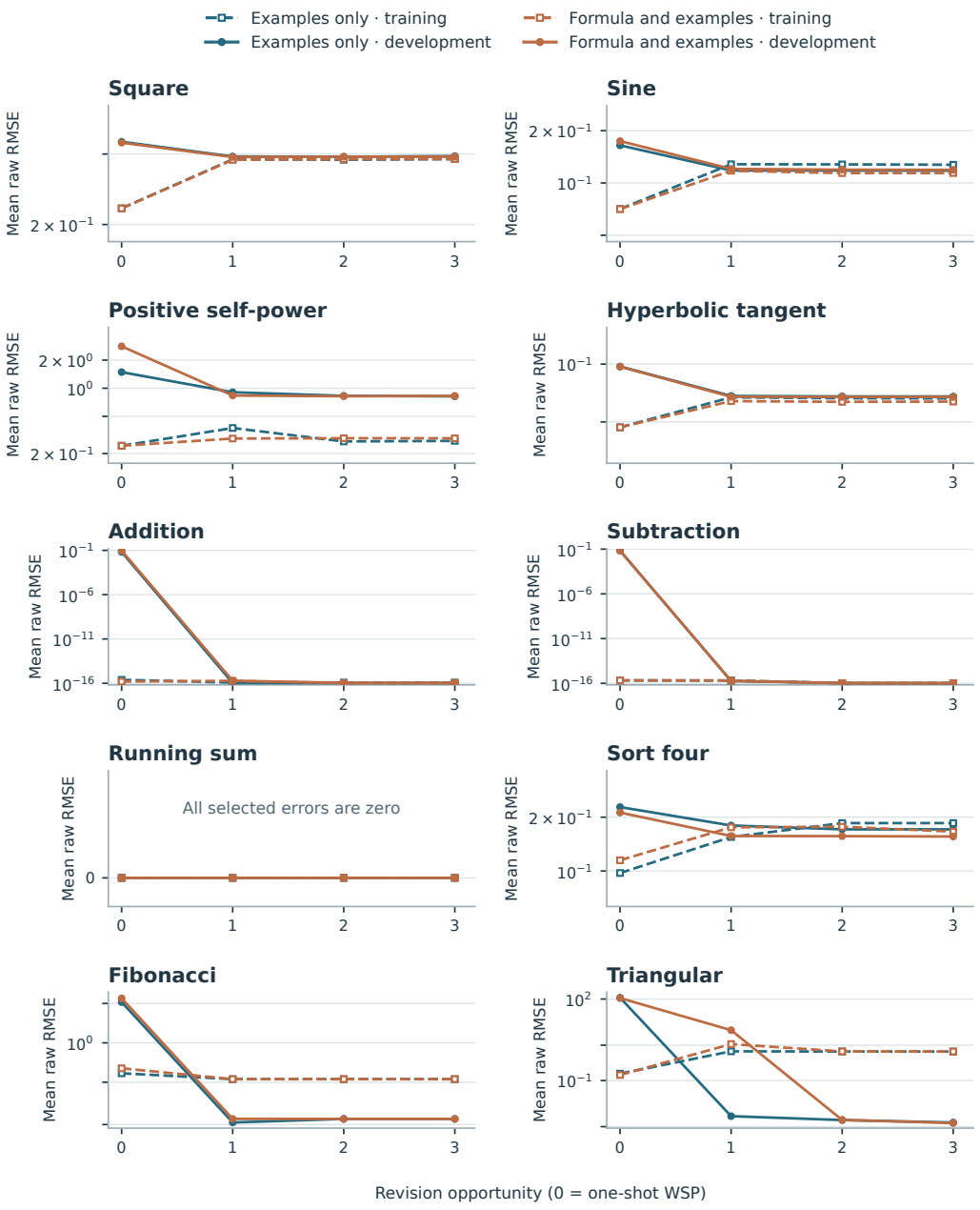


Figure 10: **Training and development curves for all ten tasks.** Each point is the mean raw RMSE over that task's tested sizes and three starts, using selected weights and carrying stopped states forward. Solid lines show development feedback; dashed lines show training. Colors distinguish examples only from formula and examples. Each task has its own logarithmic vertical range. Running sum is exact throughout and uses a linear axis at zero. Unlike the main learning curves, these within-task summaries use means, preserving the contribution of low-capacity networks that can be hidden by a zero median. The sorting panel reports errors across all output coordinates. Development data are reused for feedback and selection.

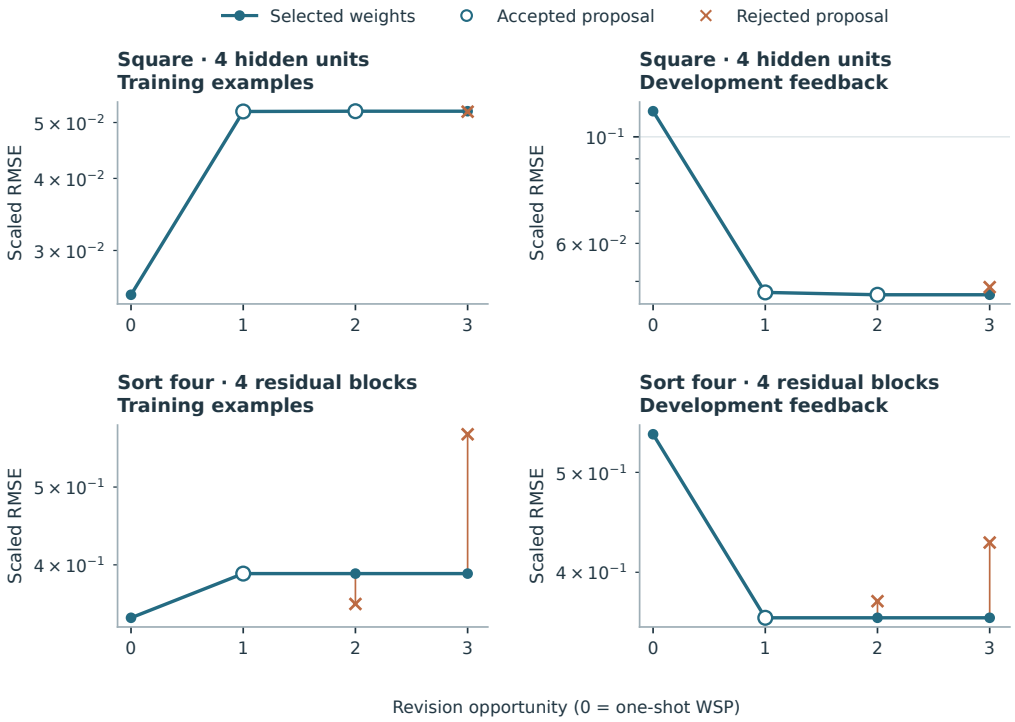


Figure 11: **A proposal is not necessarily the selected network.** Two illustrative examples-only trajectories at seed 164 show every valid proposal, the selected state, and training versus development scores. The square network has four hidden units; the sorting network has four residual blocks. Open circles mark accepted proposals and crosses mark rejected proposals. Vertical axes are logarithmic with separate ranges. Rejection leaves the selected curve flat. Selection minimizes scaled development loss first, using training loss only to break a tie. These examples explain selection behavior; Figure 10 covers the full task set.

A.3 Gradient descent and limited-memory optimization

This subsection retains the original training-only gradient comparisons. Its early integer diagnostic uses the original indices 9–32. For a comparison with iterative WSP on untouched indices 17–32, see Table 5. These fits use the same nonconvex networks, initial weights, training observations and loss as the main comparison. Plain full-batch gradient descent starts from Xavier and takes at most 800 updates. At each update it tries step size one and halves it until the training loss decreases sufficiently. Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) uses the most recent ten pairs of changes in weights and gradients to approximate curvature. We run it with no parameter bounds for up to 1,000 iterations. Its reported endpoint has the lowest training loss among the supplied starting point and its accepted iterates. Selection uses training loss throughout.

We start L-BFGS from three separate points: the original Xavier weights, the gradient-descent endpoint, or the endpoint after 800 adaptive moment estimation (Adam) updates. The Adam checkpoint in Table 8 is scored before refinement. The separate Adam–BFGS baseline applies full-memory BFGS after Adam. Each additional fit uses derivatives of the empirical loss obtained by backpropagation through the network. The complete records include optimizer termination, per-setting scores and fitted weights.

Tables 9, 10 and 11 give raw root mean squared error (RMSE) and exact matches at each tested size. Tables 4–6 give WSP and gradient results on the same outer and training inputs and on the current held-out central or early split, which uses indices 17–32 for the two integer sequences. Size denotes hidden width or the number of residual blocks for sorting. Slots count trainable scalar coefficients.

Table 8: Optimizer outcomes relative to all 87 matched settings. Near the optimum means an absolute loss gap of at most 10^{-10} to the independently certified minimum in the 12 one-hinge settings. The last column counts settings with strictly smaller outer raw RMSE for the first examples-only WSP response than for the named fit. Each setting contributes one comparison.

	Observed training set		Near the	Entire outer	WSP lower
Method	Median loss	Loss $\leq 10^{-10}$	optimum /12	set exact /87	RMSE /87
Gradient descent	0.0026	6	0	0	75
Gradient descent $\rightarrow$ L-BFGS	5.46×10^{-4}	32	1	0	75
Adam (800 updates)	0.00111	11	0	0	74
Adam $\rightarrow$ L-BFGS	5.34×10^{-4}	34	1	0	74
L-BFGS from Xavier	3.22×10^{-5}	37	3	0	71

A.4 Alternative views of the optimizer comparison

The following plots retain the earlier views of the same measured endpoints. Figure 12 displays the complete range of summed raw errors. Figure 13 summarizes the median scaled error. Figures 14 and 15 retain the original paired-comparison layouts. Their axes, legends and counts retain their original definitions.

Table 9: Exact counts and raw RMSE for gradient optimizers on the outer test. Continuous inputs have magnitudes from 3 to 10. Integer sequences use indices 33–64. Each entry is a median over the same three starts. Exact counts require every output of a stream or list, or one number at a given integer index. RMSE includes every scalar output in the target's units.

			Gradient descent		Gradient descent then L-BFGS		Adam then L-BFGS		L-BFGS from Xavier	
Target	Size	Slots	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	0/4096	51.8	0/4096	51.8	0/4096	51.8	0/4096	51.8
	4	14	0/4096	39.1	0/4096	39.1	0/4096	37.3	0/4096	40.1
	16	50	0/4096	34.8	0/4096	33.5	0/4096	29	0/4096	28.5
Sine	1	5	0/4096	2.57	0/4096	2.57	0/4096	2.18	0/4096	2.56
	4	14	0/4096	2.15	0/4096	2.45	0/4096	2.26	0/4096	2.57
	16	50	0/4096	2.33	0/4096	2.52	0/4096	2.45	0/4096	2.81
Positive self-power	1	5	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}
	4	14	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}
	16	50	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}	0/2048	1.58×10^{9}
Hyperbolic tangent	1	5	0/4096	2.23	0/4096	2.23	0/4096	1.77	0/4096	1.94
	4	14	0/4096	0.916	0/4096	0.618	0/4096	0.741	0/4096	0.551
	16	50	0/4096	0.649	0/4096	0.343	0/4096	0.307	0/4096	0.195
Addition	1	5	0/2048	9.68	0/2048	9.68	0/2048	9.63	22/2048	5.19
	2	9	0/2048	4.9	0/2048	4.2	31/2048	2.09	8/2048	5.48
	4	17	0/2048	0.266	2/2048	0.266	0/2048	0.815	2/2048	0.0279
Subtraction	1	5	0/2048	4.92	30/2048	4.77	211/2048	4.46	449/2048	4.47
	2	9	0/2048	0.831	14/2048	1.44×10^{-13}	63/2048	3.64	5/2048	6.49×10^{-14}
	4	17	0/2048	0.754	2/2048	0.00435	4/2048	5.61×10^{-13}	1/2048	6.84×10^{-13}
Running sum	1	3	0/2048	7.02×10^{-14}	0/2048	7.02×10^{-14}	0/2048	2.27×10^{-14}	0/2048	3.48×10^{-15}
	2	8	0/2048	1.01×10^{-13}	0/2048	1.01×10^{-13}	23/2048	1.95×10^{-15}	9/2048	2.72×10^{-15}
Sort four	4	32	29/2048	5	46/2048	5.86	32/2048	4.88	20/2048	4.81
	5	40	26/2048	4.44	8/2048	6.09	17/2048	8.61	30/2048	3.93
	6	48	0/2048	5.71	0/2048	5.52	0/2048	5.11	0/2048	5.77
Fibonacci	1	3	0/32	3.17×10^{12}	0/32	3.17×10^{12}	0/32	3.17×10^{12}	0/32	2.39×10^{12}
	2	8	0/32	2.44×10^{10}	0/32	0.0715	0/32	0.0503	0/32	0.0257
	3	15	0/32	4.94×10^{11}	0/32	0.0279	0/32	0.0152	0/32	0.24
Triangular	2	8	0/32	1.47×10^{9}	0/32	1.47×10^{9}	0/32	1.47×10^{9}	0/32	1.47×10^{9}
	3	15	0/32	8.30×10^{5}	0/32	7.51×10^{-8}	0/32	3.15×10^{-7}	0/32	1.99×10^{-8}
	4	24	0/32	1.59×10^{7}	0/32	6.88×10^{-7}	0/32	6.69×10^{-7}	0/32	5.65×10^{-7}

Table 10: Exact counts and raw RMSE for gradient optimizers on the central or early-index test. Continuous inputs have central magnitudes up to 3. Integer sequences use unseen indices 9–32. Each entry is a median over the same three starts. Exact counts require every output of a stream or list, or one number at a given integer index. RMSE includes every scalar output in the target's units.

			Gradient descent		Gradient descent then L-BFGS		Adam then L-BFGS		L-BFGS from Xavier	
Target	Size	Slots	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	0/2048	3.27	0/2048	3.27	0/2048	3.27	0/2048	3.27
	4	14	0/2048	1.24	0/2048	1.24	0/2048	1.05	0/2048	1.24
	16	50	0/2048	0.765	0/2048	0.575	0/2048	0.194	0/2048	0.309
Sine	1	5	0/2048	0.445	0/2048	0.445	0/2048	0.386	0/2048	0.381
	4	14	0/2048	0.242	0/2048	0.109	0/2048	0.169	0/2048	0.137
	16	50	0/2048	0.132	0/2048	0.0671	0/2048	0.111	0/2048	0.0902
Positive self-power	1	5	0/2048	5.59	0/2048	5.59	0/2048	5.59	0/2048	5.59
	4	14	0/2048	5.59	0/2048	5.59	0/2048	3.62	0/2048	5.59
	16	50	0/2048	3.81	0/2048	3.81	0/2048	4.45	0/2048	3.23
Hyperbolic tangent	1	5	0/2048	0.237	0/2048	0.237	0/2048	0.204	0/2048	0.209
	4	14	0/2048	0.114	0/2048	0.133	0/2048	0.0923	0/2048	0.0412
	16	50	0/2048	0.0421	0/2048	0.0368	0/2048	0.0266	0/2048	0.015
Addition	1	5	0/2048	2.43	0/2048	2.43	0/2048	2.44	97/2048	0.43
	2	9	0/2048	0.248	1/2048	0.0323	51/2048	0.331	25/2048	0.258
	4	17	0/2048	0.0161	9/2048	0.0156	4/2048	0.109	6/2048	1.34×10^{-13}
Subtraction	1	5	0/2048	0.304	185/2048	0.24	426/2048	0.152	539/2048	0.154
	2	9	0/2048	0.167	65/2048	2.85×10^{-14}	134/2048	0.0952	8/2048	1.67×10^{-14}
	4	17	0/2048	0.152	6/2048	0.00129	2/2048	1.11×10^{-13}	3/2048	1.50×10^{-13}
Running sum	1	3	0/2048	1.79×10^{-14}	0/2048	1.79×10^{-14}	0/2048	5.79×10^{-15}	1/2048	8.88×10^{-16}
	2	8	0/2048	2.56×10^{-14}	0/2048	2.56×10^{-14}	21/2048	5.09×10^{-16}	8/2048	7.14×10^{-16}
Sort four	4	32	21/2048	1.32	43/2048	1.58	29/2048	1.28	10/2048	1.22
	5	40	25/2048	1.11	9/2048	1.65	20/2048	2.26	35/2048	1.01
	6	48	0/2048	1.44	0/2048	1.43	0/2048	1.3	0/2048	1.5
Fibonacci	1	3	0/24	5.66×10^{5}	0/24	5.66×10^{5}	0/24	5.66×10^{5}	0/24	5.66×10^{5}
	2	8	0/24	2.73×10^{3}	0/24	7.67×10^{-9}	0/24	5.48×10^{-9}	0/24	3.12×10^{-9}
	3	15	0/24	4.93×10^{4}	0/24	3.09×10^{-9}	0/24	1.78×10^{-9}	0/24	2.54×10^{-8}
Triangular	2	8	0/24	3.00×10^{4}	0/24	3.00×10^{4}	0/24	3.00×10^{4}	0/24	3.00×10^{4}
	3	15	0/24	1.30×10^{3}	0/24	1.05×10^{-9}	0/24	5.78×10^{-9}	0/24	3.79×10^{-10}
	4	24	0/24	5.09×10^{3}	0/24	1.38×10^{-8}	0/24	1.27×10^{-8}	0/24	9.38×10^{-9}

Table 11: Exact counts and raw RMSE for gradient optimizers on the observed training set. Each setting has 12 labeled examples, or nine integer indices 0–8 for a number sequence. Each entry is a median over the same three starts. Exact counts require every output of a stream or list, or one number at a given integer index. RMSE includes every scalar output in the target's units.

			Gradient descent		Gradient descent then L-BFGS		Adam then L-BFGS		L-BFGS from Xavier	
Target	Size	Slots	Exact	RMSE	Exact	RMSE	Exact	RMSE	Exact	RMSE
Square	1	5	0/12	2.8	0/12	2.8	0/12	2.8	0/12	2.8
	4	14	0/12	0.792	0/12	0.792	0/12	0.434	0/12	1.07
	16	50	0/12	0.296	0/12	0.29	0/12	0.00464	0/12	0.0291
Sine	1	5	0/12	0.399	0/12	0.399	0/12	0.399	0/12	0.325
	4	14	0/12	0.0679	0/12	0.0327	0/12	0.0339	0/12	0.0339
	16	50	0/12	0.0272	0/12	0.00294	0/12	0.0017	0/12	4.30×10^{-4}
Positive self-power	1	5	0/12	3.03	0/12	3.03	0/12	3.03	0/12	3.03
	4	14	0/12	3.03	0/12	3.03	0/12	1.48	0/12	2.67
	16	50	0/12	0.912	0/12	0.912	0/12	2.07	0/12	0.603
Hyperbolic tangent	1	5	0/12	0.211	0/12	0.211	0/12	0.21	0/12	0.175
	4	14	0/12	0.0327	0/12	0.0279	0/12	0.0135	0/12	0.0107
	16	50	0/12	0.0097	0/12	0.00527	0/12	0.00169	0/12	0.00115
Addition	1	5	0/12	1.72	0/12	1.72	0/12	1.72	0/12	2.55×10^{-15}
	2	9	0/12	0.427	0/12	1.09×10^{-13}	0/12	6.11×10^{-15}	0/12	2.95×10^{-15}
	4	17	0/12	0.00363	0/12	2.08×10^{-14}	0/12	2.86×10^{-14}	0/12	2.27×10^{-14}
Subtraction	1	5	0/12	0.0851	2/12	1.15×10^{-15}	2/12	5.62×10^{-16}	2/12	5.48×10^{-16}
	2	9	0/12	0.106	0/12	5.02×10^{-15}	0/12	2.17×10^{-15}	0/12	1.53×10^{-14}
	4	17	0/12	0.00799	0/12	3.14×10^{-14}	0/12	4.77×10^{-14}	0/12	5.56×10^{-14}
Running sum	1	3	0/12	1.25×10^{-14}	0/12	1.25×10^{-14}	0/12	5.07×10^{-15}	0/12	8.42×10^{-16}
	2	8	0/12	2.03×10^{-14}	0/12	2.03×10^{-14}	0/12	4.84×10^{-16}	0/12	5.68×10^{-16}
Sort four	4	32	0/12	1.07	0/12	0.243	0/12	0.316	0/12	0.325
	5	40	0/12	0.279	0/12	0.238	0/12	0.135	0/12	0.271
	6	48	0/12	0.722	0/12	0.167	0/12	0.158	0/12	0.181
Fibonacci	1	3	0/9	7.53	0/9	6.78	0/9	5.68	0/9	8.91
	2	8	0/9	0.00781	1/9	6.71×10^{-15}	1/9	1.35×10^{-14}	1/9	8.14×10^{-15}
	3	15	0/9	0.0644	0/9	6.71×10^{-15}	0/9	5.76×10^{-15}	0/9	3.64×10^{-14}
Triangular	2	8	0/9	0.491	0/9	0.491	0/9	0.491	0/9	0.491
	3	15	0/9	0.18	0/9	2.71×10^{-14}	0/9	1.15×10^{-13}	0/9	5.79×10^{-14}
	4	24	0/9	0.214	0/9	2.44×10^{-13}	0/9	1.40×10^{-13}	0/9	1.01×10^{-13}

Table 12: Raw-error totals and their dominant task. The total sums each task's mean RMSE across sizes and starts. The final two columns identify the largest task contribution and its share of that method's total. The graph-choosing proposals are evaluated separately in Appendix A.6.

Rank	Method	Raw-error sum	Largest contribution	Share
1	Iterative WSP, formula + examples	2.02598×10^{9}	Positive self-power	78.12%
2	Iterative WSP, examples only	2.02598×10^{9}	Positive self-power	78.12%
3	L-BFGS from Xavier	7.97356×10^{11}	Fibonacci	99.74%
4	Trust-region least squares	8.84785×10^{11}	Fibonacci	99.77%
5	One-shot WSP, formula + examples	9.38023×10^{11}	Fibonacci	99.80%
6	One-shot WSP, examples only	1.05832×10^{12}	Fibonacci	99.80%
7	One-shot WSP + least squares, examples only	1.05966×10^{12}	Fibonacci	99.80%
8	Gradient descent, full feedback pool	2.64518×10^{16}	Fibonacci	100.00%
9	Gradient descent	7.79363×10^{22}	Fibonacci	100.00%
10	Adam, 800 updates	1.83949×10^{30}	Fibonacci	100.00%
11	Gradient descent + L-BFGS	5.88992×10^{32}	Fibonacci	100.00%
12	Gradient descent + L-BFGS, full feedback pool	2.29320×10^{33}	Fibonacci	100.00%
13	Adam + L-BFGS	2.24329×10^{83}	Fibonacci	100.00%
14	Adam + full BFGS	8.22122×10^{232}	Fibonacci	100.00%

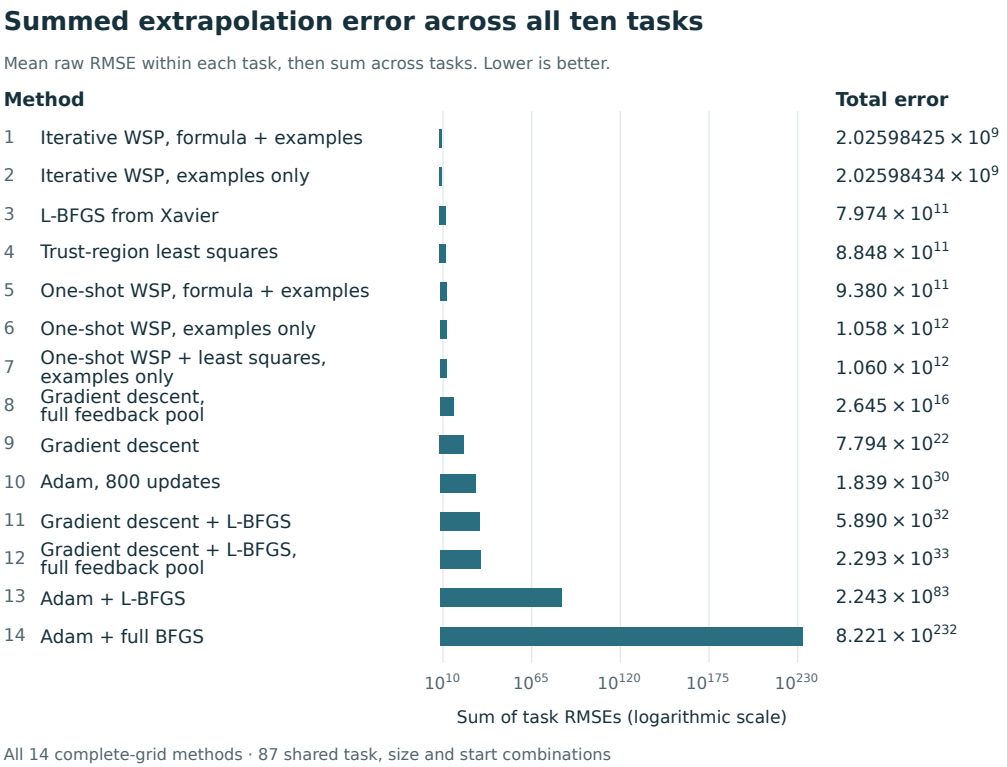


Figure 12: The full range of summed raw extrapolation errors. Each method is evaluated on the same 87 task, architecture and start combinations. For each task, we average raw root mean squared error (RMSE) across its tested sizes and three starts, then sum those ten task values. Each task contributes one mean; output scales are not normalized. Ranks use unrounded totals. The two iterative WSP totals agree to six significant figures. The full feedback-pool controls use additional labels, and WSP followed by least squares receives an additional numerical stage. Adam means adaptive moment estimation; BFGS means Broyden–Fletcher–Goldfarb–Shanno. Large-magnitude targets and individual divergent runs strongly affect this raw sum. Fibonacci contributes more than 99.8% of examples-only one-shot WSP's total. Positive self-power contributes about 78.1% of its iterative counterpart's total.

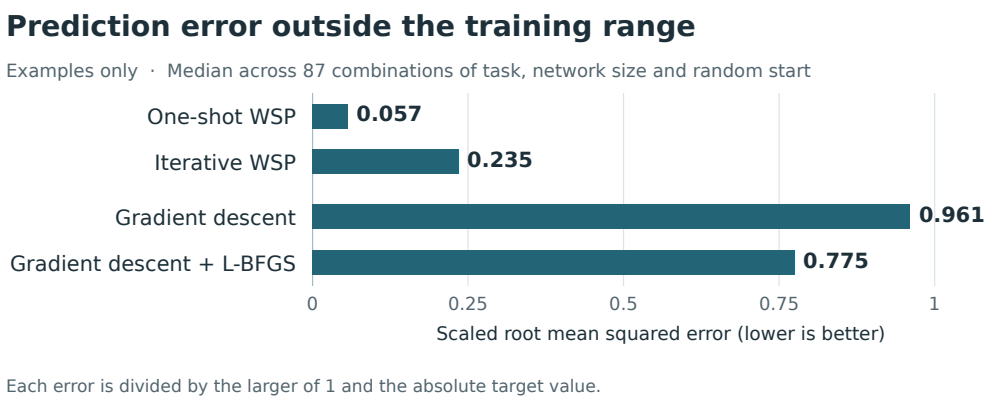


Figure 13: Extrapolation error after optimization. Each bar is the median across all 87 settings (29 task and architecture pairs with three random starts each). Both weight-space programming (WSP) methods received examples alone. Iterative WSP also received training and development feedback. Each output error is divided by the larger of one and the target magnitude before computing root mean squared error for that setting. Table 3 gives errors in the original output units for individual tasks.

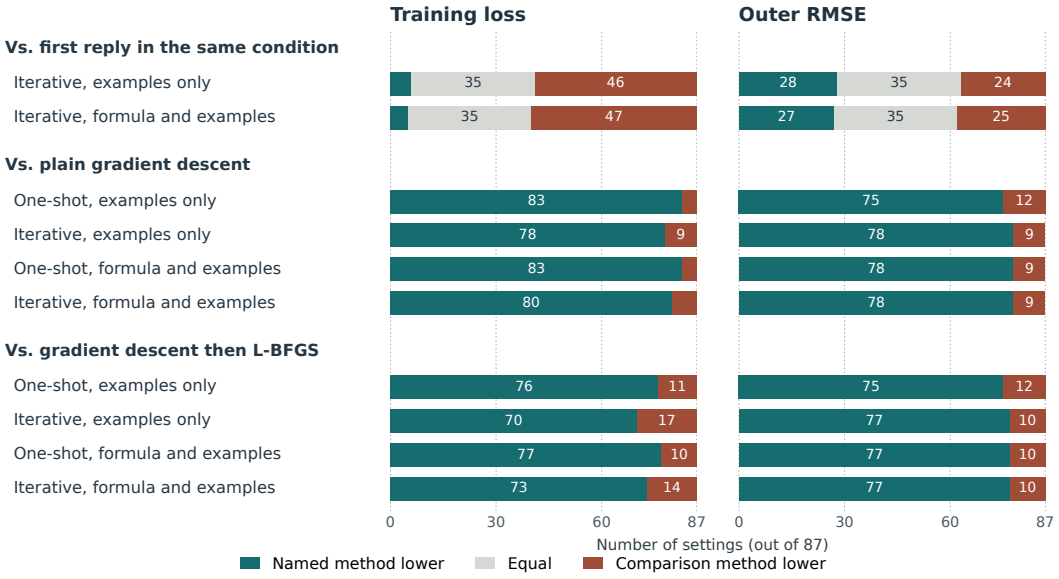


Figure 14: The earlier paired comparison in its original format. Each bar divides the same 87 settings according to whether the named WSP method has lower error, equal error, or higher error than the stated comparison. The left panel uses the scaled training loss and the right uses raw extrapolation RMSE. These counts show the direction of paired changes. Figures 1 and 12 show their magnitudes through the sum.

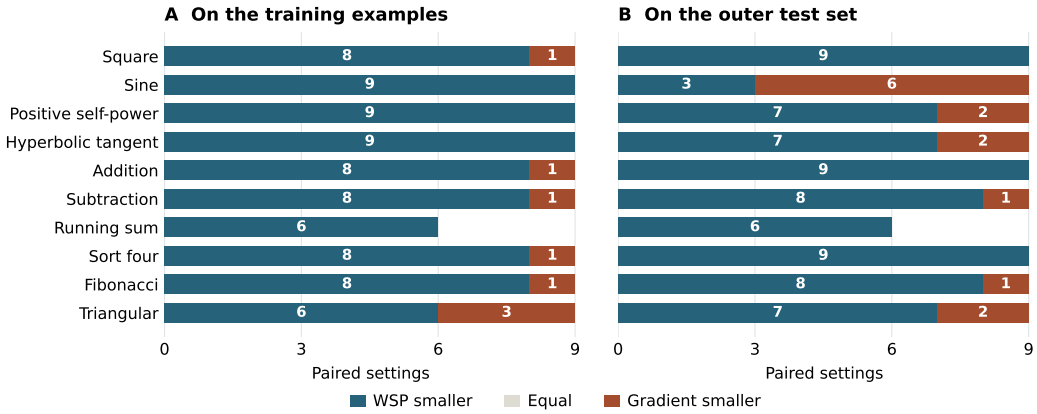


Figure 15: The first WSP reply given examples only and the Adam–BFGS gradient fit start from the same weights on the same task and architecture. Each bar counts how often one method's endpoint has smaller error or the two unrounded recorded errors agree. Even a very small strict difference is counted. Panel A uses their common loss on the labeled training examples. Panel B uses raw root mean squared error (RMSE) on held-out outer inputs in each target's own units. Each task contributes three sizes with three starts, except running sum with two sizes and three starts. Every listed setting has a certified nonconvex training objective.

A.5 Prompts and recorded interaction traces

The archive contains every original prompt and numerical reply from the 60 one-shot calls, plus every prompt, numerical reply, diagnostic, acceptance decision and event summary from the 161 iterative calls. The audit found the extrapolation instruction in all 60 original prompts and embedded in all 161 revisions. Each recorded call ended with one final model message and zero tool events.

An interaction trace here consists of the supplied prompt, returned weights, measured training and development scores, and the decision to accept or reject a proposal. The original and iterative output instructions requested only numerical arrays and explicitly forbade explanations. Their recorded responses therefore contain no explanatory derivations. Token-usage metadata includes a count labeled reasoning tokens, but it contains no text for those tokens. The new architecture-selection requests ask for a short construction summary as a public final-response field.

The complete example below is the examples-only Fibonacci request with random start 164, followed by its first revision. The original prompt requests state widths one, two and three. The latter two returned exact training and development outputs and stopped before feedback. The revision consequently requests only the one-state model. Formatting wraps the literal text for readability. The byte-exact source files, identifiers, and SHA-256 hashes are supplied in the companion archive.

Original prompt. `You are writing neural-network coefficients, in binary64. Three architectures below are independent size variants of one unknown map. Input indices are nonnegative integers up to 64. The only labeled observations provided are [[0,0.0],[1,1.0],[2,1.0],[3,2.0],[4,3.0],[5,5.0],[6,8.0],[7,13.0],[8,21.0]]. Your primary objective is to minimize the training loss on the supplied observations: half the mean squared error after dividing each scalar prediction error by max(1, abs(target)). Include every output coordinate in that mean. Among similarly fitting solutions, aim for exact outputs on unseen inputs across the stated domain. You may infer a rule from observations. Do not use any tools or external files. Return ONLY JSON of shape [vector0,vector1,vector2], each a complete numeric array of the stated flattened length. Do not return expressions, code or ellipses.`

`Variant 0 (3 coefficients): d=1; flatten A (1 by 1, row-major), then vector b, then vector v. h[0]=b, h[n+1]=A @ h[n], y[n]=v dot h[n]. Initial weights: [-1.6984773572580318, 0.1085747430453964,-0.2749483841108896]`

`Variant 1 (8 coefficients): d=2; flatten A (2 by 2, row-major), then vector b, then vector v. h[0]=b, h[n+1]=A @ h[n], y[n]=v dot h[n]. Initial weights: [0.3293508786209253, 1.0420055489082358,0.5741131105793458,-0.0012447971253200126,-0.8733483141770927, 0.4821413534074803,1.2360881054518182,-0.5097733693852923]`

`Variant 2 (15 coefficients): d=3; flatten A (3 by 3, row-major), then vector b, then vector v. h[0]=b, h[n+1]=A @ h[n], y[n]=v dot h[n]. Initial weights: [-0.9901521414641481, 0.8732758514117205,0.09902704201856238,0.5986704249125343,0.511249615668522, 0.11350660484047359,0.16969819917563278,0.016829689914695578,-0.05788416328754664, 0.2989801435980603,1.0793285516282238,1.019886067980408,1.0090474869388815,0.951521528069607, -0.6941601628334186]`

Original final response. `[[1.64203,1,0.40965],[0,1,1,1,0,1,1,0],[0,1,0,1,1,0,0,0,0,0,1,0,1,0,0]]`

First revision prompt. `You are revising previously returned numerical neural-network weights. Use no tools or external files. Here is the original task specification and the exact original initial arrays. Treat its output formatting as applying only to the original request.`

`You are writing neural-network coefficients, in binary64. Three architectures below are independent size variants of one unknown map. Input indices are nonnegative integers up to 64. The only labeled observations provided are [[0,0.0],[1,1.0],[2,1.0],[3,2.0],[4,3.0],[5,5.0],[6,8.0],[7,13.0],[8,21.0]]. Your primary objective is to minimize the training loss on the supplied observations: half the mean squared error after dividing each scalar prediction error by max(1, abs(target)). Include every output coordinate in that mean. Among similarly fitting solutions, aim for exact outputs on unseen inputs across the stated domain. You may infer a rule from observations. Do not use any tools or external files. Return ONLY JSON of shape [vector0,vector1,vector2], each a complete numeric array of the stated flattened length. Do not return expressions, code or ellipses.`

Variant 0 (3 coefficients): d=1; flatten A (1 by 1, row-major), then vector b, then vector v. h[0]=b, h[n+1]=A @ h[n], y[n]=v dot h[n]. Initial weights: [-1.6984773572580318, 0.1085747430453964,-0.2749483841108896]

Variant 1 (8 coefficients): d=2; flatten A (2 by 2, row-major), then vector b, then vector v. h[0]=b, h[n+1]=A @ h[n], y[n]=v dot h[n]. Initial weights: [0.3293508786209253, 1.0420055489082358,0.5741131105793458,-0.0012447971253200126,-0.8733483141770927, 0.4821413534074803,1.2360881054518182,-0.5097733693852923]

Variant 2 (15 coefficients): d=3; flatten A (3 by 3, row-major), then vector b, then vector v. h[0]=b, h[n+1]=A @ h[n], y[n]=v dot h[n]. Initial weights: [-0.9901521414641481, 0.8732758514117205,0.09902704201856238,0.5986704249125343,0.511249615668522, 0.11350660484047359,0.16969819917563278,0.016829689914695578,-0.05788416328754664, 0.2989801435980603,1.0793285516282238,1.019886067980408,1.0090474869388815,0.951521528069607, -0.6941601628334186]

END ORIGINAL TASK.

REVISION 1. These are the current incumbent vectors for the active architectures, in the listed order, with training and development diagnostics. The development inputs are new central observations, or the early continuation of a sequence. Each loss is half the mean of squared coordinate errors divided by max(1,abs(target)) squared. Smaller development loss improves the candidate; among identical development losses smaller training loss improves it. Exact counts require every coordinate of an example to match. For each split we show up to three worst examples with nonzero errors. Seek a rule that also generalizes to examples not shown. Revise each current vector within its specified architecture. You may keep it as is. Return ONLY a JSON array of complete numeric arrays in the active order shown below, even if only one size is active. Do not return vectors for stopped sizes, explanations or expressions. The original task's input support remains the same.

[{"size":1,"incumbent_weights":[1.64203,1.0,0.40965],
"diagnostic":{"train":{"loss":0.016460182626530297,"exact":0,"n":9,"rmse":0.2964654444894327,
"examples":[{"input":0,"target":0.0,"prediction":0.40965,"maximum_scaled_error":0.40965},
{"input":1,"target":1.0,"prediction":0.6726575895000001,
"maximum_scaled_error":0.3273424104999999},{"input":2,"target":1.0,
"prediction":1.1045239416866852,"maximum_scaled_error":0.10452394168668522}]},
"dev":{"loss":0.005861020901437812,"exact":0,"n":8,"rmse":66.51195950918358,"examples":[
{"input":16,"target":987.0,"prediction":1144.2221702520592,
"maximum_scaled_error":0.15929297897878336},{"input":15,"target":610.0,
"prediction":696.8339008739542,"maximum_scaled_error":0.14235065717041678},{"input":14,
"target":377.0,"prediction":424.37342854512656,"maximum_scaled_error":0.12565896165816062}]
}}}]

First revision final response. `[[1.61801576,1.0,0.44728357]]`

A.6 Architecture discovery protocol and resources

Each request supplies the original twelve labeled examples or nine integer indices, input/output shapes and the deployment domain. It supplies no task name, formula, initial weights or prescribed architecture. Extrapolation is the primary objective. Resource minimization is secondary. There are three independent first responses per task, and no response is selected using test data. Every graph is frozen before evaluation. The two invalid responses remain failed trials.

Allowed operators are affine maps, rectified linear units (ReLU), hyperbolic tangent, sigmoid, elementwise addition and multiplication, static tensor indexing and concatenation, reshape, transpose, matrix multiplication, softmax and one-dimensional convolution. Optional recurrence repeats a shared cell. This library permits multilayer perceptrons, recurrent neural networks, convolutional neural networks and attention networks, as well as multiplicatively gated arithmetic graphs. It bounds every proposal at 64 nodes and 4,096 scalar coefficients. A tanh node directly implements hyperbolic tangent, and a multiplication node directly implements square. Both operations therefore affect capacity and the appropriate interpretation of comparisons with the fixed ReLU architectures.

Each floating-point operation (FLOP) is a scalar addition, multiplication or division. Dot products count every multiply and accumulation, including fixed zeros and ones. Comparisons and transcendental evaluations are listed separately. The coefficient count includes every stored scalar constant, bias, matrix entry and initial-state value. Each coefficient takes eight bytes. Peak workspace counts live binary64 input, state, intermediate and output buffers under the interpreter's dependency-based lifetime rule. Graph outputs are materialized. Python, allocator and matrix-library workspaces are excluded. Serialized executable JavaScript Object Notation (JSON) reports the full graph-file size, including topology and numeric literals. This serialization metric is separate from runtime memory. For recurrence, costs use one example starting from the initial state; the listed worst case is index 64 or a six-element stream. A comparison of costs therefore requires the same operations, precision, input length and memory accounting.

Zero stored coefficients is the elementary lower bound on a nonnegative coefficient count. The square, tanh and addition proposals attain that bound in this operator library. The square graph also uses one arithmetic multiplication. A nonconstant scalar output requires at least one stored output scalar under the materialized-output convention, or eight bytes in binary64. The observed 16-byte square workspace includes input and output. These simple bounds provide no certificate that the other proposed architectures minimize coefficients, workspace or arithmetic.

The three sine constructions illustrate competing resource objectives. Their coefficient counts were 19, 14 and 10, while peak live workspaces were 32, 40 and 88 bytes and arithmetic counts were 32, 27 and 27. All three produced the same measured outer RMSE. Reducing coefficient storage therefore did not minimize workspace in this sample.

The independently drawn confirmation set uses a post-freeze random seed. It contains 2,048 new central and 2,048 new outer inputs per stochastic task. The integer tasks have fixed held-out indices 17–32 and 33–64. All fifteen exact stochastic-task proposals also matched every independently drawn confirmation output. The six exact integer-sequence proposals concern fixed indices. All three running-sum proposals also matched all 2,048 length-32 central streams. The complete scores are in the data archive.

Table 13: Every architecture proposal on the original outer test. Exact is the number of whole examples matching the reference. FLOPs, comparisons, transcendental calls and workspace use the largest input in the split. Dashes indicate a rejected executable graph. Weight bytes count stored coefficients once, including recurrent initial state. Graph bytes measure the serialized graph file, including numeric literals. Live bytes measure peak live tensor storage; all three sizes are in bytes.

Target	Seed	Exact	RMSE	Weight bytes	Graph bytes	FLOPs	Comparisons	Function calls	Live bytes
Square	164	4096/4096	0	0	109	1	0	0	16
Square	317	4096/4096	0	0	109	1	0	0	16
Square	583	4096/4096	0	0	109	1	0	0	16
Sine	164	273/4096	7.79×10^{-16}	152	1580	32	0	0	32
Sine	317	273/4096	7.79×10^{-16}	112	1688	27	0	0	40
Sine	583	273/4096	7.79×10^{-16}	80	1937	27	0	0	88
Positive self-power	164	0/2048	1.68×10^{4}	744	4926	250	0	0	1040
Positive self-power	317	invalid	∞	–	–	–	–	–	–
Positive self-power	583	invalid	∞	–	–	–	–	–	–
Hyperbolic tangent	164	4096/4096	0	0	101	0	0	1	16
Hyperbolic tangent	317	4096/4096	0	0	81	0	0	1	16
Hyperbolic tangent	583	4096/4096	0	0	101	0	0	1	16
Addition	164	2048/2048	0	0	202	1	0	0	32
Addition	317	2048/2048	0	0	220	1	0	0	32
Addition	583	2048/2048	0	0	220	1	0	0	32
Subtraction	164	2048/2048	0	16	119	3	0	0	24
Subtraction	317	2048/2048	0	16	119	3	0	0	24
Subtraction	583	2048/2048	0	16	119	3	0	0	24
Running sum	164	2048/2048	0	8	173	6	0	0	112
Running sum	317	2048/2048	0	8	173	6	0	0	112
Running sum	583	2048/2048	0	8	173	6	0	0	112
Sort four	164	1096/2048	5.23×10^{-16}	8	1546	25	5	0	96
Sort four	317	1128/2048	5.15×10^{-16}	64	1484	35	5	0	112
Sort four	583	1096/2048	5.23×10^{-16}	8	1713	25	5	0	88
Fibonacci	164	32/32	0	16	389	64	0	0	48
Fibonacci	317	32/32	0	16	389	64	0	0	48
Fibonacci	583	32/32	0	16	389	64	0	0	48
Triangular	164	32/32	0	8	243	3	0	0	24
Triangular	317	32/32	0	16	205	3	0	0	24
Triangular	583	32/32	0	16	205	3	0	0	24

A.7 Complete architecture-selection prompts and final answers

The following two complete prompts use the first data seed, 164, for square and sine. They are literal request texts, with line wrapping added for printing. The archive supplies all 30 prompts and replies. The network language and objective were fixed before dispatch. Each response contained a public construction summary and a proposed mathematical certificate. These are model-supplied final-answer explanations; no hidden internal reasoning trace was recorded. Numerical outcomes and interpreter resource counts above are independently measured. Model-supplied certificates are reproduced as claims and are not automatically promoted to proved results.

```
You are designing a compact neural network from input/output examples. Use weight-space programming (WSP)
     to choose both architecture and numerical weights. Use no tools, files, web, or external numerical
     solver. The MAIN goal is to infer the rule and extrapolate it accurately to unseen inputs throughout
     the deployment domain, including far beyond the displayed examples. Among architectures that
     implement the inferred rule equally accurately, minimize stored coefficients, batch-one inference
     workspace and scalar arithmetic operations. Favor a rule-based construction with a checkable
     certificate; fit quality only on displayed examples is insufficient. You may choose an MLP, RNN, CNN,
      transformer/attention, gated network, or another network representable in the graph language. No
     architecture or initialization is prescribed. Do not report unsupported global optimality. Input mode
      is scalar. One input has shape scalar and one output has shape scalar. Every input coordinate is a
     real value in [-10,10].
Observed input/output pairs:
[[1.6459866948837452,2.709272199734315],
[1.3890508033130673,1.9294621341846776],
[-2.8382621975164986,8.055732301851183],
[-0.3836959492604779,0.14722258147889922],
[-2.421046817864994,5.861467694294213],
[-2.7329781850724792,7.469169760082063],
[-0.9995956811732642,0.9991915258202422],
[1.1493795191426672,1.321073279024629],
[-1.959233424926871,3.838595613350677],
[2.0158703022728743,4.06373307558573],
[0.2575390480313793,0.06632636126090909],
[-1.2286324596762328,1.5095377209700698]]
Return one JSON object with exactly these fields: family (your description), inferred_rule (brief),
     construction_summary (up to 150 words), certificate (a short checkable mathematical justification,
     not a private reasoning trace), and network.
network is either:
{"kind":"graph","nodes":[...],"output":"node_name"}
or
{"kind":"recurrent","initial_state":[numeric values],"cell":[...],"state_output":"node_name","readout
     ":[...],"output":"node_name"}.
You choose all widths, depths, connections and coefficients. Name every node uniquely within its graph. A
     node has {"name":"...","op":"...",...}.
Allowed forms are:
- {"name":...,"op":"constant","value":[numbers]} (a scalar is [number]; matrices may be nested)
- {"name":...,"op":"affine","source":"tensor","weight":[[numbers],...],"bias":[numbers]} computes W @
     source.reshape(-1) + bias. Bias may be omitted.
- {"name":...,"op":"relu"|"tanh"|"sigmoid","source":"tensor"} applies elementwise.
- {"name":...,"op":"add"|"multiply"|"matmul","left":"tensor","right":"tensor"} uses NumPy broadcasting or
     matrix multiplication.
- {"name":...,"op":"concat","sources":["tensor",...]} concatenates flattened tensors.
- {"name":...,"op":"take","source":"tensor","indices":[integer,...]} indexes the flattened source,
     including repeated indices.
- {"name":...,"op":"reshape","source":"tensor","shape":[positive integers]} changes shape without
     arithmetic; total size must match.
- {"name":...,"op":"transpose","source":"tensor"} transposes a matrix.
- {"name":...,"op":"softmax","source":"tensor"} normalizes along the final axis using max subtraction.
- {"name":...,"op":"conv1d","source":"tensor","weight":[[[numbers],...],...],"bias":[numbers]} uses valid
     stride-one convolution. Source shape (length,input_channels), weight shape (output_channels,
     input_channels,kernel_width); output shape (length-kernel_width+1,output_channels). Bias optional.
Node references must be to external tensors or earlier nodes. Graph input is named "input". Recurrent cell
      external names are "state" and "input". Readout external names are "state" and "input"; it may
     directly return state without nodes. For integer index n, initialize the state, repeat the cell n
     times with input [0], then read out; n=0 reads the initial state. For streams, feed [x_t], update the
      state and read out after every element. Shared cell and readout weights are stored once. Return
     actual finite JSON numbers, never formulas or code strings in coefficient fields. The final output
     may be the external input or state. There may be at most 64 total nodes, at most 4096 stored scalar
     coefficients including constants and initial state, and at most 65536 simultaneously live activation
     scalars. Scalar outputs have one entry. All inference is binary64. No sin, exp, log, pow, sort,
     Fibonacci or triangular target-specific operation is allowed; tanh, sigmoid and softmax perform their
      named standard neural activations. Elementwise multiplication is an allowed neural gating primitive.
      Every stored coefficient, including zero/one, counts toward storage. Graph topology also consumes
     serialized storage. Do not artificially inflate an error to reduce resource cost.

Return only the JSON object. Your construction summary should explain the implementation for readers
     without giving hidden internal reasoning.
```

```
{
  "family": "Single multiplicative gate with no stored coefficients",
  "inferred_rule": "y = x^2",
  "construction_summary": "Multiply the scalar input by itself. The graph uses one node, zero stored
     coefficients, one scalar multiplication, and one output scalar, with no intermediate activations. It
     implements the inferred square rule throughout [-10,10], including beyond the observed input range.",
  "certificate": "For every real x, the graph computes x·x = x^2. The observed pairs agree with this rule to
      binary64 precision. Binary64 inference performs one rounded multiplication; on [-10,10] there is no
     overflow, and absolute rounding error is at most 2^-47. Finite examples do not uniquely determine a
     rule, so this certificate establishes implementation of the inferred rule rather than uniqueness of
     inference.",
  "network": {
    "kind": "graph",
    "nodes": [
      {
        "name": "square",
        "op": "multiply",
        "left": "input",
        "right": "input"
      }
    ],
    "output": "square"
  }
}
```


```
You are designing a compact neural network from input/output examples. Use weight-space programming (WSP)
     to choose both architecture and numerical weights. Use no tools, files, web, or external numerical
     solver. The MAIN goal is to infer the rule and extrapolate it accurately to unseen inputs throughout
     the deployment domain, including far beyond the displayed examples. Among architectures that
     implement the inferred rule equally accurately, minimize stored coefficients, batch-one inference
     workspace and scalar arithmetic operations. Favor a rule-based construction with a checkable
     certificate; fit quality only on displayed examples is insufficient. You may choose an MLP, RNN, CNN,
      transformer/attention, gated network, or another network representable in the graph language. No
     architecture or initialization is prescribed. Do not report unsupported global optimality. Input mode
      is scalar. One input has shape scalar and one output has shape scalar. Every input coordinate is a
     real value in [-10,10].
Observed input/output pairs:
[[1.6459866948837452,0.9971745358180562],
[1.3890508033130673,0.9835296938096632],
[-2.8382621975164986,-0.29870026804737076],
[-0.3836959492604779,-0.3743502255803443],
[-2.421046817864994,-0.6597949361487462],
[-2.7329781850724792,-0.3973382460129415],
[-0.9995956811732642,-0.8412524616402648],
[1.1493795191426672,0.9125103059338021],
[-1.959233424926871,-0.9255021271796147],
[2.0158703022728743,0.9025788195990063],
[0.2575390480313793,0.2547015364974582],
[-1.2286324596762328,-0.9420308371992286]]
Return one JSON object with exactly these fields: family (your description), inferred_rule (brief),
     construction_summary (up to 150 words), certificate (a short checkable mathematical justification,
     not a private reasoning trace), and network.
network is either:
{"kind":"graph","nodes":[...],"output":"node_name"}
or
{"kind":"recurrent","initial_state":[numeric values],"cell":[...],"state_output":"node_name","readout
     ":[...],"output":"node_name"}.
You choose all widths, depths, connections and coefficients. Name every node uniquely within its graph. A
     node has {"name":"...","op":"...",...}.
Allowed forms are:
- {"name":...,"op":"constant","value":[numbers]} (a scalar is [number]; matrices may be nested)
- {"name":...,"op":"affine","source":"tensor","weight":[[numbers],...],"bias":[numbers]} computes W @
     source.reshape(-1) + bias. Bias may be omitted.
- {"name":...,"op":"relu"|"tanh"|"sigmoid","source":"tensor"} applies elementwise.
- {"name":...,"op":"add"|"multiply"|"matmul","left":"tensor","right":"tensor"} uses NumPy broadcasting or
     matrix multiplication.
- {"name":...,"op":"concat","sources":["tensor",...]} concatenates flattened tensors.
- {"name":...,"op":"take","source":"tensor","indices":[integer,...]} indexes the flattened source,
     including repeated indices.
- {"name":...,"op":"reshape","source":"tensor","shape":[positive integers]} changes shape without
     arithmetic; total size must match.
- {"name":...,"op":"transpose","source":"tensor"} transposes a matrix.
- {"name":...,"op":"softmax","source":"tensor"} normalizes along the final axis using max subtraction.
```

```
- {"name":...,"op":"conv1d","source":"tensor","weight":[[[numbers],...],...],"bias":[numbers]} uses valid
      stride-one convolution. Source shape (length,input_channels), weight shape (output_channels,
      input_channels,kernel_width); output shape (length-kernel_width+1,output_channels). Bias optional.
Node references must be to external tensors or earlier nodes. Graph input is named "input". Recurrent cell
       external names are "state" and "input". Readout external names are "state" and "input"; it may
      directly return state without nodes. For integer index n, initialize the state, repeat the cell n
      times with input [0], then read out; n=0 reads the initial state. For streams, feed [x_t], update the
       state and read out after every element. Shared cell and readout weights are stored once. Return
      actual finite JSON numbers, never formulas or code strings in coefficient fields. The final output
      may be the external input or state. There may be at most 64 total nodes, at most 4096 stored scalar
      coefficients including constants and initial state, and at most 65536 simultaneously live activation
      scalars. Scalar outputs have one entry. All inference is binary64. No sin, exp, log, pow, sort,
      Fibonacci or triangular target-specific operation is allowed; tanh, sigmoid and softmax perform their
       named standard neural activations. Elementwise multiplication is an allowed neural gating primitive.
       Every stored coefficient, including zero/one, counts toward storage. Graph topology also consumes
      serialized storage. Do not artificially inflate an error to reduce resource cost.

Return only the JSON object. Your construction summary should explain the implementation for readers
      without giving hidden internal reasoning.
```


```
{
  "family": "Scalar polynomial gating network with triple-angle reconstruction",
  "inferred_rule": "The examples agree with y = sin(x), inferred over the full domain [-10,10]. Finite
     examples alone do not uniquely determine this rule.",
  "construction_summary": "Scale the input by 1/27, evaluate the degree-13 sine Taylor polynomial using
     Horner gates, and apply s → s(3−4s^2) three times. This implements sine across the deployment domain
      with near-binary64 accuracy, without a sine operation. The graph uses 23 nodes and 19 stored scalar
     coefficients. With activation reuse, inference requires only a few live scalars and 34 scalar
     arithmetic operations. No global resource optimality is claimed.",
  "certificate": "Let P(t)=t−t^3/3!+t^5/5!−t^7/7!+t^9/9!−t^11/11!+t^13/13! and F(s)=3s−4s^3. For |t|≤10/27,
     the alternating-series bound gives |P(t)−sin(t)|≤(10/27)^15/15!<2.7e−19. Also F composed three
     times at sin(t) equals sin(27t). On the intervening initial-value interval, the derivative of this
     composition has magnitude below 30, using sin(27 arcsin(s)); hence the exact-coefficient real-
     arithmetic error is below 8.1e−18 throughout [-10,10]. Stored-coefficient rounding and binary64
     arithmetic add floating-point error; the real-arithmetic bound is not a bound on that additional
     error.",
  "network": {
    "kind": "graph",
    "nodes": [
      {"name":"t","op":"affine","source":"input","weight":[[0.03703703703703035]]},
      {"name":"z","op":"multiply","left":"t","right":"t"},
      {"name":"p11","op":"affine","source":"z","weight":[[1.6059043836821613e-10]],"bias
     ":[-2.505210838544172e-8]},
      {"name":"m9","op":"multiply","left":"z","right":"p11"},
      {"name":"p9","op":"affine","source":"m9","weight":[[1]],"bias":[0.000002755731922398589 3]},
      {"name":"m7","op":"multiply","left":"z","right":"p9"},
      {"name":"p7","op":"affine","source":"m7","weight":[[1]],"bias":[-0.0001984126984126984]},
      {"name":"m5","op":"multiply","left":"z","right":"p7"},
      {"name":"p5","op":"affine","source":"m5","weight":[[1]],"bias":[0.008333333333333333]},
      {"name":"m3","op":"multiply","left":"z","right":"p5"},
      {"name":"p3","op":"affine","source":"m3","weight":[[1]],"bias":[-0.16666666666666666]},
      {"name":"m1","op":"multiply","left":"z","right":"p3"},
      {"name":"p1","op":"affine","source":"m1","weight":[[1]],"bias":[1]},
      {"name":"s0","op":"multiply","left":"t","right":"p1"},
      {"name":"q0","op":"multiply","left":"s0","right":"s0"},
      {"name":"g0","op":"affine","source":"q0","weight":[[-4]],"bias":[3]},
      {"name":"s1","op":"multiply","left":"s0","right":"g0"},
      {"name":"q1","op":"multiply","left":"s1","right":"s1"},
      {"name":"g1","op":"affine","source":"q1","weight":[[-4]],"bias":[3]},
      {"name":"s2","op":"multiply","left":"s1","right":"g1"},
      {"name":"q2","op":"multiply","left":"s2","right":"s2"},
      {"name":"g2","op":"affine","source":"q2","weight":[[-4]],"bias":[3]},
      {"name":"output_sine","op":"multiply","left":"s2","right":"g2"}
    ],
    "output":"output_sine"
  }
}
```


A.8 All public construction summaries

Each paragraph reproduces the inferred rule, implementation explanation and certificate fields returned by one request. Unicode mathematical characters are typeset equivalently. Failed executable graphs are labeled below. Their explanations remain part of the response record.

Square, seed 164, valid. **Family.** Single multiplicative gate with no stored coefficients

Inferred rule. $y = x^2$

Construction summary. Multiply the scalar input by itself. The graph uses one node, zero stored coefficients, one scalar multiplication, and one output scalar, with no intermediate activations. It implements the inferred square rule throughout [-10,10], including beyond the observed input range.

Model-supplied certificate. For every real x, the graph computes $x \cdot x = x^2$. The observed pairs agree with this rule to binary64 precision. Binary64 inference performs one rounded multiplication; on [-10,10] there is no overflow, and absolute rounding error is at most 2^-47. Finite examples do not uniquely determine a rule, so this certificate establishes implementation of the inferred rule rather than uniqueness of inference.

Square, seed 317, valid. **Family.** Parameter-free multiplicative graph

Inferred rule. $y = x^2$

Construction summary. A single multiplication node uses the scalar input for both operands. It stores zero numerical coefficients, performs one scalar multiplication, and produces one scalar output without intermediate activations. The construction implements the inferred rule throughout [-10,10], including beyond the observed inputs.

Model-supplied certificate. In exact arithmetic, the node returns $x \cdot x = x^2$ for every real x. In binary64 it returns the rounded product, with no overflow on [-10,10]; underflow is possible only for extremely small inputs. The observed outputs agree with the square rule to displayed numerical precision. Finite examples do not uniquely determine a function, so this certificate establishes implementation of the inferred rule, not uniqueness of the inference.

Square, seed 583, valid. **Family.** Parameter-free multiplicative network

Inferred rule. $y = x^2$

Construction summary. A single multiplication node uses the scalar input as both operands. The network stores zero numerical coefficients and performs one scalar multiplication, producing one output scalar with no intermediate activations. It implements the inferred square rule throughout [-10,10], including outside the observed input range.

Model-supplied certificate. In real arithmetic, the node outputs input $\times$ input $= x^2$ identically. In binary64, it outputs the correctly rounded product $\mathrm{fl}(x \times x)$; for $|x| \leq 10$, no overflow occurs and absolute rounding error is at most 2^(-47). The observed pairs are consistent with the square rule at the displayed precision. Finite examples alone do not uniquely determine the underlying function.

Sine, seed 164, valid. **Family.** Scalar polynomial gating network with triple-angle reconstruction

Inferred rule. The examples agree with $y = \sin(x)$, inferred over the full domain [-10,10]. Finite examples alone do not uniquely determine this rule.

Construction summary. Scale the input by 1/27, evaluate the degree-13 sine Taylor polynomial using Horner gates, and apply $s \to s(3-4s^2)$ three times. This implements sine across the deployment domain with near-binary64 accuracy, without a sine operation. The graph uses 23 nodes and 19 stored scalar coefficients. With activation reuse, inference requires only a few live scalars and 34 scalar arithmetic operations. No global resource optimality is claimed.

Model-supplied certificate. Let $P(t)=t-t^3/3!+t^5/5!-t^7/7!+t^9/9!-t^{11}/11!+t^{13}/13!$ and $F(s)=3s-4s^3$. For $|t|\leq 10/27$, the alternating-series bound gives $|P(t)-\sin(t)|\leq(10/27)^{15}/15!<2.7e\text{-}19$. Also F composed three times at sin(t) equals sin(27t). On the intervening initial-value interval, the derivative of this composition has magnitude below 30, using $\sin(27 \arcsin(s))$; hence the exact-coefficient real-arithmetic error is below 8.1e-18 throughout [-10,10]. Stored-coefficient rounding and binary64 arithmetic add floating-point error; the real-arithmetic bound is not a bound on that additional error.

Sine, seed 317, valid. **Family.** Scalar polynomial network with triple-angle gates

Inferred rule. The examples are consistent with y = sin(x), with x in radians. Finite examples alone do not uniquely determine this rule.

Construction summary. Scale the input by 1/27 and evaluate the degree-13 sine Taylor polynomial using Horner evaluation. Three triple-angle gates then restore the original argument. This implements the inferred sine rule throughout [-10,10], rather than interpolating the displayed samples. The graph stores 14 scalar coefficients. Sequential evaluation with activation reuse requires only a few live scalars. No global resource optimality is claimed.

Model-supplied certificate. Let p(t)=t-t^3/3!+t^5/5!-t^7/7!+t^9/9!-t^11/11!+t^13/13!, and T(s)=3s-4s^3. In exact arithmetic the graph computes T(T(T(p(x/27)))). For |t|<=10/27, the alternating-series bound gives |p(t)-sin(t)|<2.6e-19. On this range arcsin has derivative below 1.08, so |arcsin(p(t))-t|<2.808e-19. Since T(sin(a))=sin(3a), the uniform exact-arithmetic output error is below 7.6e-18. Stored coefficient rounding and binary64 arithmetic add floating-point roundoff; the exact-arithmetic bound is not a claim of correctly rounded sine.

Sine, seed 583, valid. Family. Polynomial gating network with shared triple-angle coefficients

Inferred rule. y = sin(x), consistent with every displayed pair.

Construction summary. Scale the input by 1/27, evaluate the degree-13 sine Taylor polynomial using Horner's rule, and apply $s \to s(3 - 4s^2)$ three times. This reconstructs sine across the full deployment interval without a sine operation. The graph stores 10 scalar coefficients and uses 27 scalar arithmetic operations. Sequential evaluation needs only a small constant workspace. This is a compact rule-based construction, not a claim of global optimality.

Model-supplied certificate. Let t=x/27 and P(t)=sum from k=0 to 6 of (-1)^k t^(2k+1)/(2k+1)!. For |x|≤10, |P(t)-sin(t)|≤(10/27)^15/15!<2.7e-19. The identity 3sin(u)-4sin(u)^3=sin(3u) makes the output sin(27 arcsin(P(t))). Since |sin(t)|<0.363, the mean-value theorem bounds its ideal real-arithmetic error by 27(2.7e-19)/sqrt(1-0.364^2)<8e-18 uniformly. Stored decimal coefficients and binary64 operations introduce additional rounding error; the bound above describes the approximation error, not a certified floating-point bound. Finite observations cannot uniquely determine the generating rule.

Positive self-power, seed 164, valid. Family. Rule-based scalar network with polynomial logarithm and scaled polynomial exponential

Inferred rule. The examples are consistent with y = x^x. The network approximates this rule throughout 0 < x <= 10.

Construction summary. The network computes x log(x) using a reciprocal iteration followed by a 64-term odd-power logarithm series. It then approximates the exponential with a fourth-degree Taylor polynomial and eight squarings. All coefficients are explicit. The logarithm approximation is most accurate away from zero; its absolute contribution to the exponent still vanishes as x approaches zero. This is a constructive approximation with an explicit error certificate, not a claim that finite examples uniquely identify the rule or that the architecture is globally optimal.

Model-supplied certificate. In exact arithmetic, the reciprocal after seven iterations is r=(1-e)/(x+1), where e=((5-x)/6)^128 and 0<=e<7.4e-11. Put z=(x-1)r and a=|z|. The computed exponent is L=2x sum_{k=0}^{63} z^(2k+1)/(2k+1). Its error from x log(x) is at most D=7.4e-10+2x*a^129/(129*(1-a*a)). The output is P4(L/256)^256, where P4(t)=1+t+t^2/2+t^3/6+t^4/24. Since |L|<=24, define E=exp(2*24/256)*(24/256)^5/120. Relative output error is at most exp(D+256*E/(1-E))-1. These bounds exclude binary64 rounding. The logarithm tail bound tends to zero as x approaches zero when its actual series remainder, rather than the geometric upper bound, is used.

Positive self-power, seed 317, invalid. Family. Scalar gated arithmetic network with rational quadrature and polynomial exponentiation

Inferred rule. y = x^x. The graph approximates this rule over 0 < x <= 10.

Construction summary. Approximate log(x) by 16-point Gauss–Legendre quadrature of (x-1)/(1+t(x-1)). Fourteen parallel Newton iterations compute the quadrature reciprocals. Multiply by x, evaluate a fourth-degree exponential polynomial at one 256th of the result, and square eight times. This construction follows a domain-wide mathematical rule rather than interpolating the ob-

servations. It uses 64 nodes and 59 stored scalar coefficients. Accuracy is strongest away from zero; the certificate supplies a conservative uniform relative-error bound, not a claim of machine-precision accuracy or global architectural optimality.

Model-supplied certificate. The observations are consistent with exp(x log(x)). For exact reciprocals, quadrature is exact through degree 31. Expanding the integral geometrically bounds the log error by (1-x)^32/33 for x<=1 and (1-1/x)^32/33 for x>=1. Thus the error in x log(x) is at most 1/33 throughout the domain. Each normalized reciprocal has residual (1-d)^16384; d>=0.0005299532504175, so its residual is below 0.000170. For x<=1 its additional exponent error is below 0.000170/e; for x>=1 it is negligible. The scaled exponent lies between -1/(256e) and 10 log(10)/256. Taylor's remainder bounds the final exponentiation relative error below 0.000014. Together these give a conservative uniform relative error below 0.032, with ample margin for binary64 rounding. The graph is an approximation, not an exact representation of x^x.

Positive self-power, seed 583, invalid. Family. Scalar arithmetic network with quadrature, shared vector operations, and repeated squaring

Inferred rule. y = x^x. The network approximates exp(x log x) on 0 < x <= 10, with limit 1 as x approaches zero.

Construction summary. A 32-point Gauss–Legendre quadrature approximates log(x) using an integral of reciprocal affine functions. Thirteen vectorized Newton steps approximate those reciprocals. Multiplication by x produces the exponent. A fourth-order exponential polynomial followed by twelve squarings produces the output. This is a rule-based approximation rather than an interpolation of the observations. Accuracy is strongest away from zero; the small-input region has additional quadrature and reciprocal approximation error. No global resource optimality is claimed.

Model-supplied certificate. The observations agree with x^x. Use log(x)=(x-1) integral_0^1 1/(1+(x-1)t) dt. The listed t,w are 32-point Gauss–Legendre nodes and weights. Write d_i=1+(x-1)t_i. Since 0<d_i<10, Newton iteration r'=r(2-d_i r), initialized at 0.1, gives r_i=(1-(1-0.1d_i)^8192)/d_i after 13 steps. Thus the graph's exponent is exactly z=x(x-1) sum_i w_i r_i in real arithmetic. Its output is $[1+u+u^2/2+u^3/6+u^4/24]^{4096}$, u=z/4096. These identities provide a directly checkable approximation certificate, not an assertion of exact equality to x^x. Binary64 adds rounding error.

Hyperbolic tangent, seed 164, valid. Family. Single scalar tanh activation with no learned coefficients

Inferred rule. y = tanh(x).

Construction summary. Apply the standard tanh activation directly to the scalar input. This uses one graph node, zero stored numerical coefficients, and one scalar activation evaluation. The same rule applies throughout [-10,10], including the saturation regions beyond the observed examples.

Model-supplied certificate. The observed pairs agree with y = tanh(x) to displayed numerical precision. The graph computes this inferred rule directly for every input in [-10,10], subject only to binary64 activation rounding. Finite samples cannot uniquely determine an arbitrary generating function; this certificate establishes implementation of the inferred rule, not uniqueness of the inference.

Hyperbolic tangent, seed 317, valid. Family. Parameter-free scalar tanh network

Inferred rule. y = tanh(x).

Construction summary. Apply one tanh activation directly to the scalar input. The network stores zero numerical coefficients, uses one scalar activation output, and requires one tanh evaluation with no additional arithmetic. It implements the inferred rule across the entire deployment interval [-10,10].

Model-supplied certificate. The sole node computes tanh(x) by definition, so in exact arithmetic the graph equals the inferred rule for every real x. Binary64 inference introduces only the activation implementation's numerical error. The observed pairs agree with this rule to the displayed precision; finite samples alone do not uniquely determine an underlying function.

Hyperbolic tangent, seed 583, valid. **Family.** Single scalar tanh activation with no stored coefficients

Inferred rule. $y = \tanh(x)$.

Construction summary. Apply the standard tanh activation directly to the scalar input. The network uses one node, zero stored coefficients, and one scalar activation evaluation, with no affine arithmetic or intermediate layers. It implements the inferred rule throughout [-10,10], including the saturation regions beyond the observed inputs.

Model-supplied certificate. The graph directly evaluates tanh(x), so its output equals the inferred rule over the entire deployment domain, subject only to binary64 activation rounding. The displayed pairs agree with this rule to their shown numerical precision. Finite observations do not uniquely determine an underlying function; this certificate establishes implementation of the inferred rule rather than uniqueness of inference.

Addition, seed 164, valid. **Family.** Parameter-free additive network

Inferred rule. $y = x_1 + x_2$.

Construction summary. Extract the two input coordinates and add them. The graph stores no numerical coefficients and performs one binary64 addition, with no hidden layers or nonlinearities. It implements the inferred rule throughout the deployment domain.

Model-supplied certificate. For every input (a,b) in $[-10,10]^2$, the take nodes return [a] and [b], so the output is [fl(a+b)], where fl denotes binary64 rounding. The observed outputs agree with this rule to displayed precision. Finite observations do not uniquely determine an unrestricted underlying function.

Addition, seed 317, valid. **Family.** Parameter-free additive graph

Inferred rule. $y = x_0 + x_1$.

Construction summary. Select the two input coordinates and add them. This implementation stores no numerical coefficients and performs one binary64 addition. It requires no hidden layers or nonlinear activations and applies throughout the deployment domain.

Model-supplied certificate. The take nodes return $[x_0]$ and $[x_1]$, so the output is $[\mathrm{fl}(x_0 + x_1)]$, where fl denotes binary64 rounding. The observed outputs agree with this rule to floating-point precision. The same identity holds for every input in $[-10,10]^2$; the examples support, but do not uniquely determine, the inferred rule.

Addition, seed 583, valid. **Family.** Parameter-free additive graph

Inferred rule. $y = x_0 + x_1$

Construction summary. Extract the two input coordinates and add them directly. The graph stores no numerical weights, biases, or constants and performs one scalar addition. It uses two one-entry selections and a one-entry output. This construction implements the inferred rule throughout $[-10,10]^2$, including inputs far beyond the observed examples.

Model-supplied certificate. Every displayed output agrees with $x_0 + x_1$ to binary64 rounding precision. For any input, the take nodes return x_0 and x_1 exactly, so the output is $\mathrm{fl}(x_0 + x_1)$, the correctly rounded binary64 sum. Thus the graph implements the inferred addition rule over the full domain. Finite observations alone do not uniquely determine the underlying rule.

Subtraction, seed 164, valid. **Family.** Single linear neuron

Inferred rule. Output the first input coordinate minus the second: $y = x_1 - x_2$.

Construction summary. One bias-free affine node with weights [1, -1] computes the difference. It stores two scalar coefficients and produces one activation scalar, with no hidden layers or nonlinearities. The affine expression requires two multiplications and one addition; multiplication by ± 1 can be implemented as identity and sign change.

Model-supplied certificate. For every input (a,b) in $[-10,10]^2$, $[1,-1]\cdot[a,b] = a-b$. This matches the displayed outputs to binary64 rounding precision and implements the inferred rule throughout the domain. Multiplication by ± 1 is exact in binary64, leaving only the rounding of the subtraction.

Subtraction, seed 317, valid. **Family.** Single bias-free linear neuron

Inferred rule. $y = x_0 - x_1$

Construction summary. A single affine node uses weights [1, -1] and no bias. It stores two scalar coefficients and produces one output activation, with no hidden layers or intermediate activation tensors. The same subtraction rule applies throughout the deployment domain.

Model-supplied certificate. For every input, $[1, -1] \cdot [x_0, x_1] = x_0 - x_1$. This matches all displayed outputs to binary64 rounding precision. Multiplication by ± 1 is exact on the specified domain, so evaluation yields the correctly rounded subtraction throughout $[-10,10]^2$. The examples support this rule but cannot uniquely determine an unrestricted function.

Subtraction, seed 583, valid. **Family.** Single-layer linear network

Inferred rule. $y = x_1 - x_2$.

Construction summary. One affine node subtracts the second coordinate from the first using weights [1, -1] and no bias. It stores two scalar coefficients and produces one activation scalar, with no hidden layers. The same rule applies throughout $[-10,10]^2$.

Model-supplied certificate. For every input (a,b), the network computes $[1,-1]\cdot[a,b] = a-b$ in exact arithmetic. All displayed outputs agree with this rule to binary64 rounding precision. The weights are exactly representable in binary64. This certifies implementation of the inferred rule across the domain; finite observations alone cannot uniquely determine an arbitrary unseen target.

Running sum, seed 164, valid. **Family.** Single-state additive recurrent network

Inferred rule. Each output is the running sum of the inputs through that position.

Construction summary. Maintain one scalar state initialized to zero. Add each incoming scalar to the state and emit the updated state directly. The network stores one scalar coefficient, uses one addition per element, and requires constant scalar workspace apart from input/output storage. No learned weights or nonlinearities are needed, and the same cell handles arbitrary sequence lengths.

Model-supplied certificate. In exact arithmetic, s_0=0 and s_t=s_{t-1}+x_t imply by induction that s_t=sum_{i=1}^t x_i. Thus the output is the prefix sum for every sequence over [-10,10], regardless of length. In binary64, the cell implements sequential rounded summation, matching the observed rule up to floating-point rounding.

Running sum, seed 317, valid. **Family.** Scalar linear recurrent accumulator

Inferred rule. Each output is the inclusive prefix sum: y_t = x_1 + ... + x_t.

Construction summary. Use one scalar state initialized to zero. Each cell adds the current input to the state, and the readout returns the updated state directly. The network stores one coefficient (the initial zero), performs one addition per input, and needs constant workspace independent of sequence length, excluding retained outputs. It applies to arbitrary sequence lengths and inputs throughout [-10,10], without clipping or saturation.

Model-supplied certificate. In exact arithmetic, s_0 = 0 and s_t = s_{t-1} + x_t imply by induction that s_t = sum_{i=1}^t x_i. Returning s_t therefore gives every inclusive prefix sum. In binary64, the same cell implements sequential floating-point prefix summation, with its ordinary rounding and representational limits.

Running sum, seed 583, valid. **Family.** Single-state additive recurrent network

Inferred rule. Each output is the inclusive prefix sum of the input sequence: y_t = x_1 + ... + x_t.

Construction summary. Initialize one scalar state to zero. At each step, add the current input to the state and emit the updated state directly. The network stores one scalar coefficient, uses one addition per element, and requires constant streaming workspace, excluding retained inputs and outputs. No

trained weights or nonlinearities are needed. The same recurrence applies to arbitrary sequence lengths and all inputs in [-10,10].

Model-supplied certificate. Initially s_0 = 0. If s_(t-1) equals the sum of the first t-1 inputs, the cell gives s_t = s_(t-1) + x_t, the sum of the first t inputs. Induction proves the prefix-sum rule in exact arithmetic. Binary64 inference computes the corresponding left-to-right floating-point sums, subject to rounding and representable range.

Sort four, seed 164, valid. Family. Sparse ReLU sorting network with five compare-exchange units

Inferred rule. Return the four input coordinates in ascending order.

Construction summary. Three stages compare two disjoint pairs, compare their respective minima and maxima, then order the two middle candidates. Each compare-exchange uses subtraction, ReLU, and addition, implemented with one shared stored coefficient, -1. Indexing and concatenation route values without dense matrices. The construction implements sorting throughout [-10,10]^4, including ties and unseen permutations, subject to binary64 rounding. No global resource optimality is claimed.

Model-supplied certificate. For a pair (a,b), let t=ReLU(a-b). Then (a-t,b+t)=(min(a,b),max(a,b)) in exact arithmetic. After the first stage, write the sorted pairs as l0<=h0 and l1<=h1. The second stage produces L=min(l0,l1), U=min(h0,h1), V=max(l0,l1), H=max(h0,h1). L is the global minimum, H the global maximum, and U,V are the remaining elements. Sorting U,V therefore sorts all four values. Binary64 introduces only rounding error; on the stated domain a conservative absolute error bound per output relative to exact sorting is 1e-13.

Sort four, seed 317, valid. Family. Three-stage ReLU sorting network

Inferred rule. Return the four input coordinates in ascending order.

Construction summary. Five compare-exchange units sort four values in three stages. Each unit computes r = ReLU(a - b), then returns (a - r, b + r). The first stage sorts two input pairs; the second compares their minima and their maxima; the final stage sorts the middle two values. Parallel units share convolution coefficients, and all stages share the correction vector. The graph stores eight scalar coefficients, including zeros and ones, and uses no biases. It implements sorting throughout the deployment domain, with only binary64 rounding error.

Model-supplied certificate. In exact arithmetic, (a-ReLU(a-b), b+ReLU(a-b)) = (min(a,b),max(a,b)). After stage one, write the pairs as $l0 \le h0$ and $l1 \le h1$. Stage two produces [min(l0,l1),max(l0,l1),min(h0,h1),max(h0,h1)]. Its first and last entries are the global minimum and maximum. Sorting its two middle entries therefore sorts all four coordinates. This proof includes ties and holds for all real inputs. Binary64 subtraction and addition can introduce rounding; exact bitwise preservation is not asserted.

Sort four, seed 583, valid. Family. Five-comparator ReLU sorting network with one shared scalar coefficient

Inferred rule. Return the four input coordinates in ascending order.

Construction summary. Three stages perform five compare-and-exchange operations: two on the original pairs, two on their minima and maxima, and one on the middle candidates. Each comparator computes r = ReLU(a - b), then returns (a - r, b + r). Vector operations execute independent comparators together. The only stored numerical coefficient is a shared -1; indexing and concatenation specify the wiring. Arithmetic comprises 15 scalar additions, 10 scalar negations implemented as multiplications, and 5 ReLUs. Intermediate tensors have at most four entries. No fitting or domain-specific thresholds are used.

Model-supplied certificate. In real arithmetic, r=max(a-b,0) gives a-r=min(a,b) and b+r=max(a,b). After stage one, let the sorted pairs be (p,P) and (q,Q). Stage two produces L=min(p,q), u=min(P,Q), v=max(p,q), H=max(P,Q). L is the global minimum, H is the global maximum, and the remaining values are u,v. Sorting u,v therefore sorts all four inputs, including ties. This identity holds for every real input. Binary64 evaluation introduces ordinary subtraction/addition roundoff; the stated domain cannot overflow.

Fibonacci, seed 164, valid. **Family.** Two-state additive recurrent network

Inferred rule. Fibonacci sequence: F(0)=0, F(1)=1, and F(n+2)=F(n+1)+F(n).

Construction summary. Initialize the state to [0,1]. Each iteration replaces [a,b] with [b,a+b]. After n iterations, return the first coordinate. The network stores only two numerical initialization coefficients and uses one scalar addition per iteration, with constant-size inference workspace. Indexing and concatenation require no scalar arithmetic. This implements the inferred recurrence across the entire deployment domain rather than approximating the displayed values.

Model-supplied certificate. Induction gives state=[F(n),F(n+1)] after n iterations: initialization establishes n=0, and the cell implements the defining recurrence. For n<=64, every state entry and addition result is a nonnegative integer at most F(65)=17167680177565<2^53, so all arithmetic is exact in binary64.

Fibonacci, seed 317, valid. **Family.** Two-state additive recurrent network

Inferred rule. Fibonacci sequence: F(0)=0, F(1)=1, and F(n+2)=F(n)+F(n+1).

Construction summary. Initialize two state entries to [0,1]. Each iteration replaces [a,b] with [b,a+b]. After n iterations, return the first entry. The network stores only two numerical coefficients, uses one scalar addition per iteration, and requires constant workspace. Selection and concatenation require no scalar arithmetic. This construction implements the inferred recurrence throughout the deployment domain rather than interpolating the examples.

Model-supplied certificate. Induction gives state=[F(n),F(n+1)] after n iterations: initialization establishes the base case, and the update establishes the induction step. For n<=64, every computed integer is at most F(65)=17167680177565<2^53, so all additions and outputs are exact in binary64. The observed examples agree with this rule, although finitely many examples cannot uniquely determine an extrapolation.

Fibonacci, seed 583, valid. **Family.** Two-state additive recurrent network

Inferred rule. Fibonacci sequence: F(0)=0, F(1)=1, and F(n+2)=F(n+1)+F(n).

Construction summary. Initialize the state to [0,1]. Each iteration replaces [a,b] with [b,a+b]. After n iterations, return the first entry. The construction stores two numerical coefficients in the initial state and uses one scalar addition per iteration, with constant-size workspace and no learned weights. It implements the inferred recurrence throughout the deployment domain rather than interpolating the examples.

Model-supplied certificate. The initial state is [F(0),F(1)]. If the state is [F(k),F(k+1)], the cell produces [F(k+1),F(k)+F(k+1)]=[F(k+1),F(k+2)]. Induction proves the output is F(n). For n<=64, every state entry is a nonnegative integer at most F(65)=17167680177565<2^53, so binary64 additions and outputs are exact.

Triangular, seed 164, valid. **Family.** Scalar quadratic network with multiplicative gating

Inferred rule. The triangular-number continuation: y(n) = n(n + 1)/2.

Construction summary. Square the input, add the original input, and scale by one half. This graph stores one scalar coefficient and uses three scalar arithmetic operations with constant-size workspace. It implements the inferred rule across the entire deployment domain without iteration or a lookup table.

Model-supplied certificate. The observed successive differences are 1,2,...,8, consistent with y(n)=sum_{k=1}^n k. The graph computes $(n^2+n)/2$, which equals that sum for every nonnegative integer n. For $0 \le n \le 64$, $n^2 \le 4096$ and $n^2+n \le 4160$; these integers and multiplication by 0.5 are represented exactly in binary64. The finite observations do not uniquely determine the continuation.

Triangular, seed 317, valid. **Family.** Scalar polynomial network with multiplicative gating

Inferred rule. Triangular numbers: y(n) = n(n + 1)/2.

Construction summary. An affine node computes (n + 1)/2, then a multiplicative gate multiplies it by n. This construction uses two nodes, two stored scalar coefficients, and three scalar arithmetic operations per input, with constant workspace and no iteration.

Model-supplied certificate. The output is n(0.5n + 0.5) = n(n + 1)/2. It is zero at n=0 and its successive difference is n, matching all observed pairs. For every integer $0 \leq n \leq 64$, all intermediate values and outputs are exactly representable in binary64, so evaluation is exact. The examples support this rule but do not uniquely determine an arbitrary unseen function.

Triangular, seed 583, valid. **Family.** Scalar affine network with multiplicative gating

Inferred rule. Triangular numbers: y(n) = n(n + 1)/2.

Construction summary. An affine node computes (n + 1)/2, and a multiplicative gate multiplies it by n. The network stores two scalar coefficients and uses two scalar multiplications and one addition per inference, with constant workspace and no iteration.

Model-supplied certificate. The output is n(0.5n + 0.5) = n(n + 1)/2. This gives y(0) = 0 and y(n + 1) - y(n) = n + 1, matching the observed successive increments. For every integer 0 <= n <= 64, all intermediate values are exactly representable in binary64, so the inferred rule is computed exactly. Finite examples do not uniquely determine extrapolation.

A.9 Nonconvexity witnesses for the fixed networks

How we checked nonconvexity. A convex training loss at the midpoint of any two weight vectors is at most the average of their losses. We check this inequality on the actual training examples. A linear recurrent network is linear in its input for fixed weights, but its input and readout weights multiply each other. For example, switch off the state transition and use just one state coordinate. Let its input weight be a and its readout weight be b. Each tick then outputs ab times the current input. The pairs $(a, b) = (1, 4)$ and $(4, 1)$ have the same product, 4, while their midpoint $(5/2, 5/2)$ has product $25/4$. Reversing both readout signs if needed makes increasing the magnitude of this product increase the squared training loss. To see why, the initial slope of the loss as a function of the product at zero can have either sign. Choose the direction with nonnegative slope. Along that direction the loss is a strictly convex quadratic because at least one observed input is nonzero. The midpoint therefore has strictly greater loss than either endpoint.

The same construction works with the bias and readout of one active scalar rectifier, the input and output weights of one sorting block, or the initial state and readout of an autonomous sequence generator. All unused paths can be set to zero. In the sorter the original input still passes through its fixed identity connection. For the two autonomous tasks, a transition of one on the active coordinate keeps the output constant across the observed indices. Exact rational evaluation of the stored training data certifies 36 smooth-function settings (four tasks, widths 1, 4 and 16), six running sums (widths 1 and 2), nine four-input sorters (four, five or six blocks), and 18 sequence generators (Fibonacci widths 1–3 and triangular widths 2–4). Each size uses three starts. The sequence generators share their training labels across starts. Replaying the training code gives midpoint violations from 0.872 to 16.014 in units of training loss. The original 18 affine arithmetic settings and three memoryless sums minimize a weighted sum of squares of affine functions of their weights, a convex quadratic. Addition and subtraction on trainable rectifier networks of widths 1, 2 and 4 are a separate 18-setting experiment with their own midpoint certificates. A midpoint violation establishes the shape of the objective. Whether a solver reaches a better fit is a separate measurement.

A.10 What the WSP responses encode

What Astra returned. Each original one-shot model response was restricted to numeric arrays, with one full array per network size. Across 60 calls we recovered 174 arrays, all the requested shapes. The recording for each call contains one final message and zero tool actions. Relative to the supplied initialization, 3,142 of 3,252 stored scalars changed value. The prompts, original responses, run metadata and evaluated arrays permit an independent check of the computations encoded by the weights. The internal procedure by which Astra chose their numbers is unobserved.

For example, an examples-only addition request initialized a two-input, two-unit network with nine stored coefficients. Its returned array was $(1, 1, -1, -1, 0, 0, 1, -1, 0)$. Read in the prompt's order, its hidden input rows are $(1, 1)$ and $(-1, -1)$, both hidden biases are zero, the output weights are $(1, -1)$ and the output bias is zero. For inputs $(7, -9)$ the two rectified hidden values are $(0, 2)$ and the output is -2. For inputs $(9, -7)$ the hidden values are $(2, 0)$ and the output is 2. The complete array thus implements $R(t) - R(-t)$ for the sum t. All 12 width-two arithmetic replies across the two tasks, both information conditions and three starts implement this sign pairing. The coefficient audit in the research bundle gives the prompt and reply paths for every case.

To check whether the scalar networks preserved a known symmetry, we evaluated each stored binary64 weight as an exact rational number. A one-layer scalar ReLU network gives an affine function between consecutive hidden-unit breakpoints. Its deviation from odd symmetry, $f(x) + f(-x)$, is therefore affine between those breakpoints and their reflections. Its maximum absolute value on $[-10, 10]$ occurs at a breakpoint or an interval endpoint. None of the 18 sine arrays or 18 hyperbolic-tangent arrays was exactly odd on this interval. For sine specifically, 16 of 18 networks gave a nonzero value at zero. The remaining two failed the sign-reversal test at 1 and -1. The analogous check for the even square target found zero exactly even arrays out of 18. These are symmetry tests on the returned coefficients, distinct from prediction tests in rounded arithmetic. The experiment varied task, architecture and formula access. The effect of a rule's familiarity to Astra remains open.

Does feedback restore symmetry? Among the 18 sine networks across both information conditions, the exact symmetry test passed for none of the one-shot networks and 8 of the iteratively selected networks. For square, 0 one-shot and 11 iterative networks were exactly even; for hyper-

bolic tangent, 0 and 8 were exactly odd (out of 18 each). These statements concern the real function represented by the stored weights over $[-10, 10]$.

B Numerical conventions for the additional studies

The primary benchmark uses the named NumPy numerical-library operation or an integer recurrence converted to binary64 as its reference. In the supplementary elementary-function experiments, the MPFR library for multiple-precision floating-point computation supplies correctly rounded mathematical values [22]. *Correct rounding* returns the representable number nearest the exact mathematical result, choosing the one with an even last significand bit in a tie. The *significand* holds the significant binary digits and the exponent gives their scale. Normal binary64 numbers have 53 significant bits. A dyadic rational is an integer divided by a power of two: for example, $3/8$ is exactly 0.011 in binary. Every finite binary64 number has this form.

A *unit in the last place* (ULP) is one step in the last stored digit at a given scale [7]. We measure ULP distance by counting adjacent representable values, treating both signs of zero as equal. At 1, for example, the next larger number is $1 + 2^{-52}$ and the next smaller number is $1 - 2^{-53}$. Both have distance one under this convention. The mean scaled absolute error averages each absolute prediction error divided by the larger of one and the absolute reference value. A parameter *slot* stores one scalar even if it is zero. An *edit* changes one stored scalar from its initialized value. An *operation* is an executed arithmetic or activation step in the stated graph. A real-arithmetic proof assumes exact algebra; a binary64 proof specifies where rounding occurs.

C Why fitting a short running sum can fail on the next step

A recurrent neural network (RNN) carries a state vector from one input to the next. Let $d \geq 1$ be the number of state coordinates. At step $t \geq 1$ let x_t be the scalar input, h_t the column of d real state values, and y_t the scalar output. Choose a coordinate j from 1 to d, let e_j be the column with a one at j and zeros elsewhere, and write $c = e_j^\top$ for its transpose. Multiplication by the fixed row c reads coordinate j. With a d-by-d transition matrix A and an input-weight column b, the network is

$$h_0 = 0, \qquad h_t = Ah_{t-1} + bx_t, \qquad y_t = ch_t.$$

There is no additive offset. Holding A, b fixed over time gives the same update rule at every step.

Specify the target by a nonzero sequence of real coefficients $g = (g_0, g_1, \ldots)$. Its response at step t is the convolution $\sum_{k=0}^{t-1} g_k x_{t-k}$, a weighted sum of the current and past inputs. The *unit impulse* is the stream beginning with 1 followed by zeros. The network's successive outputs on it are $cA^k b$ for integers $k \geq 0$, with A^0 the identity matrix. By linearity, matching all streams for the first H steps is equivalent to the H identities $cA^k b = g_k$ for $0 \leq k < H$. We call H the horizon and the first H coefficients the prefix.

Let S be the shift on sequences, defined by $(Sg)_k = g_{k+1}$. For a polynomial $p(z) = \sum_{i=0}^{m} p_i z^i$ of degree m in a variable z, write $(p(S)g)_k = \sum_{i=0}^{m} p_i g_{k+i}$. The *recurrence order* r is the smallest degree of such a polynomial with leading coefficient one for which $p(S)g$ vanishes at every index. Thus r terms determine the next by a fixed linear rule. For fixed g and j, let $E(d, r, H)$ be the fewest entries of the initialized pair (A, b) that must change to match the prefix. There are $D = d^2 + d$ editable entries, each costing one edit if changed.

Theorem 1 (Minimum edit count)**.** *Fix the target independently of the initialization. Assume $d \geq r$, and draw (A, b) from a joint distribution with a density on the real parameter space. With probability one, for every integer $H \geq 1$,*

$$E(d, r, H) = \min\{H, d + r\}.$$

Matching every horizon costs $d + r$ edits. For $d < r$, no such network matches every horizon.

Here *with probability one* excludes a set of initial weights of volume zero. Continuous Xavier draws satisfy this assumption. The theorem provides a benchmark for finite binary64 draws, which have a discrete distribution. At $H < d + r$ a minimum-edit exact fit must fail on the next coefficient, since that coefficient requires one more edit. Conversely, let $\chi_A(z) = \det(zI - A)$ be the characteristic polynomial of A, where I is the identity matrix and $\det$ the determinant. The Cayley–Hamilton identity $\chi_A(A) = 0$ makes the network coefficients obey a recurrence of degree at most d. Their difference from the target obeys the recurrence given by $\chi_A(S)p(S)$ of degree $d + r$. Its first $d + r$ zeros force all subsequent terms to vanish.

For a running sum $g_k = 1$ and $r = 1$. One construction sets $cA = c$ and $cb = 1$ by writing the observed row and one input weight. A second sets $b_j = 1$ and changes column j until $Ab = b$, so all input contributions lie on the line spanned by b. Both use $d + 1$ edits at almost every starting point, or 33 of 1,056 entries at $d = 32$. In the general construction, changing entries of b matches a short prefix. Beyond d coefficients, selected entries in one column of A can also change its characteristic polynomial. Appendix E gives the construction and lower bound, including networks whose output cannot observe every state direction.

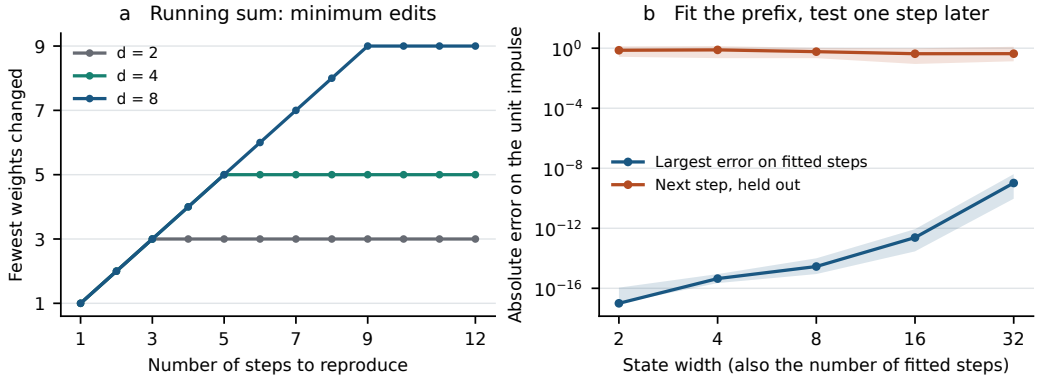


Figure 16: A linear recurrent network stores d numbers and returns one fixed state coordinate. At left, choose a number of steps on the horizontal axis. A point gives the fewest individual weights that must change to reproduce all those running sums, for the indicated state width. For example, at width four the cost stops at five edits. These points are exact theoretical values. At right, the horizontal axis is the number of state coordinates. We give each network an input of one followed by zeros and fit its input weights to the first d outputs, all equal to one. The vertical axis gives absolute error on this sequence. Only the right panel aggregates trials: each point is a median across 64 random starts. The bands show the middle 50% of these runs. The fitted error is the largest absolute error on the prefix. The unseen error is the absolute error on its next coefficient. Values below 10^{-17} appear at 10^{-17} on the logarithmic axis.

D Can a language model find the predicted edits?

For a target state s_t at step t, choose a multiplier λ on its previous value and an input gain γ. Starting at $s_0 = 0$, the target rule is $s_t = \lambda s_{t-1} + \gamma x_t$. A running sum has $\lambda = \gamma = 1$. In each of six pairs we sent Astra the same initialized network, rule, fixed output coordinate and ordering of entries in the array. One request specified just a prefix and allowed scratch arithmetic. The other specified all horizons and prohibited tools. Each edit was an index and a replacement value. Neither request received test feedback. Table 14 reports all six pairs.

Table 14: Paired WSP requests for $s_t = \lambda s_{t-1} + \gamma x_t$, and a BP control given the same editable entries. Here d is state width and H is the short horizon and short edit count. Fit exact counts matching impulse coefficients out of H. Next gives the absolute error on coefficient g_H, also its single-output RMSE. WSP and BP agree to the three decimal places shown for Next. The last column counts all-horizon WSP edits.

			Short WSP		Short BP			All-horizon
(λ, γ)	d	H	Fit exact	Fit RMSE	Fit exact	Fit RMSE	Next	WSP edits
$(1, 1)$	4	2	1/2	7.85×10^{-17}	1/2	1.57×10^{-16}	0.156	5
$(1, 1)$	5	3	2/3	1.28×10^{-16}	0/3	1.24×10^{-15}	1.528	6
$(1, 1)$	6	4	2/4	3.14×10^{-16}	0/4	2.10×10^{-14}	0.938	7
$(-1/2, 2)$	4	2	1/2	2.21×10^{-14}	1/2	2.20×10^{-15}	0.630	5
$(1/2, -1)$	5	3	2/3	3.20×10^{-17}	0/3	1.57×10^{-16}	1.514	6
$(-1, 1/2)$	6	4	2/4	6.21×10^{-17}	0/4	9.36×10^{-15}	0.018	7

All six all-horizon replies set $cA = \lambda c$ and $cb = \gamma$. Substitution into the update proves the target recurrence for every real stream. Each also reproduced all 512 central streams of length eight and all 512 outer streams of length 64 with zero RMSE. The short replies used H edits. Their decimal values met the requested absolute tolerance of 10^{-9} on every fitted impulse coefficient, and each missed the next coefficient. The theorem predicts their edit *count* for a real exact fit. The numerical replay measures agreement after rounding.

Two checks separate this result from the behavior of one solver. At width eight, a numerical solve for the input weights on eight impulse coefficients gave a median maximum prefix error of 2.8×10^{-15} and a median next error of 0.588 over 64 starts (Figure 16b). An adaptive gradient method also learned an exact running sum. In a scalar cell with transition weight a, input weight b and output equal to its state, a unit gradient step on half the squared final error for the labeled stream (1) sets $b = 1$ in real arithmetic. Next choose the input stream $(1, -a)$ and hold its values fixed while differentiating. The output has derivatives $(1, 0)$ with respect to (a, b) and its target is $1 - a$. A second unit step sets $a = 1$. Repeating each update once to accommodate rounding gave exactly $(a, b) = (1, 1)$ in 256 of 256 binary64 Xavier draws, with zero RMSE on 512 outer streams per draw.

For a matched numerical control, we disclosed the locations of the WSP edits to BP, froze all other entries and used 800 Adam steps followed by up to 1,000 BFGS iterations. The short control received the same H impulse labels. For the all-horizon control we supplied the first $d + 1$ impulse labels, enough to certify an *exact* fit for an order-one target at width d. On each of the six outer tests WSP attained 512/512 complete streams with zero RMSE. BP attained 0/512 in each case, with shell RMSE between 1.02×10^{-13} and 1.44×10^{-11}. The archive includes all six pairs at full precision.

E Proof of Theorem 1

We count changes to the $D = d^2 + d$ real entries of the transition matrix A and input column b of a width-d linear recurrent network. The output row c reads one fixed state coordinate. Renumbering the coordinates preserves edit counts, so take $c = e_1^\top$, where e_1 is the first standard basis column and $^\top$ denotes transpose. Here, the real numbers form the field of coefficients for every calculation in the proof. A unit impulse is input 1 on the first tick and 0 thereafter. Index its desired scalar outputs by $g_0, g_1, \ldots$ and fix the number $H \geq 1$ of required outputs. For a positive integer s, define the *observation matrix* $O_s(A)$ by stacking the s rows $c, cA, \ldots, cA^{s-1}$. The rank of a matrix is the number of independent columns (equivalently, rows). Its kernel comprises the vectors that it sends to zero. The product of $O_s(A)$ with b gives the network's first s impulse outputs. Write $g_{0:s-1} = (g_0, \ldots, g_{s-1})^\top$.

Let S be the forward shift on sequences, so that $(Sg)_i = g_{i+1}$. A monic polynomial has leading coefficient one. For a monic polynomial of degree m, written $q(z) = z^m + \sum_{j=0}^{m-1} q_j z^j$ in the indeterminate z, set

$$(q(S)g)_i = g_{i+m} + \sum_{j=0}^{m-1} q_j g_{i+j}.$$

Fix the smallest-degree monic polynomial p for which $p(S)g = 0$ at all indices. Its degree r is the recurrence order of the nonzero target. We assume $d \geq r$ until the final step. A *Hankel matrix* is constant on each antidiagonal. The leading r-by-r Hankel matrix $G_r = (g_{i+j})_{0 \leq i,j < r}$ is invertible. Otherwise a nonzero linear combination of its columns would give a nonzero polynomial of degree less than r whose filtered sequence has r initial zeros. Filtering commutes with $p(S)$, so the recurrence propagates those zeros to every index, contradicting minimality. All columns of the infinite Hankel matrix after the first r obey the same recurrence, giving overall rank r.

Controlling the characteristic polynomial. Call (c, A) *observable* if $O_d(A)$ has rank d, so the first d scalar observations distinguish all initial states. Let I be the d-by-d identity, let $\chi_A(z) = \det(zI - A)$ be the characteristic polynomial, and let $\operatorname{adj}(M)$ denote the transpose of the cofactor matrix of a square matrix M. For a freely chosen column $u \in \mathbb{R}^d$, changing A to $A + uc$ changes only column one. The rank-one determinant identity gives

$$\chi_{A+uc}(z) - \chi_A(z) = -c \operatorname{adj}(zI - A)u.$$

For observable (c, A), the displayed map from u to polynomials of degree below d is a linear bijection. Indeed, if the right side vanishes, so does $c(zI - A)^{-1}u$ wherever $zI - A$ is invertible. Its power series in $1/z$ near infinity has coefficients $cA^k u$ for integers $k \geq 0$. They all vanish, hence $O_d(A)u = 0$ and $u = 0$. Both spaces have dimension d, proving bijectivity. The new matrix remains observable because $c(A+uc)^k$ equals cA^k plus a linear combination of preceding observation rows. Finally, the determinant of $O_d(A)$ is a polynomial which equals one for the matrix with ones on its first superdiagonal and zeros elsewhere. Its zero set has Lebesgue measure zero. Thus the required observability holds with probability one under the initialization in Theorem 1.

Constructing an edit. If $H \leq d$, select H linearly independent columns of $O_H(A)$. Holding the remaining entries of b fixed, solve the resulting invertible system for the selected entries of the new input vector b' to obtain $O_H(A)b' = g_{0:H-1}$. This takes at most H edits.

For $H > d$, put $k = H - d$. Seek a monic degree-d polynomial q such that the first k terms of $q(S)g$ vanish. These linear equations in the coefficients of q are consistent because a monic degree-d multiple of p exists. Their k-by-d Hankel coefficient matrix has rank $\min(k, r)$, since it contains the first $\min(k, r)$ rows of the invertible G_r and the infinite matrix has rank r. The bijection above lets us impose the equations on $q = \chi_{A+uc}$ with the same rank in u. Select that many independent columns of this linear system and use just their coordinates of u. Then solve $O_d(A + uc)b' = g_{0:d-1}$, using at most d more edits. The Cayley–Hamilton identity says a square matrix satisfies its characteristic polynomial. Consequently the network impulse outputs satisfy the recurrence q, and so agree by induction with the target through the first H outputs. If $k \geq r$, the first r terms of $q(S)g$ vanish, and the recurrence p propagates those zeros to all subsequent terms. The total cost is at most $d + \min(k, r)$.

Bounding the size of the solution set. Codimension is the ambient dimension minus the dimension of the subset. We show that the set of (A, b) realizing the prefix has codimension at least $\min(H, d + r)$, allowing for unobservable realizations. Let $m = \operatorname{rank} O_d(A)$, and write $U = \ker O_d(A)$ for the subspace of initial states invisible to the output. Because $c \neq 0$, we have $1 \leq m \leq d$. The subspace U has dimension $d - m$, lies in $\ker c$, and satisfies $AU \subseteq U$ by Cayley–Hamilton. The family of possible subspaces U of this dimension inside the $(d - 1)$-dimensional space $\ker c$ has dimension $(d - m)(m - 1)$. Locally such a subspace is the graph of a linear map between spaces of dimensions $d - m$ and $m - 1$. For fixed U, the condition $AU \subseteq U$ forces a block of $m(d - m)$ entries to vanish in a basis adapted to U. Subtracting the freedom to choose U leaves codimension at least $d - m$. Identifying states that differ by an element of U gives an m-dimensional quotient space. Denote its induced transition and output by $\bar{A}, \bar{c}$. This quotient is observable.

If $m \geq H$, the first H observation rows are independent. Once an observation row belongs to the span of earlier rows, multiplying the relation by A prevents every subsequent row from increasing that span. Hence the H impulse equations impose H constraints on b for fixed A, as required. If $m < H$, the first m impulse equations impose m independent constraints on b, leaving $d - m$ free coordinates. Put $k = H - m$. The other equations require $q = \chi_{\bar{A}}$ to satisfy $(q(S)g)_i = 0$ for $0 \leq i < k$. The coefficient matrix $B = (g_{i+j})_{0 \leq i < k,\, 0 \leq j < m}$ has rank $\operatorname{rank} B$. When consistent, these equations impose that many independent conditions on $\bar{A}$. To see this, perturb it by $\bar{u}\bar{c}$ for a quotient-space column $\bar{u}$. The characteristic coefficients vary freely by the bijection above, and every such perturbation lifts to a matrix that preserves U. Thus the total codimension for this value of m is at least $(d - m) + m + \operatorname{rank} B = d + \operatorname{rank} B$.

If $m \geq r$, then $\operatorname{rank} B = \min(k, r)$ and $d + \operatorname{rank} B \geq \min(H, d + r)$. If $m < r$ and $k \geq r$, propagation of the initial zeros of $q(S)g$ would give an annihilating polynomial of degree $m < r$, contradicting the definition of r. Finally if $m < r$ and $k < r$, deleting $r - k$ rows and $r - m$ columns of G_r leaves B, so $\operatorname{rank} B \geq k + m - r = H - r$. Then $d + \operatorname{rank} B \geq H$, as required.

From dimension to scalar edits. Fix a set of s coordinates allowed to change. Initial networks which can reach the prefix solution set lie in its sum with the s-dimensional subspace supported on those coordinates. The solution set is defined by polynomial equations in real parameters. Its sum with this subspace is semialgebraic, meaning that it can be described using finitely many real polynomial equalities and inequalities and Boolean combinations. Its dimension is at most $D - \min(H, d + r) + s$ because it is the image under addition of a product of that dimension at most. If $s < \min(H, d + r)$, this is less than D, so the sum has Lebesgue measure zero. There are finitely

many choices of coordinates and countably many positive integers H, giving the lower bounds simultaneously with probability one. At $H = d + r$, the upper and lower constructions give the all-horizon result. For $d < r$, Cayley–Hamilton would give an annihilating polynomial of degree d, ruling out all-horizon realization. □

Exact arithmetic verification. We treated each binary64 Xavier coefficient as its exact rational value. The three test impulse sequences were the constant $g_n = 1$, the Fibonacci numbers $F_0 = 0, F_1 = 1, F_{n+2} = F_{n+1} + F_n$, and the triangular numbers $T_n = n(n+1)/2$ for nonnegative integers n. We tested five initialization seeds and widths $1, 2, 3, 4, 6, 8$ whenever $d \geq r$. Across 480 cases with $1 \leq H \leq d + r$, every write changed exactly the predicted number of coordinates. Each of the 405 shorter-horizon writes failed at the next coefficient g_H. At $H = d + r$, all 75 characteristic polynomials of the edited matrices were divisible by p, and their first 32 impulse coefficients matched directly. Divisibility and agreement on the first d coefficients then certify every later coefficient.

F A comparator that preserves binary64 operands

A comparator takes two inputs and returns their smaller and larger values in that order. Let a, b be finite binary64 inputs in $[-10, 10]$, let $R(t) = \max(t, 0)$ be the rectified linear unit, and take fixed constants $K = 2^{537}$, $M = 10$. Round each displayed operation separately to the nearest binary64 value, choosing the one with even last significand bit at a tie. Gradual underflow retains reduced-precision values below the smallest positive normal number. Numerical equality identifies the two signed zeros.

First extract the positive difference r, amplify it to u, and compute the auxiliary signals v, B, S. Then compute the keep and swap selectors k, s, respectively, in this order:

$$r = R(a - b), \quad u = Kr, \quad v = u - R(u - 1), \quad B = 1 - R(1 - u), \quad S = 1 - R(1 - Kv),$$
$$k = R((1 - B) - S), \qquad s = 1 - k.$$

For an operand x and an enable bit $g \in \{0, 1\}$, set $G(x, g) = R(x - M(1-g)) - R(-x - M(1-g))$. The comparator returns $(G(a, k) + G(b, s),\ G(a, s) + G(b, k))$, reusing the shifts Mk, Ms.

When $a \leq b$, $u = B = S = 0$ and $k = 1$. When $a > b$, the rounded positive difference is at least the smallest representable positive value 2^{-1074}. For $0 < u \leq 1$, $v = u$ and $Kv \geq 1$, hence $S = 1$. For $u \geq 1$, $B = 1$. In that case the rounded value of $u - 1$ lies between $u - 2$ and u, since u itself is representable and is one unit of absolute distance from the exact difference. Therefore $0 \leq v \leq 2$, which also shows that the product Kv remains finite. Both B, S lie in $[0, 1]$ and at least one equals one, so $k = 0$. For $g = 1$ the gate G returns the original operand, while for $g = 0$ both rectifiers vanish. Composition along a sorting schedule therefore preserves all numerical operands. The stored comparator contains 43 edge weights and seven biases, with Xavier initialization for the weights and zero initialization for the biases.

G Scalar construction and size sweeps

For a real input $x \in [-16, 16]$, set $t = |x|/16$. Define the tent map $T(t) = 2t - 4R(t - 1/2)$ on $[0, 1]$, where $R(t) = \max(t, 0)$. For a positive integer refinement depth m, let $T^{\circ i}$ denote i successive applications of T. The depth-m square interpolant is

$$256 \left[t - \sum_{i=1}^{m} 4^{-i} T^{\circ i}(t) \right].$$

At each refinement, the added term corrects the midpoint error of the preceding linear interpolant on intervals with endpoints at multiples of a power of two. The maximum absolute error of the resulting interpolant on $[-16, 16]$ in real arithmetic is $64 \cdot 4^{-m}$. The compact model starts at hidden width two, uses width three in its remaining layers, and has an affine output and $12m + 5$ parameter slots.

For the compensated version let q denote the running unscaled sum and u its next term to subtract. Compute the rounded difference $s = q - u$, then $d = q - s$, then $e = d - u$, using the compensated subtraction transform [4]. Accumulate $R(e)$ and $R(-e)$ separately. Add their difference to the last

q before scaling the output. Copying nonnegative intermediate results through rectifiers implements the stated arithmetic with maximum hidden width six and $203m + 35$ slots. In exact real arithmetic $e = 0$. Setting only the two weights that read the error accumulators to zero tests the numerical contribution of the compensation.

To construct difficult shell inputs, express a positive normal input as $x = N\,2^{e-52}$, where the integer exponent e places x in the interval $[2^e, 2^{e+1})$ and N is its 53-bit integer significand. The low bits of N^2 determine which side of a rounding midpoint contains x^2. Let $k \in \{52, 53\}$ count the discarded bits and take $j \in \{7, 15, \ldots, 255\}$. Find integers satisfying $N^2 \equiv 2^{k-1} - j \pmod{2^k}$ by extending the low bits one at a time. Retain significands with the specified discard count and shell magnitude, then include both signs. This gives 304 inputs from 152 magnitudes. Networks with 35, 48 and 55 refinements match respectively $0, 40, 304$ of these inputs. All three match all 131,072 fresh uniform shell inputs. On 501 small powers $x = 2^n$, $-500 \le n \le 0$, the same networks match 32, 45 and 52 inputs. The first affine segment near zero gives large relative errors despite small absolute errors.

H Historical controls with an extrapolation-first WSP prompt

This preliminary study used an extrapolation-first prompt and included affine arithmetic and memoryless sum controls. All counts in this appendix refer to that earlier design. A network's *architecture* specifies which operations it applies and where coefficients can enter. For the scalar tests, an *affine* function multiplies input values by fixed coefficients, sums them and adds a constant. A *rectified linear unit* (ReLU) is the scalar function $R(u) = \max(0, u)$ of a real input u. Let w count the hidden units, let a, b denote the slope and offset of a direct input-to-output path (a *skip*), and give hidden unit j input slope u_j, offset v_j and output weight c_j. The shallow network computes

$$f(x) = ax + b + \sum_{j=1}^{w} c_j R(u_j x + v_j).$$

Each hidden unit contributes a *hinge*, a possible change of slope at its zero crossing (a *knot*). This architecture has $3w + 2$ scalar slots.

We tested the square, sine in radians, positive self-power x^x, hyperbolic tangent $\tanh x = (e^{2x} - 1)/(e^{2x} + 1)$ with e the base of the natural logarithm, addition, subtraction, running sum, sorting four inputs in ascending order, Fibonacci numbers, and triangular numbers. At integer index $n \ge 0$, Fibonacci numbers start at $F_0 = 0, F_1 = 1$ and obey $F_{n+2} = F_{n+1} + F_n$. The triangular number is $T_n = n(n + 1)/2$. Self-power uses positive real inputs. Appendix L handles the separate negative-integer extension.

For each task and each of three random starts, two independent Astra requests saw the same three architectures, weights, deployment envelope and labeled examples. One request supplied *the formula and examples*. The other supplied *examples only*, omitting the task name and rule. Each request returned three complete arrays. This gives 60 calls and 180 WSP arrays. All 180 could be executed. The three arrays from a single call are dependent. This protocol allows a complete overwrite. Only Section C minimizes edits. For each of the 90 task, size and start combinations we also fit all slots using BP with 800 Adam updates followed by up to 1,000 BFGS iterations. Each solver saw 12 central examples except on the integer-sequence tasks, which used indices 0 through 8. WSP was asked to prioritize the *largest* scaled error. BP minimized half the *mean squared* scaled error. Architectures, examples and initial arrays were paired. Objectives, prior knowledge and computational budgets differed.

For this preliminary study, Table 15 uses the largest tested shallow scalar width (16 hinges) and the smallest tested size known to suffice over the reals for each algorithmic task. Both WSP conditions achieved perfect outer scores on addition, subtraction, running sum, Fibonacci and triangular numbers at those sizes. The examples-only condition still exposed a task-specific graph and recognizable labels. At width 16 all four smooth targets had zero exact outer matches for either WSP condition and for BP. On square, however, examples-only WSP had RMSE 0.633 versus 29.5 for BP. For sorting, formula-and-examples WSP matched only 53.5% of complete vectors, but its RMSE was 5.2×10^{-16}. With a gradient fit the corresponding values were 0.6% and 5.95. Appendix I retains the scale-adjusted results, which reveal a different aspect of errors on rapidly growing x^x.

Table 15: Held-out outer predictions on the same networks. Each method has an exact-match percentage and a raw root mean squared error (RMSE). Entries are medians over three starts. Formula and examples and examples only are the two weight-space programming (WSP) conditions. The backpropagation condition receives examples only. Scalar scores use 2,048 inputs per available shell side. Addition, subtraction, running sum and sorting each use 2,048 inputs or lists. A list must match at every position. Each Fibonacci or triangular score uses the 32 indices from 33 through 64 of one trajectory.

		WSP Formula and examples		**WSP** Examples only		**Backpropagation** Examples only	
Target	Slots	Exact %	RMSE	Exact %	RMSE	Exact %	RMSE
x^2	50	0.0	0.644	0.0	0.633	0.0	29.5
$\sin x$	50	0.0	0.0335	0.0	0.0324	0.0	2.46
x^x, positive	50	0.0	1.3×10^{8}	0.0	1.3×10^{8}	0.0	1.6×10^{9}
$\tanh x$	50	0.0	0.00154	0.0	0.0015	0.0	0.52
Addition	2	100.0	0	100.0	0	27.2	2.3×10^{-15}
Subtraction	2	100.0	0	100.0	0	1.3	4.7×10^{-15}
Running sum	3	100.0	0	100.0	0	0.0	2.3×10^{-14}
Sort four	40	53.5	5.2×10^{-16}	53.6	5.0×10^{-16}	0.6	5.95
Fibonacci	8	100.0	0	100.0	0	0.0	0.269
Triangular	15	100.0	0	100.0	0	0.0	2.1×10^{-8}

How close was each fit to the best training loss? The training loss is half the mean of $[(\hat{y}_i - y_i)/\max(1, |y_i|)]^2$ over observed scalar outputs. For the 12 one-hinge settings (four smooth functions and three starts), we computed the globally smallest *real-arithmetic* loss on the stored binary64 labels by enumerating all intervals in which the knot can lie. For a second gradient-based optimizer, *trust-region least squares* fits a local linear approximation of all output errors and restricts the update to a neighborhood where that approximation is useful. We applied it both at the original weights and after WSP.

Using an absolute training-loss gap of 10^{-10} to the certified optimum, Adam succeeded in 0/12 settings, Adam followed by BFGS in 1/12, and trust-region least squares from the same Xavier weights in 5/12. Starting least squares from examples-only WSP raised that to 9/12. Among the 90 settings, least squares from Xavier obtained six completely exact outer sets, all on addition or subtraction. In contrast, both WSP conditions obtained 30/90. The stronger optimizer improved a central objective without guaranteeing transfer. Refining the formula-and-examples WSP arrays reduced training loss materially in 54 settings. It improved outer mean scaled absolute error in seven and worsened it in 47. Figure 17 gives the paired counts. Appendix N reports the per-task RMSE comparisons, solver budgets and optimality argument. Appendix O repeats the comparison with hidden units frozen so that the output weights admit an exact least-squares characterization.

How did conventional recurrent networks fare? A hyperbolic-tangent recurrent cell applies $\tanh$ to its updated state. A gated recurrent unit (GRU) uses gates, numbers between zero and one computed from the current input and previous state, to regulate state retention and updating. We trained both on running sums, Fibonacci and triangular recurrences with two variable starting values, and list sorting in which a shared cell first reads a list and then emits its sorted elements. All 102 task, cell, width and initialization settings were fit at two budgets. None of the resulting 204 fits returned a complete exact example on any of the four held-out sets (central, changed magnitude, changed length, or both). A different one-state $\tanh$ cell, with its hidden scale fixed to 2^{-40} and two weights learned from ordinary central labels, learned the exact running sum in three of five starts. Each successful start matched all 1,024 outer length-32 streams, with zero RMSE. Appendix P gives the update equations, unscaled errors and the separate results for the WSP state scaling.

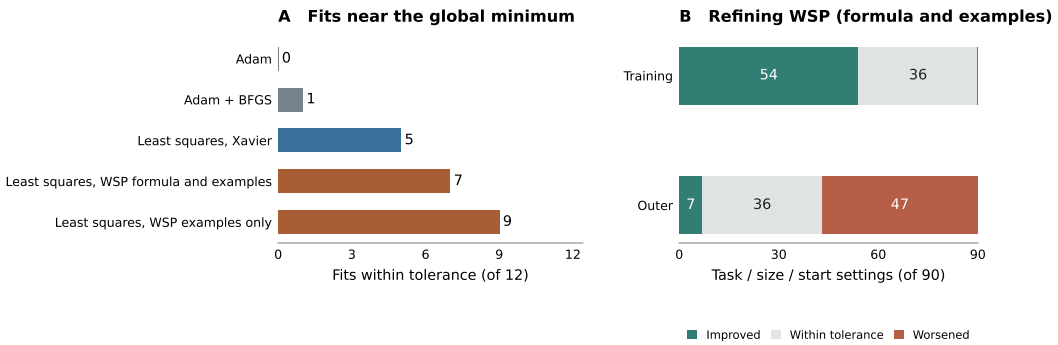


Figure 17: Optimization of the central labels and transfer to the outer shell. Panel a counts fits whose observed loss lies within 10^{-10} of the certified real minimum across 12 one-hinge settings. Panel b starts from formula-and-examples WSP and refines each of 90 arrays by least squares. Lower is better. An absolute change of at most 10^{-10} is classified as within tolerance. Training uses half the mean squared scaled error, and outer evaluation here uses mean scaled absolute error. Table 15 gives raw RMSE in output units.

I Additional value and sequence extrapolation controls

The extrapolation-first experiment in Appendix H reports error in target units. Table 16 gives that experiment's scaled absolute errors so that large targets, such as positive self-power near ten, can also be interpreted on a relative scale.

Table 16: The same ten architectures and outer inputs as Table 15. Entries are mean absolute errors after dividing each error by the larger of one and the absolute reference value, followed by the median over three starts.

	Weight-space programming (WSP)		Backpropagation
Target	Formula and examples	Examples only	Examples only
Square	0.0113	0.0112	0.419
Sine	0.0289	0.0279	1.83
Positive self-power	0.0774	0.0804	0.979
Hyperbolic tangent	0.00133	0.00132	0.367
Addition	0	0	3.38×10^{-16}
Subtraction	0	0	1.05×10^{-15}
Running sum	0	0	2.46×10^{-15}
Sort four	3.91×10^{-17}	3.72×10^{-17}	0.776
Fibonacci	0	0	8.43×10^{-14}
Triangular	0	0	1.06×10^{-11}

Analytic choices of hinge positions. For the earlier scalar study, two WSP constructions choose knot locations from the known target expression. Central WSP uses only knots in the central interval. Full-envelope WSP uses formula values outside it. Both use 63 hinges plus a direct affine path (191 slots). The BP comparator sees central labels and uses Adam followed by limited-memory BFGS, which estimates curvature from a short history of changes in weights and gradients. Table 17 shows mean absolute error (MAE), the average of $|\hat{y}_i - y_i|$, alongside RMSE.

A WSP sine construction first folds the input using its known period and reflection symmetry, then fits its final spline to 1,024 central labels. Outer MAE was 4.75×10^{-5} at 211 slots. For self-power, reallocating knots and centering the error using the formula reduced maximum relative shell error to 1.01%. A single native hyperbolic-tangent activation between identity maps matches the same library function by construction. Ten Gauss–Newton steps, which approximate nonlinear least squares by successive linear least-squares updates, gave 100% shell accuracy from all three starts and 89.1–93.2% inner accuracy. A two-weight affine model learned addition and subtraction exactly

Table 17: Outer errors for the analytic WSP and gradient controls. Mean absolute error (MAE) is the average absolute prediction error. All methods had 0% exact on all four tasks at this width. Evaluation uses 4,096 samples per available shell side. BP columns give medians over three starts. The two signed sides are pooled before computing RMSE.

	Central WSP		Full-envelope WSP		Backpropagation	
Target	MAE	RMSE	MAE	RMSE	MAE	RMSE
Square	16.6	22.2	0.00737	0.00829	16.9	22.6
Sine	3.26	4.08	0.00486	0.00602	3.26	4.07
Hyperbolic tangent	0.0332	0.0395	1.12×10^{-5}	3.22×10^{-5}	0.0919	0.109
Positive self-power	4.32×10^{8}	1.46×10^{9}	9.55×10^{6}	3.54×10^{7}	4.32×10^{8}	1.46×10^{9}

in three starts with 64 full-batch gradient updates. Per-split errors for these variants are in the accompanying data.

Fixed sequence interfaces. Table 18 compares WSP and BP on the original multistep tasks. In an autonomous linear network, a column v reads out starting state h_0 after n transitions as $v^\top A^n h_0$. Its state width must be at least the recurrence order. Fibonacci and triangular numbers require widths two and three, respectively, in this homogeneous model (no additive state offset). A one-state running sum uses unit input, transition and readout weights. Setting the scalar input weight to one and adjusting the readout to preserve their product leaves the real input–output function unchanged. On the same 256 training streams this parameterization yielded exact weights in four of five fresh starts, or three of five starts matched for initial function. Each successful run matched all tested streams at both lengths.

Table 18: Exact rate and raw root mean squared error (RMSE) for WSP and backpropagation. Sums use 2,048 central length-eight streams and 2,048 outer length-32 streams. Sorting uses 2,048 four-item lists per split. These two exact rates require complete outputs. Fibonacci and triangular rates count indices on one trajectory, the nine *fitted* indices 0–8 versus the 32 *held-out* indices 33–64. Values are medians over five starts.

			WSP		Backpropagation	
Target	Slots	Split	Exact %	RMSE	Exact %	RMSE
Running sum	3	Central	100.00	0	0.93	5.42×10^{-16}
Running sum	3	Outer	100.00	0	0.00	6.94×10^{-15}
Fibonacci	8	Central	100.00	0	11.11	5.67×10^{-15}
Fibonacci	8	Outer	100.00	0	0.00	0.0531
Triangular	15	Central	100.00	0	0.00	2.36×10^{-14}
Triangular	15	Outer	100.00	0	0.00	1.02×10^{-8}
Sort four	15	Central	65.14	1.21×10^{-16}	0.00	1.22
Sort four	15	Outer	53.96	5.12×10^{-16}	0.00	4.94

The sorter in this table fixes the order of five pair comparisons and stores three adjustable gains per comparison (15 slots). Tying these five sets uses three stored gains. The 40-slot sorter in the main comparison instead learns eight coefficients per step on four wires, including which wires to combine. The operand-preserving comparator stores 50 coefficients in one block shared across all comparisons. From central labels, a search over 9,331 schedules of up to five pair comparisons found 12 shortest sorting schedules. A rational linear solve recovered Fibonacci and triangular recurrences from nine central labels. These controls supply mathematical structure to the solver.

J First-response WSP pilot across nine tasks

We made nine independent requests, one per task listed below, beyond the twelve recurrence requests of Appendix D. Each prompt gave Astra the formula, required domain, architecture and exact initialized weights, and requested one complete vector of replacement weights. The model could perform scratch arithmetic. It received no test feedback. For square, sine and positive self-power,

the graph used a seven-hinge hidden layer and a direct affine input–output connection (23 stored scalars). Hyperbolic tangent used the native activation between two affine transformations. Addition and subtraction had two coefficients each. The homogeneous recurrent networks for Fibonacci and triangular numbers used widths two and three respectively. The sorting graph used five prespecified pairs of wires and reused three scalar coefficients on every pair. Untying those coefficients at each comparison gives the 15-slot sorter in Table 18. Five freely routed four-wire residual blocks use 40 slots (Appendix M). The 50-slot sorter stores one operand-preserving block for repeated use (Appendix F).

Scoring the nine WSP responses. For continuous inputs, the central and outer tests each contained 4,096 examples. Outer input signs were sampled within this total whenever both signs belonged to the task. Number-sequence rows score the nine individual central indices $0, \ldots, 8$ and 56 later indices $9, \ldots, 64$ from one sequence. Four-item sorting requires all four entries of a vector for an exact match. RMSE includes every scalar output on the split. Scaled absolute error for prediction $\hat{y}$ and label y is $|\hat{y} - y| / \max(1, |y|)$. The binary64 reference is the specified NumPy operation, or the integer valued sequence. All entries below are for one response per task. The matched WSP/BP comparison is in Section 3.

Table 19: First WSP response on each of nine tasks. Exact percentages and raw RMSE use the same central and outer observations. Addition and subtraction have separate, identical scores.

		Exact (%)		Raw RMSE	
Target	Slots	Central	Outer	Central	Outer
x^2	23	0	0	.514	.547
$\sin x$	23	0	0	.0985	.0934
$x^x,\ x > 0$	23	0	0	2.65	5.24×10^{8}
$\tanh x$	4	100	100	0	0
Addition and subtraction (each)	2	100	100	0	0
Fibonacci	8	100	100	0	0
Triangular	15	100	100	0	0
Sort four	3	64.33	55.13	1.18×10^{-16}	4.91×10^{-16}

The square response used skip slope -17.5, intercept -75.78125, unit hinge input slopes, knots at $-7.5, -5, -2.5, 0, 2.5, 5, 7.5$ and slope increases of 5. This shifted equal-spacing spline attains the width-seven uniform-error optimum $50/64 = 0.78125$ over the real interval. The sine response had maximum absolute error 0.13822, positive self-power had maximum scaled error 0.48922, and the maximum coordinate error of the sorter was 3.55×10^{-15}. The exact arithmetic weights were $(1, 1)$ for addition and $(1, -1)$ for subtraction. The tanh response used identity maps around the native activation.

For the *initial suite*, the archive contains all 21 prompts and replies. Metadata record the requested service alias `gpt-6-astra`, reasoning effort setting, completed event types and token counts. Requests used fresh isolated chats. The provider may change the deployment behind this alias. Archived arrays can be replayed independently in the specified numerical environment.

K Experimental detail and reproducibility

Hardware and arithmetic. The initial suite used binary64 arithmetic on an Apple M5 central processing unit. For the initial scalar experiments, the environment used Python 3.13.12. The main sequence, frontier and archive scoring environments used Python 3.12.12. Both used NumPy 2.5.4 and the macOS Accelerate implementation of the Basic Linear Algebra Subprograms (BLAS), with SciPy 1.18.1 for optimization. Exact rational computations, which retain integer numerators and denominators, use Python's `Fraction`. Stream sums use successive rounded additions in temporal order. In this suite the transcendental labels are the platform NumPy results. Tests compare to that implementation. Correct rounding of transcendental functions is a separate question. Re-running with a different BLAS or math library can affect last bits and optimizer paths.

Initialization and budgets. Let n_{in} and n_{out} be the input and output coordinate counts of a layer (its fan-in and fan-out). Xavier initialization draws each weight uniformly between the

negative and positive values of $\sqrt{6/(n_{\text{in}} + n_{\text{out}})}$. Biases start at zero. Nominal tensor slots include zeros. We count distinct stored trainable coefficients separately from unrolled uses and count edits separately from either. Scalar widths are $0, 1, 3, 7, 15, 31, 63$. Compact tent depths are $1, 4, 8, 12, 16, 20, 24, 27, 30$. The compensated depths are $16, 24, 27, 30, 35, 40, 48, 55$. In the local and full-envelope WSP spline controls, all coefficients were overwritten using the specified formula. Let t_j be an interior knot and s_{j-1}, s_j the slopes of the adjacent secants. The corresponding hinge has input weight $w_j = 1$, bias $c_j = -t_j$ and output weight $v_j = s_j - s_{j-1}$. A linear skip carries the leftmost secant. No samples from the outer test arrays enter training or checkpoint selection.

Adam, the adaptive moment estimation optimizer, updates weights using running estimates of the first and second moments of the training gradients. The quasi-Newton optimizer BFGS is named for Broyden, Fletcher, Goldfarb and Shanno. Its limited-memory version with support for bounds is L-BFGS-B. A minibatch is a sampled subset of training examples used for one update. The shallow scalar gradient runs use three seeds (11, 22, 33), 1,024 in-range training points, 500 minibatch Adam updates of size 64 with learning rate 0.01, then at most 250 full-batch L-BFGS-B iterations. We retain the endpoint with smaller training mean squared error (MSE), the mean of squared scalar prediction errors. The compact deep square runs use 300 Adam updates and at most 100 L-BFGS-B iterations. Inner scoring uses 4,096 points and outer scoring uses 4,096 per sign where defined. Tanh polishing uses ten least-squares solves with the full analytic Jacobian. For the affine control let X collect the training input rows and N count the rows. The gradient step size is the reciprocal of the largest eigenvalue of $X^\top X/N$. Phase folding uses six rectifiers to implement the piecewise-linear triangular wave $\arcsin(\sin x)$, where $\arcsin$ is the inverse sine taking values in $[-\pi/2, \pi/2]$, on the declared envelope. Its readout uses a least-squares fit on folded central examples. Identical-shape end-to-end training yields median shell mean absolute error (MAE) 3.169.

Sequence sweeps use widths $1, 2, 4$ for rolling sum and $1, 2, 3, 4$ for the two autonomous sequences. Slots total $d^2 + 2d$ including a learned readout. The ordinary four-item sorter sweeps three, four, and five untied gates and an additional full tied model. Training uses 256 inner streams of length eight or 256 four-item sort examples. Test sets use 2,048 examples per split. The autonomous tasks use one integer trajectory, first nine labels for training. Each of the 15 configurations uses five independent Xavier seeds, 800 full-batch Adam updates at 0.02 with gradient-norm cap 5, then BFGS up to 1,000 iterations. Objective is half mean squared residual. For a target sequence value g_n at index n, integer-sequence residuals are divided by $\max(1, |g_n|)$ before squaring. We checked reverse-mode gradients, obtained by propagating output derivatives backwards through the graph, against centered finite differences obtained from small positive and negative parameter perturbations. We report medians across seeds and retain each seed's scores, convergence status and final tensors. With shared three-gain sorting, individual BP fits range from zero to 46.39% whole-shell equality. With $\epsilon = 2^{-52}$ denoting the gap above 1 in binary64 and y the reference coordinate, the best fit passes the whole-vector $8\epsilon \max(1, |y|)$ test on every four-item example.

Activation and magnitude controls. Writing x_t for the current input, h_t for the scalar state after tick t and y_t for its output, a conventional one-state tanh RNN $h_t = \tanh(h_{t-1} + 2^{-40} x_t)$, $y_t = 2^{40} h_t$, reproduced every measured held-out prefix on the same binary64 tests. For the small arguments encountered on these samples, the library tanh returns its input value. The output is bounded, and scaling subnormal values can erase inputs outside the tested conditions. The matched five-start gradient model had median outer length-32 RMSE 0.452 and zero whole-stream accuracy at the median. The ordinary sorting control with gains fixed to one recovers every lattice vector because differences and outputs stay representable. The 50-weight selection comparator is trained on the same sorting task in a separate matched-architecture run. Its median outer RMSE is 4.35, compared with zero for the WSP comparator on that set. For the full-width, depth-48 square we tested the effect of setting the compensation readout coefficients to zero. The associated gradient experiment trained the compact deep family.

Reproducible outputs. The accompanying code includes raw per-split tables, fixed generators and seed-level results, exact-rational validations, the individual Astra responses and functions to apply them. A replay of the rational weights obtained by setting the transition matrix characteristic polynomial after rounding them to binary64 produced 32/32 exactly matching impulse values in 15 of 75 cases. These were 14/30 for prefix sum, 1/25 for Fibonacci and 0/20 for triangular numbers under fixed-order separate multiply/add. This is specific to that construction and its reduction order.

The isolated-row implementation exploits a different coordinate support. All observations in this initial suite are synthetic. The later transcendental studies also use published difficult-input corpora.

Width sweep of shallow scalar networks. For each hidden width, the first table gives mean absolute error on the outer range. Its WSP network uses uniform knots and formula values throughout the deployment envelope. BP uses central data only and its cell is the median of three independently initialized fits. A stored slot is one scalar coefficient, including a coefficient set to zero. The number of slots is three times the hinge count plus two.

Table 20: Mean absolute error (MAE) on the outer shell for shallow networks of increasing hidden width. WSP has access to the formula throughout the envelope. BP uses central labels.

		x^2		$\sin x$		x^x	
Hinges	Slots	WSP	BP	WSP	BP	WSP	BP
0	2	23.3	43.0	0.692	2.23	6.08×10^9	4.32×10^8
1	5	8.67	28.1	0.692	2.60	3.15×10^9	4.32×10^8
3	11	1.82	21.9	0.755	2.21	1.37×10^9	4.32×10^8
7	23	0.485	20.8	0.212	2.75	4.92×10^8	4.32×10^8
15	47	0.121	18.2	0.0726	3.05	1.42×10^8	4.32×10^8
31	95	0.0296	16.4	0.0195	3.30	3.69×10^7	4.32×10^8
63	191	0.00737	16.9	0.00486	3.26	9.55×10^6	4.32×10^8

The second table reports raw RMSE in the same target units on the same split, pooling scalar observations from the two shell sides before computing the seed median. Positive self-power uses the positive side alone. The closely spaced WSP knots at larger widths reduce absolute errors. The positive self-power target reaches approximately 10^{10} in this shell.

Table 21: Root mean squared error (RMSE) for the same shallow networks and outer draws. WSP uses full-envelope knots, and BP is the median of three fits.

		Square		Sine		Self-power		Tanh	
Hinges	Slots	WSP	BP	WSP	BP	WSP	BP	WSP	BP
0	2	26.7	50.5	0.817	2.31	6.34×10^9	1.46×10^9	0.404	2.13
1	5	9.41	34.8	0.817	3	4.24×10^9	1.46×10^9	0.404	1.68
3	11	2.05	28.1	0.97	2.82	2.56×10^9	1.46×10^9	0.122	0.336
7	23	0.54	26.9	0.258	3.47	1.26×10^9	1.46×10^9	0.00249	0.174
15	47	0.135	24	0.0933	3.83	4.60×10^8	1.46×10^9	8.06×10^{-4}	0.14
31	95	0.0332	22	0.0239	4.13	1.33×10^8	1.46×10^9	1.14×10^{-4}	0.156
63	191	0.00829	22.6	0.00602	4.07	3.54×10^7	1.46×10^9	3.22×10^{-5}	0.109

L Signed indices and the negative shell

The central integer set is $\{-3, \dots, 3\}$. The outer sets are $\{-10, \dots, -4\}$ and $\{4, \dots, 10\}$. For negative binary64 inputs, each noninteger has a reduced rational representation with an even denominator. Raising a negative base to that exponent has no conventional real value. On the 21 integers from -10 to 10, by contrast, n^n is rational, with $0^0 = 1$ by continuity from the positive side. NumPy's result equals the once-rounded rational reference at every index. A 572-slot rectifier lookup uses one triangular hat per integer. A hat equals one at its own integer and zero at all other integers. Setting its readout coefficient to the target value matched all seven central integers, seven negative-shell integers and seven positive-shell integers. Among the linear output weights that fit the seven central labels, the solution of smallest Euclidean norm in the same feature map missed all 14 shell integers. On the seven negative and seven positive outer integers, WSP gave RMSE zero for both signs. BP from a Xavier readout on the same frozen hats gave outer RMSE .314 and 3.78×10^9, respectively. The minimum-norm readout instead gave .00148 and 3.78×10^9. Each central-only fit missed every outer index. The full WSP lookup stores the specified answers.

For integer index n, let F_n denote the Fibonacci value and $T_n = n(n+1)/2$ the triangular value. For $n > 0$ the signed Fibonacci convention is $F_{-n} = (-1)^{n+1} F_n$. The input consists of $|n|$ unit ticks and a supplied sign selecting the forward or inverse transition. Each model stores forward and inverse transition matrices, an initial state, and a readout. The width-two Fibonacci and width-

three triangular models therefore use 12 and 24 scalar slots. A direct write and a rational recurrence inferred from only $n = -3, \ldots, 3$ both matched 7/7 central, 7/7 negative-shell, and 7/7 positive-shell labels, compared with independent integer formulas. In teacher-forced training, the true preceding values are fed into each one-step update. A gradient fit on those same central windows and the supplied shift structure achieved 14/14 shell accuracy in one of five starts for each task. WSP gave zero RMSE for each task and each sign. The median BP RMSE over five starts on the negative and positive shells, respectively, was 3.93×10^{-14} and 3.49×10^{-14} for Fibonacci and 7.21×10^{-14} and 1.33×10^{-13} for triangular numbers.

Observed sizes versus certified minima. The minimum state widths are one for the driven linear accumulator and two and three for the homogeneous, autonomous Fibonacci and triangular generators, respectively. For four-input real sorting, 5 ReLU gates are both sufficient and necessary, while 65 executions suffice for the bounded binary64 guarantee. The binary64 gate minimum is unknown. The shallow square has a sharp real-error frontier at every width. For the compensated deep square, the lowest tested step count reaching 100% on the ordinary shell was 35 (7,140 slots). The first to pass all ordinary inner inputs was 48, and the first tested to pass the midpoint suite was 55. Those are sweep outcomes, not lower bounds for the architecture. The residual sine/self-power models reach positive approximation error at the assessed budgets, with no global optimum claimed. For a finite real-arithmetic ReLU network, restriction to a line is piecewise affine with finitely many pieces. Exact agreement on an open interval of curvature with any of the smooth nonlinear targets is impossible. Finite binary64 and finite test sets require their own arguments.

M Protocol for the preliminary ten-task comparison

Each request provided one task, its central examples and three candidate architectures with their initialized coefficients. The response had to supply one complete array of replacement weights for each architecture. We crossed ten tasks, three random-number seeds (164, 317, 583) and two information conditions, yielding 60 independent calls and 180 arrays. Three arrays in one response share a prompt. A seed changes the initialization and the sampled examples, except that the two canonical number sequences always use the same indices. The *Formula and examples* condition supplied the task definition and central input–output pairs. The *Examples only* condition omitted the formula and task name but retained informative choices of architecture, indexing and deployment range. Every saved command-line interface trace recorded one response and zero tool actions. Each measurement uses the first returned array.

For the four smooth targets, the choices were one, four or 16 scalar hinges with an affine skip from input to output (5, 14 or 50 adjustable scalars). For addition and subtraction, the three models saw only the first input, both inputs without bias, or both inputs with a bias (1, 2 or 3 scalars). For running sum, the models were a memoryless linear map or recurrent networks with one or two hidden coordinates (1, 3 or 8 scalars). For sorting, let $h \in \mathbb{R}^4$ denote the current four values and let $v, w \in \mathbb{R}^4$ be learned columns. A block updates h to $h + v\,\mathrm{ReLU}(w^\top h)$, using eight unconstrained coefficients. Four, five or six such blocks use 32, 40 or 48 coefficients and permit different routing in every block. For a sequence generator of width d, the adjustable entries are a d-by-d transition, d initial values and d readout weights. The tested widths were $1, 2, 3$ for Fibonacci and $2, 3, 4$ for triangular numbers. Table 15 selects the 16-hinge smooth network, the two-input affine map, and the smallest known sufficient real-arithmetic configurations for the remaining tasks.

The WSP and backpropagation conditions received 12 central labeled examples per task, except for the canonical sequence generators, which used the nine indices $0, \ldots, 8$. Input lists had length four for sorting and six for sums. Fixed-length tests held the lengths constant and changed the input magnitudes. A scalar shell side contributed 2,048 inputs, and positive self-power used only the right side. Each vector test comprised 2,048 independent examples. Sums also used separate length-32 tests on central and outer values. Canonical number sequences have an early transfer set at indices 9–32 and a later outer set at 33–64. An exact percentage there counts correct individual numbers along one trajectory. For a vector or stream, exactness requires equality at every coordinate or tick within that example.

Numerical labels come from the stated NumPy binary64 operation on those inputs. The separate smooth-function studies use the GNU MPFR library for correctly rounded multiple-precision arithmetic, which supplies an independent reference for transcendental functions. Training used 800 full-

batch updates of adaptive moment estimation (Adam) and up to 1,000 Broyden–Fletcher–Goldfarb–Shanno (BFGS) iterations using an approximate inverse curvature matrix. For each scalar reference y, the residual is the prediction minus y, divided by $\max(1, |y|)$. Training minimizes half the mean of the squared residuals. Each gradient at initialization was checked by finite differences, comparing it to a symmetric small perturbation of each parameter. The WSP prompt prioritized the largest absolute scaled residual. To compare both methods under one criterion, Appendix N scores all returned weights under the same training loss and gives outer root mean squared error (RMSE) in the original output units.

N How well did the optimizers fit the displayed observations?

Fix one of the 90 preliminary combinations of task, size and starting weights in Appendix H. Collect the observed inputs into an array X, write θ for the candidate parameter vector, and let $f_\theta(X)$ be the resulting predictions. Flatten the predictions and scalar reference labels into the same order. If N is their common length and y_i is label i, the training loss is

$$L(\theta) = \frac{1}{2N} \sum_{i=1}^{N} \left(\frac{[f_\theta(X)]_i - y_i}{\max(1, |y_i|)} \right)^2 .$$

We used identical observed labels and initial weight arrays to score both WSP information conditions and gradient fits. Replaying 800 full-batch Adam updates followed by BFGS reproduced all 90 archived training and outer scores. Our additional least-squares solver (LS) computed the residual Jacobian, the matrix of all residual derivatives with respect to the parameters. It restricted each candidate parameter step to a region where a local linear model of the residuals was reliable. The budget allowed 250 solver evaluations of the residual vector, followed by at most ten Gauss–Newton updates that decreased the loss. Each such update solves a linearized least-squares problem. Four starting points were the original Xavier weights, the Adam+BFGS result, and the two WSP arrays. The outer scores were computed after fitting. For integer sequences the first held-out indices are 9–32.

Training loss and exact predictions. Each row below uses equally weighted settings. *Near best* counts settings within 10^{-10} in additive training loss of the best measured feasible binary64 candidate at that setting. *Global* counts one-hinge settings within the same tolerance of the independently certified real-arithmetic optimum. *Outer perfect* counts settings with equality for every held-out output, including every coordinate of vector outputs. An arrow means that the solver starts from the weights on its left. A raw WSP row scores the array as returned.

Table 22: Training loss and exact outer predictions for the matched optimizer study. Near best and outer perfect use all 90 configurations. Global uses the 12 one-hinge fits with certified real minima.

Method or start → solver	Median L	Near best /90	Global /12	Outer perfect /90
Xavier	.773	0	0	0
Xavier → Adam	.00190	34	0	0
Xavier → Adam+BFGS	.000823	48	1	0
Xavier → LS	.000613	53	5	6
Adam+BFGS endpoint → LS	.000822	49	1	5
WSP, Examples only	.000234	36	0	30
WSP, Examples only → LS	.0000223	72	9	30
WSP, Formula and examples	.000273	36	0	30
WSP, Formula and examples → LS	.0000307	66	7	30

The six perfect outer sets from Xavier-start LS occurred on addition and subtraction.

Prediction error in original units. For each configuration we computed outer RMSE over all test examples and output coordinates. The table reports the median RMSE over three sizes and three seeds per task, so large and small target magnitudes from different tasks remain separate. The WSP rows name the information supplied to the model, *Formula and examples* or *Examples only*. For each condition we score the returned weights before and after refinement by the same LS solver. The three BP rows start from Xavier weights. The Adam+BFGS row includes 800 initial Adam

steps. A zero is a zero median across the nine configurations. The exact rates, all 945 sets of weights and their unrounded training, central and outer RMSEs are retained in the release.

Table 23: Outer RMSE in target units, with one median over nine size–start combinations per task. WSP rows give the information supplied to the model. An arrow to LS means the returned weights were refined by trust-region least squares.

Method	Square	Sine	Self-power	Tanh	Addition
WSP, Formula and examples	7.06	0.596	1.20×10^{9}	0.0169	0
WSP, Examples only	7.07	0.605	1.20×10^{9}	0.0167	0
WSP, Formula and examples → LS	9.55	1.3	1.20×10^{9}	0.184	0
WSP, Examples only → LS	10.6	1.14	1.20×10^{9}	0.184	0
Xavier → Adam	37.3	2.25	1.58×10^{9}	0.902	1.48×10^{-11}
Xavier → Adam+BFGS	37.3	2.36	1.58×10^{9}	0.741	8.13×10^{-15}
Xavier → LS	33.5	2.5	1.58×10^{9}	0.701	4.30×10^{-18}
Method	Subtraction	Running sum	Sort	Fibonacci	Triangular
WSP, Formula and examples	0	0	5.23×10^{-16}	0	0
WSP, Examples only	0	0	4.98×10^{-16}	0	0
WSP, Formula and examples → LS	0	0	5.32×10^{-16}	0	0
WSP, Examples only → LS	0	0	1.52×10^{-15}	0	0
Xavier → Adam	5.63×10^{-13}	9.43×10^{-14}	4.62	3.56×10^{9}	8.09×10^{7}
Xavier → Adam+BFGS	1.57×10^{-14}	2.27×10^{-14}	6.17	0.273	2.61×10^{-7}
Xavier → LS	1.29×10^{-18}	2.05×10^{-15}	4.57	0.0102	3.13×10^{-5}

A global optimum against which to judge a local solver. For a single hidden scalar ReLU with affine skip, the model is a continuous two-piece line. Let a, b be the skip slope and intercept, let t be the hinge location and v its change in slope, and let $(u)_+ = \max(0, u)$. Its output at input x is $ax + b + v(x - t)_+$. A hinge of the other orientation is absorbed by the affine skip. Sort the observations. For each split with at least two sites on each side, let ℓ_1, ℓ_2 denote independent weighted least-squares lines fitted to the observations on the left and right, with weights $1/\max(1, |y_i|)^2$. If the lines intersect inside the gap, their joint optimum is feasible. Otherwise an optimum on that gap occurs at a boundary, unless the two lines coincide. To see this, let $D(x) = \ell_1(x) - \ell_2(x)$. If l, r are the adjacent sample locations on either side of the gap, intersection is equivalent to $D(l) \le 0 \le D(r)$ or $D(l) \ge 0 \ge D(r)$. Each alternative is a closed polyhedral set in the four line coefficients. Its quadratic minimizer is either the unconstrained fit or lies on an equality boundary, which fixes the knot to a sample. If one side contains only one site, a line through that site and the adjacent knot attains its unconstrained one-site fit. We therefore also solve at all interior sample knots and include the global affine fit. We evaluate these small linear systems and losses over exact rationals using the stored binary64 inputs and labels as rationals. This certifies the optimum for the real-arithmetic model. Minor last-bit differences arise when evaluating its coefficients in binary64. We also solve the weighted normal equations for 21 linear cases and construct interpolants on the 12 width-16 smooth cases, for 45 analytical comparison writes in total.

A specific failure and recovery. On the width-one square network, seed 164, five of twelve inputs initially activated the hinge. After Adam, the maximum input to the rectifier on the observations was -0.04148. The sole hinge therefore contributed zero on all observations. The loss 0.2376944 equaled the affine optimum, and BFGS took zero steps. The margin keeps the unit inactive in a neighborhood of this solution. Exhaustive rational minimization gives minimum training loss $L_* \approx 0.02293119$ with a knot at -0.140592. Trust-region LS from the identical Xavier vector reached this optimum. LS from the inactive Adam solution remained at its affine loss. The raw WSP proposal under *Examples only* had $L = 0.06706$ and mean outer scaled error 0.58786. Refining it to the certified optimum raised the outer error to 0.66849, even as the mean inner scaled error fell from 0.30945 to 0.22533.

Across all 90 settings, BFGS after Adam reduced L by more than 10^{-10} on 42, and increased mean outer scaled error by more than 10^{-10} on 13. Refining the WSP starts under *Examples only* materially reduced L on 54 settings, while outer error improved on six and worsened on 48. For WSP starts under *Formula and examples*, the respective counts were 54, seven and 47. These counts separate failure to fit a given objective from the usefulness of that objective for extrapolation. The archive stores every train loss and gap, inner and outer scores, evaluation counts, and selected parameter vectors. Over all 90 fresh Xavier-start LS runs, there were 4,849 residual evaluations and 2,613

Jacobian evaluations. Adam evaluated 72,000 joint losses and gradients. Computing a full residual Jacobian and computing a scalar loss gradient have different costs.

O When the optimizer reaches the training optimum

We froze the hidden rectifiers of a shallow scalar network and fitted only its output coefficients, including the direct input slope and constant offset. For each target (square, sine, positive self-power and hyperbolic tangent), we used seven hidden widths $0, 1, 3, 7, 15, 31, 63$ and three random starts. The frozen features were rectifiers with Xavier slopes and zero biases, rectifiers with equally spaced central knots, or rectifiers with equally spaced knots throughout the deployment range. A knot is the input where a rectifier changes slope. Each arrangement used the same initial output weights for a given width and start. Fits used 1,024 independent labeled examples sampled uniformly from either the central range or, in a separate full-information control, the deployment range. Independent test draws comprised 4,096 central examples and 4,096 per applicable outer side.

For observation i, let x_i be its input, y_i its label and $s_i = \max(1, |y_i|)$ its loss scale. Stack its frozen feature values, x_i and the constant 1 into a row and divide it by s_i. Let M be the resulting design matrix, and let column b have entries y_i/s_i. Write θ for the output coefficients, θ_0 for their initial values, and $^\top$ for matrix transpose. The training residual vector is $M\theta - b$. The Moore–Penrose pseudoinverse M^+ maps any residual vector to its minimum-length least-squares coefficient vector. The nearest global minimizer in Euclidean distance to θ_0 is

$$\theta_* = \theta_0 - M^+(M\theta_0 - b).$$

At θ_*, the residual is perpendicular to every design column, which certifies the minimum in real arithmetic.

The singular value decomposition (SVD) expresses M as orthogonal input and output directions with nonnegative gains called singular values. Let σ_j be gain j, $\sigma_{\max}$ the largest, m, n the numbers of rows and columns, and $\epsilon = 2^{-52}$ the spacing immediately above 1 in binary64. We keep directions with $\sigma_j > \sigma_{\max} \max(m, n)\epsilon$ when computing the pseudoinverse. The release records the largest absolute entry of $M^\top(M\theta - b)$ divided by m. This is zero when the normal equations hold exactly. In exact arithmetic, one gradient update on half the summed squared residual, preconditioned by $(M^\top M)^+$, gives θ_*. We also compute an idealized 5,000-step trajectory of full-batch gradient descent (GD) on that summed loss with step size $1/\sigma_{\max}^2$. In each retained singular direction its residual component is multiplied by $(1 - \sigma_j^2/\sigma_{\max}^2)^{5000}$. Evaluating this formula avoids 5,000 separate matrix multiplications. The reported training loss divides the summed loss by the number of observations.

Sixty-three centrally placed hinges. The table reports the median of three starts. Training loss uses the scaled labels above. Outer mean scaled error averages $|\hat{y} - y|/\max(1, |y|)$ over held-out scalar predictions $\hat{y}$ with references y. Outer root mean squared error (RMSE) is in original target units. GD and SVD are two solvers for the same frozen-feature fit. The preconditioned gradient calculation agrees with SVD to the digits shown.

Table 24: Central loss, mean scaled outer error and unscaled outer RMSE on frozen features. Each result is the median of three starts for 63 central hinges. GD uses 5,000 spectral steps and SVD uses the numerical least-squares solution.

	Central training loss		Outer scaled error		Outer RMSE	
Target	GD	SVD	GD	SVD	GD	SVD
Square	.00149	8.68×10^{-8}	.506	.275	34.0	22.3
Sine	.00113	2.59×10^{-8}	2.06	3.27	3.05	4.09
Positive self-power	.00159	2.69×10^{-7}	.932	.882	1.43×10^{9}	1.43×10^{9}
Tanh	.000920	9.32×10^{-9}	.952	.0333	1.20	.0396

The globally optimized central sine fit extends its last observed slope into the shell, producing higher outer RMSE than finite-step GD. With knots spread through the full envelope, central sine labels give a numerical design rank of 21 out of 65 directions. Permitting labels throughout the envelope raises the rank to 65 and lowers mean outer scaled error from 5.43 to .00194 with the same optimizer. In

the zero-bias Xavier arrangement all initial hinges meet at zero, so the model has at most two affine rays and design rank at most three. End-to-end BP can move those hinges.

P Passive training of recurrent networks

Recurrent cells and task interfaces. Passive training here means learning from fixed central input–output examples. We compare a recurrent neural network (RNN) that uses the hyperbolic tangent $\tanh(a) = (e^a - e^{-a})/(e^a + e^{-a})$ with a gated recurrent unit (GRU). At integer tick t, let x_t be the input vector, let $h_t \in \mathbb{R}^d$ be the hidden state of width d, and let $v \in \mathbb{R}^d, b_y \in \mathbb{R}$ be the readout weights and bias. Both cells predict $\hat{y}_t = v^\top h_t + b_y$. The tanh cell applies $\tanh$ coordinatewise to the sum of affine transforms of x_t and h_{t-1}.

For the GRU, use subscripts r, z, n for reset, update and candidate, respectively. For each subscript j, let $I_j(x) = W_j x + b_j$ and $H_j(h) = U_j h + c_j$ be learned affine maps to d coordinates. The parameters W_j, U_j are matrices and b_j, c_j biases. The sigmoid $\sigma(a) = 1/(1 + e^{-a})$ converts a real input, or logit, into a gate between 0 and 1. With $\odot$ denoting coordinatewise multiplication, the reset and update gates r_t, z_t, the candidate $\tilde{h}_t$, and the updated state are

$$r_t = \sigma(I_r(x_t) + H_r(h_{t-1})), \qquad z_t = \sigma(I_z(x_t) + H_z(h_{t-1})),$$
$$\tilde{h}_t = \tanh(I_n(x_t) + r_t \odot H_n(h_{t-1})), \quad h_t = (1 - z_t) \odot \tilde{h}_t + z_t \odot h_{t-1}.$$

For running sums it reads one scalar per tick, starts from zero and predicts the sequential binary64 sum after each update. For Fibonacci and triangular sequences we draw two initial values (a, b) for each example and train on a family of trajectories. Fibonacci starts at $(y_0, y_1) = (a, b)$ and follows $y_{t+2} = y_{t+1} + y_t$. In the triangular family let u_t denote the increment to add at tick t. We initialize $(y_0, u_0) = (a, b)$ and update

$$(y_{t+1}, u_{t+1}) = (y_t + u_t, u_t + 1).$$

Let E be the learned d-by-2 input encoder matrix and e its bias column. Both models initialize $h_0 = \tanh(E(a, b)^\top + e)$, read it out at tick zero and receive zero inputs thereafter. The classical sequence in each family uses $(a, b) = (0, 1)$. Sorting a list of length T takes $2T$ ticks. The cell first reads $(x_i, 0)$ for each list element, then receives $(0, 1)$ on each of T emission ticks. The readout must emit the sorted values in order. Only the final encoder state carries information about the list into emission.

Necessary state dimensions over the reals. Define the Jacobian of a map as its matrix of first derivatives, and its rank as the number of independent output directions. The map from (a, b) to the first two outputs has Jacobian determinant one in both seeded tasks. It factors through $h_0 \in \mathbb{R}^d$, so the chain rule requires $d \geq 2$ for exact reproduction on an open set of seeds. For sorting, restrict the inputs to the open region $x_1 < \cdots < x_T$. The desired output map is the identity there and has Jacobian rank T. All emission ticks factor through one state, so $d \geq T$ is necessary. This accounts for the width-four boundary for the four-input encoder. The tanh and GRU maps are smooth, while real sorting has a kink when two coordinates cross. Width alone therefore cannot certify continuum sorting for these cells. Both rank arguments concern exact real maps on open sets. Finite binary64 tests have different implications.

Data and optimization. Each training length has 512 training and 160 independent validation examples. The lengths are $4, 6, 8$ for sums and seeded trajectories and $2, 3, 4$ for sorting. Training inputs and initial seeds are uniform on $[-3, 3]$. For each task we evaluate four independently generated sets of 1,024 examples. We cross the largest training length (8, or 4 for sorting) with a longer length (32, or 8 for sorting), and the central value distribution with independently signed magnitudes from $(3, 10]$. Range shift for a seeded trajectory acts on (a, b).

Widths are $1, 2, 4, 8$, with width 16 added for sorting. Three Xavier initializations per cell and width yield $2 \cdot 3 \cdot (3 \cdot 4 + 5) = 102$ fits per budget. Each GRU gate block has its own Xavier draw. Biases start at zero. Both budgets use the same starts and data. Each Adam update samples a length uniformly, then 64 examples with replacement. Root mean square (RMS) is the square root of the arithmetic mean of squared values. Squared residuals are divided by the squared empirical training RMS of the corresponding output position, with RMS floored at one. We clip the gradient norm at 5

and use learning rate 0.01 for the first three quarters of the updates and 0.003 for the rest. Every 50 updates we score the central validation lengths with equal weight. Starting at the saved weights with the lowest validation loss, we run the limited-memory BFGS optimizer (L-BFGS) with a line search that requires a sufficient decrease in loss and a bound on the directional derivative (the strong Wolfe conditions) and retain whichever checkpoint has lower validation loss. The two budgets allow 1,000 or 5,000 Adam updates and 35 or 200 L-BFGS iterations, respectively.

Let Y and $\hat{Y}$ be the rectangular arrays of reference and predicted outputs over the test examples and emission ticks. Write $\|M\|_F = (\sum_{i,j} M_{ij}^2)^{1/2}$ for the Frobenius norm of a matrix M. Normalized root mean squared error (NRMSE) here means $\|\hat{Y} - Y\|_F / \|Y\|_F$, which divides the raw RMSE by the reference RMS. All denominators in these tests are positive.

Table 25: Median normalized root mean squared error for BP from three Xavier starts, with 5,000 Adam updates and up to 200 L-BFGS iterations. The metric is $\|\hat{Y} - Y\|_F / \|Y\|_F$ over all examples and ticks. Central and range retain length eight (four for sort). Length and joint use length 32 (eight for sort). Range and joint use outer magnitudes. Slots include both recurrent bias vectors, the readout, and the seed encoder where present.

Task	Cell	Width	Slots	Central	Range	Length	Joint
Sum	tanh	1	6	.00248	.0514	.0467	.3094
Sum	GRU	1	14	.00547	.0999	.0837	.4288
Fibonacci	tanh	2	19	.2283	.6581	.99998	.9999955
Fibonacci	GRU	2	39	.2163	.6557	.99998	.9999955
Triangular	tanh	2	19	.1316	.5141	.9046	.9297
Triangular	GRU	2	39	.1437	.4972	.9072	.9165
Sort	tanh	4	37	.3031	.7769	.6959	.8945
Sort	GRU	4	101	.1975	.6235	.4747	.6827

Raw RMSE by width. The table below reports the same NumPy/PyTorch binary64 sequential targets in their original units, pooling the squared errors across all 1,024 examples and their emitted ticks before taking a square root. For each cell, WSP gives the directly written state at that width, and BP gives the median over three starts after 5,000 Adam steps and up to 200 L-BFGS iterations. Central uses learned-range magnitudes and the largest training length. Joint uses outer magnitudes and the longer test length. A dash means this experiment contained no WSP checkpoint on that graph. The published feedforward sorters use a separate interface. Both budgets, every start, all four test conditions and the canonical sequence checks are available as per-checkpoint raw RMSE and exact-match fractions in the release.

Table 26: Raw central and joint-shift RMSE for the conventional recurrent cells at every tested width. A dash marks an unavailable WSP comparison. BP reports the three-start median at the longer budget.

		Tanh RNN				GRU			
		WSP		BP		WSP		BP	
Task	Width	Central	Joint	Central	Joint	Central	Joint	Central	Joint
Sum	1	0	0	0.009	8.66	0	0	0.0199	12
	2	0	0	0.028	18.2	0	0	0.0556	21.4
	4	0	0	0.0606	22.3	0	0	0.0513	22.3
	8	0	0	0.0782	22.8	0	0	0.0321	22.4
Fibonacci	1	—	—	0.778	2.38×10^6	—	—	6.25	2.38×10^6
	2	0	0	2.7	2.38×10^6	0	0	2.56	2.38×10^6
	4	0	0	3.43	2.38×10^6	0	0	2.62	2.38×10^6
	8	0	0	2.02	2.38×10^6	0	0	1.09	2.38×10^6
Triangular	1	—	—	4.68	228	—	—	4.67	228
	2	0	0	1.65	225	0	0	1.8	222
	4	0	0	0.71	226	0	0	0.225	218
	8	0	0	0.518	225	0	0	0.128	216
Sort	1	—	—	1.41	6.68	—	—	0.638	5.13
	2	—	—	0.636	5.38	—	—	0.464	5.04
	4	—	—	0.524	6.06	—	—	0.342	4.62
	8	—	—	0.415	5.59	—	—	0.243	5.86
	16	—	—	0.241	5.79	—	—	0.141	5.29

Across both budgets, every one of the 204 unconstrained fits had zero complete, numerically exact sequences on each test split. Approximation quality varied considerably (Table 25). For example, increasing sorting GRU width from 4 to 16 reduced central NRMSE from .1975 to .0816 but increased joint NRMSE from .6827 to .7811. We can diagnose a specific failure using the fitted output head. Both cells keep every state coordinate within $[-1, 1]$. Let $B = \sum_{j=1}^{d} |v_j| + |b_y|$, an upper bound on the output magnitude. With $(a)_+ = \max(a, 0)$ applied entrywise, the relative RMS error is bounded below by

$$\frac{\|(|Y| - B)_+\|_F}{\|Y\|_F}.$$

For width-two triangular models, the median floor is .9234 for tanh and .8947 for GRU, close to the observed .9297 and .9165. The width-four tanh sorting floor is only .1783, leaving more of its observed error unexplained by the output range.

WSP cells and a passive gradient success. Let q be a fixed power of two that scales the stored state. For sums we set $h_t = \tanh(h_{t-1} + qx_t)$ and $\hat{y}_t = h_t/q$ with $q = 2^{-40}$. For the two seeded families the active state encodes the two recurrence registers at scale q, with $q = 2^{-52}$ for Fibonacci and $q = 2^{-40}$ for triangular numbers. The Fibonacci transition has rows $(0, 1)$ and $(1, 1)$. The triangular transition has rows $(1, 1)$ and $(0, 1)$ and adds q to its second coordinate. In each GRU we use the same candidate transition and set reset and update logits to $+1000$ and -1000, which give exactly one and zero in the measured binary64 implementation. These weights reproduce every trajectory on the four test splits for sums and both seeded tasks. The finite horizon matters. The WSP Fibonacci cell first departs from the integer reference at classical index 39 and has NRMSE 3.73×10^{-6} over indices 33–64.

Finally, fix the scale and readout and train only the two Xavier scalars (A, C) in

$$h_t = \tanh(Ah_{t-1} + qCx_t), \qquad \hat{y}_t = h_t/q.$$

The data, validation rule and longer optimizer budget are the same as above. With $q = 1$, none of five starts matches a complete joint-test stream. With $q = 2^{-40}$, three of five reach $(A, C) = (1, 1)$ exactly and match every output on all four splits. The other two attain joint NRMSE 3.04×10^{-16} and 1.60×10^{-15}. For $q = 2^{-40}$ the median raw joint RMSE across all five fits is zero and the other two values are 8.52×10^{-15} and 4.48×10^{-14}. At $q = 1$, median raw joint RMSE is 27.8. Over those same 1,024 joint streams the deterministic WSP weights $(A, C) = (1, 1)$ give RMSE 27.8 and zero exact streams at $q = 1$, and RMSE zero and 1,024 exact streams at $q = 2^{-40}$.

Q A sharp binary64 size threshold for square on a lattice

For an integer grid index n, let $D_0 = \{n/16 : -160 \le n \le 160,\ n \in \mathbb{Z}\}$, $D_{\text{in}} = \{x \in D_0 : |x| \le 3\}$, and $D_{\text{out}} = D_0 \setminus D_{\text{in}}$. Fix a residual scalar ReLU with w hinges and $3w + 2$ stored scalars. Every coefficient is binary64 with magnitude at most 128. Write fl for binary64 rounding to nearest, ties to even, with gradual underflow. For input x, let a, b be the skip slope and bias. For each hinge $j = 1, \ldots, w$, let u_j, c_j, v_j be its input slope, input bias and output weight, respectively. Let s_0 be the rounded skip, h_j the rounded rectifier value, s_j the sum after hinge j, and $F(x)$ the binary64 output. The forward pass is

$$s_0 = \text{fl}(\text{fl}(ax) + b), \qquad h_j = \text{ReLU}\left(\text{fl}(\text{fl}(u_j x) + c_j)\right),$$
$$s_j = \text{fl}\big(s_{j-1} + \text{fl}(v_j h_j)\big), \qquad F(x) = s_w.$$

Every target $x^2 = n^2/256$ on D_0 is exactly representable.

Theorem 2 (Lattice square capacity)**.** *For any $D \in \{D_0, D_{\text{in}}, D_{\text{out}}\}$ of cardinality N, the maximum number of sites with $F(x) = x^2$ is $\min\{N, 2(w + 1)\}$. Exactness throughout D first becomes possible at $w = \lceil N/2 \rceil - 1$.*

Proof. Let f evaluate the same stored coefficients in real arithmetic. It has at most $w + 1$ affine pieces. We first control $F - f$ for $w \le 160$. Both the input product and its biased sum in a hinge (or the skip) have magnitude below 2^{11}, giving absolute error below 2^{-40} per rounding. The rectifier changes its output by at most the magnitude of a change in its input, so the hidden error is at most 2^{-39}. The output-weight product has magnitude below 2^{18} and its rounding costs at most 2^{-34}.

Propagating the hidden error through a weight of magnitude at most 128 gives absolute term error below 2^{-31}. The running sum remains below 2^{26} in magnitude, since it starts below 2^{11} and adds at most 160 terms of magnitude below 2^{18}, with less than one unit of absolute rounding error per addition. Each such addition therefore costs at most 2^{-26}. Consequently

$$\sup_{|x|\le 10} |F(x) - f(x)| \le 2^{-39} + 160(2^{-31} + 2^{-26}) < 2^{-18}$$

on binary64 inputs.

Take three sites $x_1 < x_2 < x_3$ within one affine piece. The secant, the affine line joining the two outer square values, exceeds x_2^2 at x_2 by $(x_2 - x_1)(x_3 - x_2) \ge 1/256$. For a common error bound E, if $|f(x_i) - x_i^2| \le E$ at all three sites, linear interpolation of the endpoint residuals bounds that excess by $2E$. Thus at least one real residual has magnitude at least $1/512$. Three exact machine hits would instead give $E < 2^{-18}$. Assigning a site at a knot to either adjacent piece, each piece accounts for at most two hits. This also gives a machine-error witness of at least $1/512 - 2^{-18}$ whenever there are more than $2(w + 1)$ sites.

For attainment, choose the first $\min\{N, 2(w + 1)\}$ sorted sites and pair them consecutively. If the selected count is odd, use the last two sites for the final pair, sharing its first site with the preceding pair. Write the secant through each pair. Consecutive slopes strictly increase. For disjoint pairs each secant lies below the square on the other pair, so their intersection lies in the intervening gap. For the overlapping pair it is the shared site. Store the first secant in the skip and, at each intersection t, add the slope increment times $\mathrm{ReLU}(x - t)$. This realizes the desired secant on every selected pair. On a uniformly spaced run, an intersection between disjoint pairs is the midpoint of the intervening gap. The outer set has 112 sites on each side, so the secants on either side of its central gap meet at zero. Every knot is a multiple of $1/32$, every slope increment a multiple of $1/16$, and every secant intercept a multiple of $1/256$. The coefficients obey the cap. Each intermediate on a grid site is a multiple of $1/512$ of magnitude below 2^{26}, and therefore each operation is exact in binary64. For larger widths, append hinges with zero output weight. This construction reaches all sites by width 160 even for the largest set, so the trivial upper bound N covers every remaining width. □

Capacity and information. The full, inner and outer sets contain 321, 97 and 224 sites. Their minimum widths are 160, 48 and 111, requiring 482, 146 and 335 slots. The full-grid WSP construction has RMSE zero on the grid, but misses all 320 intervening midpoints (RMSE .00218) and all 4,096 random outer examples (RMSE .00117). The inner-only secant fit has zero exact hits on the 224 outer sites and raw RMSE 22.4. These counts concern the displayed witnesses. The theorem bounds the best achievable score on the designated lattice under its coefficient cap and arithmetic contract.

A separate control specifies a polynomial of degree at most two and supplies only the labels $(-1, 1), (0, 0), (1, 1)$ and the deployment grid. For each node, its quadratic Lagrange basis function has value one there and zero at the other two nodes. Evaluating the three functions on the training nodes therefore gives the identity matrix. A unit gradient step on half-summed squared loss recovers their nodal coefficients in real arithmetic. Two binary64 steps from each of five Xavier draws give the same coefficients $(1, 0, 1)$. Evaluating the fitted polynomial offline and compiling its resulting ordinates yields the same ReLU coefficients as the full-grid witness, byte for byte, and recovers all 224 outer sites with RMSE zero. This control attributes the continuation to the supplied degree bound.

Proposition 3 (Finer grids and a lower bound on the entire interval)**.** *Under the same coefficient cap and binary64 operation order, choose an integer grid denominator q and let $D_q = \{n/q : n \in \mathbb{Z},\ |n| \le 10q\}$ for $q \in \{16, 32, 64, 128\}$. Exactly $10q$ hinges suffice and are necessary to hit all $20q + 1$ sites. Consequently, any network in this class that correctly rounds the square at every binary64 input in $[-10, 10]$ requires at least 1,280 hinges, or 3,842 stored scalar slots.*

Proof. Let $E(w)$ denote an upper bound on the absolute deviation between the binary64 calculation and the spline evaluated with the same weights in real arithmetic. The preceding proof bounds each computed hidden value by $1408 + 2^{-39}$ in magnitude and its rounded output-weight product by 180225. The initial skip has magnitude below 1410. For $w \le 1280$, the pre-rounded sum at addition k has magnitude at most $1410 + 180226k < 2^{28}$. Indeed, each preceding addition in this range contributes less than one unit of absolute rounding error. Its stronger individual rounding

bound is $(1411 + 180226k)2^{-53}$, which also covers subnormal values. Summing these bounds and the propagated term errors gives

$$E(w) = 2^{-39} + w2^{-31} + 2^{-53}\left(1411w + 180226\frac{w(w+1)}{2}\right).$$

At the largest width, $E(1280) < 1.701 \times 10^{-5} < 1/(2 \cdot 128^2)$. For all four grids, three distinct sites in the same affine piece give a secant excess of at least $1/q^2$. Hence three exact machine hits in that piece would require $2E(w) \geq 1/q^2$, which the bound excludes when $w \leq 10q$. If $w < 10q$, at most $2(w+1) \leq 20q$ sites can be hit.

The consecutive-pair secant construction attains the bound at $w = 10q$. The knots, slopes and intercepts have denominators dividing $2q$, q and q^2, respectively. The coefficient magnitudes stay below 128. At grid sites and their intervening midpoints every intermediate is a multiple of $1/(2q^2)$ of magnitude below 2^{28}. The corresponding integer numerator has magnitude below 2^{43}, so the specified operations are exact. In particular the 1,280-hinge witness hits all 2,561 sites of D_{128} and misses all 2,560 intervening midpoints in the numerical audit (midpoint RMSE 3.41×10^{-5}). Containment of D_{128} in the binary64 interval gives the stated necessary width for the entire interval. □

R From matching examples to certifying a computation

What can one hidden layer achieve? For a positive real L, consider the shallow hinge network from Section 3 on $[-L, L]$. Its *uniform error* on square is the supremum (least upper bound) of $|f(x) - x^2|$ over that interval. With w hinges the graph has at most $w + 1$ affine pieces. The smallest possible uniform error over real weights is exactly $L^2/[2(w+1)^2]$. To see the lower bound, take a segment of length ℓ on which the network is affine. For a function on its two endpoints and midpoint, the second difference is its endpoint sum minus twice its midpoint value. It is zero for an affine function and $\ell^2/2$ for square. If all three errors have magnitude at most η, their second difference has magnitude at most 4η. At least one segment has length $2L/(w + 1)$ or greater. Conversely, connect equally spaced square values by lines and shift the result down by half its largest interpolation error. This piecewise-affine function, or *spline*, attains the bound. At $L = 10$, a uniform tolerance of 0.01 first becomes possible at 70 hinges. A fixed absolute tolerance of 2^{-52} requires 474,531,328 hinges. Binary64 step sizes vary with the input and output magnitude.

A separate threshold holds for binary64 arithmetic. Limit all coefficient magnitudes to 128 and evaluate products and sums in the fixed order of Appendix Q, rounding after each operation. Exactly 1,280 hinges (3,842 slots) are necessary and sufficient for correct square outputs on all 2,561 equally spaced sites in $[-10, 10]$ at spacing $1/128$. The attaining network misses all 2,560 midpoints between those sites. Any network in this class that is correct on the entire binary64 interval must also be correct on the grid. The result gives a lower bound for that class and an exact minimum for the grid.

A sampled success, a shell proof, and an interval proof for square. For a contrast with shallow approximation, we constructed a multilayer perceptron (MLP), a sequence of affine maps and nonlinear activations, whose repeated tent-shaped map halves a grid spacing at each stage [28]. A correction path stores the small residual lost by rounding a subtraction. With 48 stages the corrected graph matched binary64 multiplication on 131,072 fresh outer inputs. On 304 outer inputs designed to put the exact square near a rounding boundary, it matched just 40. Figure 18 compares exactness with the correction present and removed. Appendix G specifies the construction.

In a different experiment Astra received the square rule and could choose both the connections and coefficients of an affine/ReLU graph. It received no rounding-boundary failures before answering. One of three first replies computed the correctly rounded square for *every* binary64 input in the two outer intervals using 2,543 input-dependent operations. Its graph extracts the significant digits, forms their integer product in three fixed-size pieces, and retains enough discarded information to round once at the end. Appendix Y proves the claim and gives an interior counterexample. We then added a rounding path for *subnormal* results, the small-magnitude values below the smallest positive normal number in magnitude, with uniform spacing 2^{-1074}. The engineered WSP graph uses 3,352 operations and is correct on all binary64 inputs in $[-10, 10]$ (Appendix Z). Both size counts are achieved upper bounds. The minimum for a deep graph remains open.

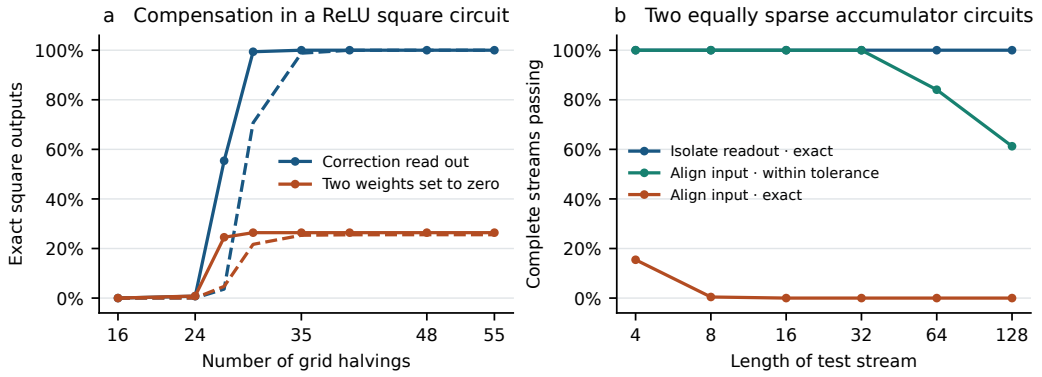


Figure 18: Two ways that a real-arithmetic identity can lose exact numerical agreement. Panel a varies the number of square refinements. Dashed lines use 4,096 central inputs, solid lines 8,192 outer inputs. The two networks differ only in whether the readout uses the accumulated rounding residual. Panel b compares two running-sum updates of width eight from Section C. At each length, 64 initial networks are evaluated on the same 64 outer streams. Exact requires equality at every step. The tolerance curve requires $|\hat{y} - y| \leq 10^{-10}(1 + |y|)$ at every step, where y is the reference prefix sum and $\hat{y}$ the predicted sum.

Sine, self-power and hyperbolic tangent. Our engineered WSP circuits for sine and positive self-power use affine operations and ReLUs. Their respective 500 and 1,352 stored coefficient words are individually addressable binary64 values. At inference both circuits matched the correctly rounded mathematical target on 32,768 fresh outer inputs, giving zero RMSE on those draws. We also used CORE-MATH, a project publishing difficult inputs for correctly rounded mathematical functions. The sine circuit missed 12 of 30,326 hard cases in its design interval $[-16, 16]$ and one of the 1,884 in the requested shell. The self-power circuit missed 450 of 896 inputs chosen near output rounding boundaries just below one. Each miss was one adjacent representable value. The WSP hyperbolic tangent matched all 60,574 signed CORE-MATH cases in $[-10, 10]$. Appendices S and T give run sizes and raw errors. On the deployed self-power and hyperbolic-tangent graphs, none of three BP fits per graph came within one ULP on an outer example after 100 Adam updates. The best of the three hyperbolic-tangent fits still achieved outer RMSE about 0.095. Tested hard cases establish measured accuracy for these circuits. Global correctness of their last output digit remains open.

Sorting without perturbing an input. For each ordering of four distinct real inputs, sorting applies a different permutation of the four coordinates. On the region corresponding to that ordering the derivative matrix (Jacobian) is the associated permutation matrix. A feedforward affine/ReLU graph with g scalar ReLUs has at most 2^g patterns of active units and therefore at most 2^g distinct Jacobians on open regions. As $4! = 24$, at least five gates are necessary. Five suffice over the reals: a *comparator* sorts a pair (a, b) by forming $t = R(a - b)$ and returning $(a - t, b + t)$. Applying it to five prescribed pairs sorts four inputs [2]. Subtracting a rounded difference can perturb one operand in binary64. In the fixed-schedule outer test this WSP sorter matched 1,105 of 2,048 lists with RMSE 5.12×10^{-16}.

A different comparator uses 13 ReLUs to select and transmit an original operand. It has 50 stored coefficients shared across comparisons. Sorting four inputs uses 65 ReLU executions, with 250 coefficient uses. Appendix F proves that the comparator returns both original numerical values in order for every binary64 pair in $[-10, 10]^2$, with signed zeros treated as equal. The resulting four-item WSP sorter had zero RMSE on the outer test. Five matched BP fits had median RMSE 4.35 and zero exact outer lists at the median. This gives a binary64 upper bound on gate count. The five-gate lower bound above concerns real arithmetic.

S How the sine and self-power circuits compute their outputs

Circuits and reference values. We designed one circuit for the sine of a scalar input x in radians on $[-16, 16]$ and one for the self-power function $x \mapsto x^x$ on $[0, 128]$, setting $0^0 = 1$ by continu-

ity. Each design uses the formula and its domain. In this engineered weight-space programming (WSP) condition, we set the weights of each graph before evaluating any test input. The circuit is a directed acyclic graph, meaning a list of operations in which each result depends only on earlier operations. Its operations are scalar affine maps (weighted sums plus offsets) and rectified linear units, $\mathrm{ReLU}(v) = \max(0, v)$ for scalar v. We checked that at least one argument of every multiplication depends only on fixed coefficients. The graph uses separate binary64 operations, each rounded to nearest with ties to even. This fixes the order of addition and multiplication and permits subnormal results, the values spaced 2^{-1074} apart below the smallest positive normal number 2^{-1022}.

Our mathematical reference is the exact function rounded once to binary64. We computed it with GNU MPFR version 4.2.2, a multiple-precision library with correctly rounded elementary functions [22], at 53 significant bits and cross-checked challenge values by rounding higher-precision evaluations. NumPy, the numerical library used in the earlier tests, differs from MPFR on 321 of the 8,192 ordinary outer sine inputs and 3 of the 4,096 ordinary outer self-power inputs. Here an exact match means agreement with MPFR. For the unit-in-the-last-place (ULP) distance we count representable steps between prediction and reference, identifying signed zeros.

Choosing a bit using rectified linear units. Let r be a nonnegative binary64 remainder, k an integer exponent used by the circuit and $a = 2^k$ a normal threshold. Its nearest smaller binary64 value is $a - 2^{k-53}$. Define the threshold gate

$$B_k(r) = \mathrm{ReLU}\big(1 - \mathrm{ReLU}\big(2^{53-k}(a - r)\big)\big) .$$

At $r \geq a$ the inner rectifier returns zero. At $r < a$ the gain 2^{53-k} scales the difference to at least one. Thus the output is precisely the bit for the decision $r \geq a$. To copy a signed value v only when a bit b equals one, choose a bound $M \geq |v|$ and evaluate $\mathrm{ReLU}(v - M(1-b)) - \mathrm{ReLU}(-v - M(1-b))$. The result is v when enabled and zero otherwise. The circuit uses these operations to select terms of known sums. Its two-word variant represents an intermediate by the sum of a leading binary64 value and a smaller correction. Compensated arithmetic retains part of the rounding error of each sum or constant product in that correction [4].

Sine. Starting with angle zero, we store its sine and cosine, zero and one. We inspect the powers 2^k in $|x|$ from $k = 4$ downward. Whenever B_k selects a power, we subtract it from the remainder and apply the sine and cosine angle-addition formulas for a rotation of 2^k radians. Its sine and cosine are coefficients computed before inference by Taylor series for angles up to one radian and angle doubling for larger angles. The 83-stage graph ends at $k = -78$ and restores the sign of x. For $|x| \leq 2^{-26}$ it returns x directly, because the bound $|x - \sin x| \leq |x|^3/6$ is below half the distance to the next binary64 value toward zero.

Self-power. For $x \geq 2^{-64}$, choose an integer exponent e and fractional remainder u with $x = 2^e(1 + u)$ and $0 \leq u \leq 1$. Write $\log$ for the natural logarithm. At fractional digit j, starting at $j = 1$, let $d = 2^{-j}$. When the leading word of u reaches d, multiply $u - d$ by the stored coefficient $1/(1 + d)$ and add the stored $\log(1 + d)$ to a running logarithm initialized at $e \log 2$. The identity $(1 + u)/(1 + d) = 1 + (u - d)/(1 + d)$ explains the update. After the last digit, add the remaining u as a first-order approximation to $\log(1 + u)$. Next form $y = 1 + x \log x$ by adding scaled copies of that logarithm for the selected input bits. Finally, start at $\exp(-1)$, where $\exp$ is the natural exponential, and multiply by the stored $\exp(2^k)$ for each selected bit of y. The exact-arithmetic identity is $\exp(-1)\exp(y) = x^x$. For $x < 2^{-64}$ the circuit returns one. These digitwise logarithm and exponential methods have precedents in elementary-function algorithms (Bajard, Kla and Muller, 1994).

Comparing sizes. We count separately a stored coefficient, an occurrence of a coefficient in an expanded affine operation, and an executed rectifier. The engineered WSP graphs for 83-stage sine and 84-fractional-bit self-power store 500 and 1,352 coefficient words. Each word overwrites a scalar Xavier draw, uniform on $[-\sqrt{3}, \sqrt{3}]$. The topology and fixed signs already encode the algorithm. Expanding repeated coefficient uses gives 25,395 and 35,908 affine occurrences and 1,506 and 2,618 rectifiers. The affine count charges two coefficients for a sum of two variables or a variable plus an offset and one for scaling. The stored graphs also have four and seven fixed structural constants respectively. Figure 19 identifies the first tested sizes to pass the ordinary outer samples. Smaller unswept graphs could have the same accuracy.

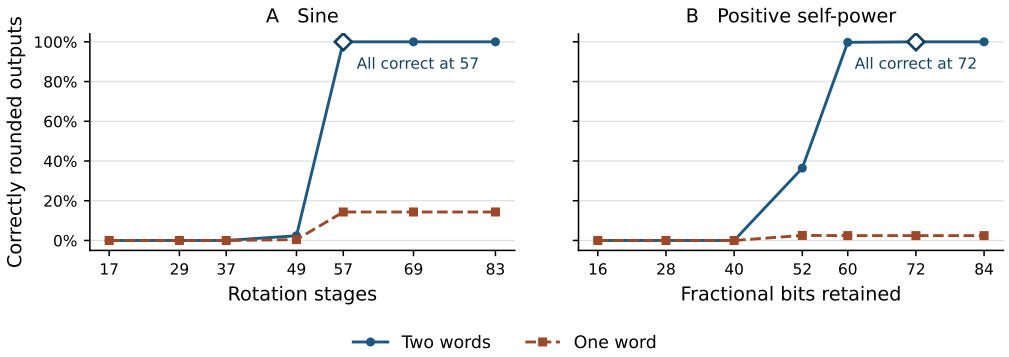


Figure 19: Fraction of inputs for which the network returns the correctly rounded mathematical value. The sine panel pools 4,096 inputs from each of $[-10, -3]$ and $[3, 10]$. The self-power panel uses 4,096 inputs from $[3, 10]$. Each curve evaluates the same inputs at seven sizes. The solid curve stores a leading word plus a rounding correction, and the dashed curve uses one word per intermediate. An open diamond marks the first sampled size with no misses in each panel. At 60 fractional bits, the two-word self-power circuit still misses 10 of 4,096 inputs.

Where agreement fails. At 57 sine stages (344 stored words) and 72 fractional bits for self-power (1,208 words), the two-word circuits first matched their respective full ordinary outer samples. The two largest WSP circuits also matched all 4,096 central and 32,768 independently drawn outer inputs per task, plus 8,192 inputs per task with exponents sampled uniformly into the subnormal range. Both had zero raw root mean squared error (RMSE) on the ordinary central and outer sets. Sine matched 102 values near nonzero multiples of π (half a turn in radians) and 4,096 farther inputs in $[10, 16]$. Self-power matched 213 values at or adjacent to powers of two and 4,096 farther inputs in $[10, 128]$.

CORE-MATH is a public project that distributes inputs whose function values are hard to round correctly. On its sine inputs the largest graph misses 12 of 30,326 distinct values in $[-16, 16]$, including one of 1,884 in the requested outer intervals. Each miss is one representable step. Across all 30,326 hard inputs the RMSE is 7.31×10^{-19}, and on the 1,884 hard outer inputs it is 2.50×10^{-21}. We write inputs as hexadecimal significands times powers of two, with `p` introducing the base-two exponent. At $x =$ `0x1.9239c4bbb44a4p+1` the absolute error is 1.08×10^{-19} and the absolute error divided by $|\sin x|$ is 1.36×10^{-16}. The archive records the source revision and file hash for the public input corpus.

For self-power, values just below one are spaced 2^{-53} apart. For integers $j = 0, \ldots, 127$, we located the small positive input satisfying $x \log x = \log(1 - (j + 1/2)2^{-53})$. Its exact output is a midpoint between two binary64 values. We tested the closest binary64 input and its three neighbors on either side. Of 896 outputs, 446 matched and the remaining 450 were one step away. At $x =$ `0x1.8e2e5fb51604cp-60` the network returns one. The reference is the next value below one, an absolute difference of 2^{-53}. The RMSE across the 896 near-midpoint inputs is 7.87×10^{-17}.

Gradient and product-gate controls. On the same 83-stage sine graph, we trained all 500 slots with adaptive moment estimation (Adam), using gradients propagated backward from the loss. We gave the optimizer 1,024 central labels, three independent Xavier starts, 100 updates at learning rate 0.01 and 32 observations per update. Respectively two, two and zero slots changed, and each run matched zero outer outputs exactly. Outer RMSE in seed order was 16.51, 8.14 and 15.96, compared with zero for the same WSP graph. A separate comparison allowed products of two input-dependent values. We fitted a degree-65 odd polynomial by full-batch Gauss–Newton updates (least-squares updates to the locally linear predictions) in a Chebyshev basis (polynomials orthogonal on a rescaled central interval under the Chebyshev weight). We evaluated both this fit and the formula-written Taylor coefficients with Horner's rule, which alternates multiplication by the argument with addition of a coefficient. Using two words for each of its 33 odd-power coefficients, the formula-written network matched all 8,192 ordinary outer sine values but missed 68 of the 102 probes near roots. The central-label fit matched none of the 8,192 and had outer RMSE 1.13×10^{35}, compared with

zero for the Taylor coefficients. The electronic tables in `revision/numeric_metrics/` retain raw RMSE and exact counts for each size, split and fitted control.

T Computing the hyperbolic tangent with affine maps and rectifiers

For a real input x, the hyperbolic tangent is $\tanh x = (e^{2x} - 1)/(e^{2x} + 1)$, with e the base of the natural logarithm. The native one-neuron control in the earlier controls (Appendix I) reproduces the NumPy library's implementation. Here both NumPy and a weight-space programming (WSP) circuit are compared with the mathematical function correctly rounded by MPFR. The WSP circuit receives the formula and the domain $[-10, 10]$.

Building a ratio from the input bits. For an integer exponent k, set the angle $\delta = 2^k$. Before inference we compute the coefficient $t_k = \tanh \delta$ at 256-bit precision. We use $t_k = \operatorname{expm1}(2\delta)/(2+\operatorname{expm1}(2\delta))$, where $\operatorname{expm1}(z) = e^z - 1$ is evaluated accurately for small z. Let K be the number of fractional input-bit positions inspected. Read the bits of $|x|$ for $k = 3, 2, \ldots, -K$. Keep a numerator N and denominator D initialized to zero and one. When the bit for δ is one, update both using their previous values,

$$(N, D) \longleftarrow (N + t_k D,\ D + t_k N).$$

The addition formula makes N/D the hyperbolic tangent of the sum of selected angles in exact arithmetic. To copy a signed intermediate z with $|z| \le 128$ when a bit b is one, the circuit uses $\operatorname{ReLU}(z-128(1-b)) - \operatorname{ReLU}(-z-128(1-b))$. Each exact update grows the maximum magnitude of the coordinates by at most $1 + t_k$. The factors for the four nonnegative exponents give a product at most 16. The factors for negative exponents give a product at most $\exp(\sum_{k<0} 2^k) = e$. These bounds motivate the choice of 128.

Quotient digits and rounding. Start with a remainder $R = N$. At fractional position $j = 1, 2, \ldots$, the next quotient digit is one if $R \ge 2^{-j} D$. Subtract $2^{-j} D$ from R whenever this comparison succeeds. The final circuit keeps the first 53 significant quotient bits beginning with the first one, and inspects the following bit, called the guard bit. If it is one, the circuit adds one unit at the last retained position. To see why the mathematical target has no tie at nonzero binary64 inputs, suppose $\tanh x$ were the midpoint between two consecutive representable values, which is a rational number. Then $e^{2x} = (1 + \tanh x)/(1 - \tanh x)$ would be rational. The Hermite–Lindemann theorem says the exponential of a nonzero algebraic number is transcendental, meaning it cannot be a root of a nonzero polynomial with integer coefficients. The nonzero rational number $2x$ meets the theorem's hypothesis. Finite-precision comparisons can still produce an incorrect bit.

For $|x| \le 2^{-27}$, a final gate copies x because $0 \le |x| - \tanh |x| \le |x|^3/3$ lies below half the spacing toward zero. For larger magnitudes the circuit restores the sign of x after extracting the quotient. The 84-fractional-bit construction uses 82 quotient positions. The input and quotient digit counts are distinct. Each coordinate uses three binary64 words representing a leading part and two corrections. The circuit combines rounded sums, constant products and their rounding residuals, recording successive corrections across the words. It stores 525 coefficient words and uses 2,146 rectifiers. Expanding repeated uses and fixed unit edges gives 75,973 affine coefficient occurrences. A traversal of the graph checks that every product has an input-independent factor.

Evaluation. The ordinary central set contains 4,096 inputs in $[-3, 3]$. The ordinary outer set contains 4,096 per side in $[-10, -3]$ and $[3, 10]$. A fresh draw contains 32,768 additional outer inputs. To construct targeted tests, take adjacent representable output values, compute their midpoint, and invert it at 240-bit precision using the inverse hyperbolic tangent, atanh. Test the nearest input and its two neighbors with both signs. These give 4,608 central and 4,608 outer examples. Finally, use the independently selected CORE-MATH hard inputs. The 53-bit MPFR results agree with the cross-check at 224 bits rounded to binary64.

For the three methods in Table 27, raw root mean squared error (RMSE) on the ordinary outer inputs is 5.38×10^{-17}, zero and zero, respectively. On the hard outer inputs it is 5.47×10^{-17}, 2.65×10^{-18} and zero. The electronic results give exact counts and raw RMSE for every method, size and input set. In the earlier two-word variant, summing 120 quotient digits gives 55,896 of 57,060 hard central and 3,512 of 3,514 hard outer matches. In the three-word size sweep, $K = 79$ misses two hard

Table 27: Exact matches to correctly rounded hyperbolic tangent. Every miss is one representable step. The hard columns use signed CORE-MATH inputs. The last two rows use 84 fractional input positions and direct guard-bit rounding.

Method Inputs	Central 4,096	Outer 8,192	Fresh outer 32,768	Midpoint 9,216	Hard central 57,060	Hard outer 3,514
Native NumPy	3,276	6,268	25,285	5,458	42,710	2,660
Two-word WSP	4,096	8,192	32,768	9,216	57,024	3,512
Three-word WSP	4,096	8,192	32,768	9,216	57,060	3,514

central inputs and two inputs near selected boundaries. At $K = 80$ all tested outputs match using 503 stored words and 2,068 rectifiers. This establishes an observed size threshold within the tested family. A proof covering every in-domain binary64 input remains open.

U How closely can gradients fit the same self-power and tanh circuits?

We exposed all 1,352 stored coefficients of the positive self-power circuit and all 525 of the three-word hyperbolic tangent circuit to backpropagation (BP). The wiring, operation order and fixed numerical constants were those of the deployed WSP circuits. The three trials initialized every exposed coefficient by a scalar Xavier draw with seeds 11, 22 and 33. The objective was mean squared error (MSE), the mean squared difference between prediction and correctly rounded output on 1,024 central inputs. A training update sampled 32 pairs with replacement. Reverse-mode differentiation propagated derivatives of the separate operations back to the weights, taking derivative zero at the rectifier's kink. Adam made 100 updates with step size 0.01. Its decay rates for first and second gradient moments were $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The stabilizer in the update denominator was $\epsilon = 10^{-8}$. The targets came from MPFR, with higher-precision cross-checks. We scored the final iterate.

Table 28: Training all exposed slots with BP and Adam. Training MSE is evaluated over all 1,024 central examples before and after fitting. A moved slot differs from its initial value. The maximum gradient is the largest absolute partial derivative of the batch objective over all slots and updates. Outer RMSE is in the output's original units against MPFR (4,096 self-power or 8,192 tanh inputs).

Target	Seed	Initial MSE	Final MSE	Moved slots	Max. gradient	Outer RMSE
Self-power	11	68.2141	59.0062	2	59.6	1.46×10^{9}
	22	59.0175	59.0123	2	0.0822	1.46×10^{9}
	33	68.3098	67.7551	74	11.7	1.46×10^{9}
Tanh	11	75.9257	1.72681	439	2.44×10^{6}	1.50
	22	11,772.6	6,168.13	373	2.52×10^{6}	103.83
	33	26.2714	0.062064	292	1.75×10^{4}	0.0953

On these same examples, both WSP circuits had zero training MSE and zero outer RMSE, matching every reference output. Each of the six gradient fits in Table 28 returned zero exact outer outputs and zero within one ULP. Every update had finite loss, gradients and optimizer state. In each run the slots whose gradients were ever nonzero were exactly those whose final values changed. With seed 33 the tanh network reduced its outer mean absolute error to 0.0711 while moving 292 slots. This comparison measures achieved accuracy at three starts and the stated training budget.

V A recurrent network that executes branching instructions

A register machine stores a nonnegative integer in each register and executes one instruction at a time. Its program counter identifies the active instruction. Let r index a register, let z_r be its current value, and let k, k_0, k_+ index destination instructions. The increment instruction $\mathrm{INC}(r, k)$ adds one to z_r and jumps to k. The conditional decrement $\mathrm{DECJZ}(r, k_0, k_+)$ jumps to k_0 if $z_r = 0$. Otherwise it subtracts one and jumps to k_+. The halt instruction holds the complete state fixed. With unbounded integer counters and time, these instructions include a universal two-counter machine,

Table 29: Outer predictions on the same graph and the same inputs for each target. WSP uses the stored coefficients of the deployed circuit. BP and Adam fit these slots from Xavier draws using 100 updates. The reported BP RMSE is the median of the three seed-specific raw RMSE values. Every seed has the exact-match count shown.

Target	Outer inputs	WSP exact	BP exact	WSP RMSE	BP RMSE
Sine	8,192	8,192	0	0	15.96
Self-power	4,096	4,096	0	0	1.46×10^9
Hyperbolic tangent	8,192	8,192	0	0	1.50

which can simulate any Turing-computable procedure under an appropriate encoding [5]. Binary64 cannot distinguish the adjacent register values 2^{53} and $2^{53}+1$. We examine both an exact-arithmetic instruction identity and finite binary64 executions. Related work characterizes when learned rectifier cells count exactly [6].

One update per instruction. Let R be the number of registers, P the number of instructions, G the number of conditional decrements, and $d = R + P$ the number of stored coordinates. Append to the register values a one-hot vector of P instruction indicators, with entry p_j equal to one for the active instruction j and zero for the others. Write $z \in \mathbb{R}^d$ for the resulting state and z' for the next state. Let D be a $d \times d$ matrix that directly updates the state, W a $G \times d$ matrix testing conditional instructions, and E a $d \times G$ matrix routing the branch corrections. With a componentwise rectifier, one update is

$$z' = \mathrm{ReLU}\big(Dz + E\,\mathrm{ReLU}(Wz)\big).$$

The register block of D copies each register. Its column for an increment adds one to the named register and places the active bit at the destination. Its halt column retains the current instruction bit. For a conditional decrement of register r at instruction j, the corresponding hidden unit computes $\mathrm{ReLU}(p_j - z_r)$. This equals one precisely when this instruction is active and the register value is zero. The corresponding column of D attempts a decrement and selects the positive destination. A unit hidden bit switches the destination through E. It can also cancel the attempted decrement. Alternatively the outer rectifier clips the attempted -1 to zero. In both cases every legal integer state advances exactly one instruction under exact arithmetic. Induction gives the specified states at every finite time.

Table 30: Five instruction lists compiled independently of the Astra trials. A dense slot is one matrix entry, including zero entries. The last column sums the number of executed instructions, or ticks, over all outer inputs. Every intermediate state and final scalar matched the integer interpreter, giving raw RMSE zero on each of these five outer sets.

Target	Registers	Instructions	Branch gates	Dense slots	Outer inputs	Total ticks
Addition	2	3	1	35	49	735
Truncated subtraction	2	3	2	45	49	644
Multiplication	4	7	3	187	49	13,083
Triangular number	3	8	3	187	7	1,008
Fibonacci number	4	9	4	273	7	2,860

The integer interpreter in Table 30 determined how many updates to score. The recurrent cell evolved autonomously from each initial state. Inputs ranged over all tuples from $\{0, 1, 2, 3\}$ centrally and $\{4, \ldots, 10\}$ on the outer set. For nonnegative integers a, b, truncated subtraction returns $\max(a - b, 0)$. In a separate test, six fresh Astra calls each received a relabeled two-register, four-instruction list and Xavier-initialized matrices. Across the calls we changed two branch wirings and permuted the state coordinates. All six replies gave the clipped-decrement construction up to gate and coordinate permutations. Each passed every one of 196 outer one-step states and 64 traces of 128 steps, with zero state RMSE. The six trials share two instruction structures.

Learning the transitions. For addition, truncated subtraction and multiplication, we trained every continuous weight on central one-step input–output labels from three Xavier starts. On each task, zero of the three BP fits produced any entirely exact outer trace. For the final scalar on each of the 49 outer inputs, median raw RMSE across starts was 10.60, 2.00 and 45.99 respectively. The

corresponding WSP compiler and a solver that inferred the discrete instruction table from the central labels both had zero RMSE. One fitted multiplication cell produced the right rounded next state on all 16,807 outer one-step cases, while none of its 49 autonomous final scalar outputs matched. We rounded the single-step prediction only for this diagnostic. Autonomous execution passed the unrounded state to the next update. The electronic tables distinguish one-step state error, state error pooled across ticks and coordinates, and final scalar error.

W Choosing both the operations and the weights

We asked whether Astra could implement square, sine or positive self-power when it could choose the circuit topology, meaning the operations and their connections. A domain-specific language (DSL) restricted the allowed operations. Each reply was an instruction list written in JavaScript Object Notation (JSON), a text format for structured data. A wire carries the value computed at one graph node. An input-dependent wire can vary with the input, while an input-independent wire carries a fixed constant. The affine/rectifier operator set permits addition, subtraction, multiplication by an input-independent constant, division by a constant and rectification. A second operator set also permits products of two input-dependent wires. In both sets, constants can use sine, logarithms and other elementary functions when the graph is built. For each unseen input, inference executes only the resulting fixed circuit. Every operation that depends on the input counts once toward the 4,096-operation cap after expansion of loops with fixed bounds. Precomputed constants, the size of the instruction text and the number of wires are separate costs.

Each prompt supplied the target formula, input interval, arithmetic rules and operation cap. Eighteen independent requests covered three targets, two operator sets and three replicate labels (9213, 2451 and 8282). The labels distinguish calls with the same instruction, without changing their inputs. The model returned weights and topology without a given Xavier array, tools, examples or evaluation feedback. In the affine/rectifier self-power condition two requests timed out after 300 seconds. We repeated those two before scoring, with a 900-second limit. One reply had invalid JSON syntax and the other exceeded the operation cap. Neither produced an evaluable circuit. All 16 responses returned within the original budget compiled. Table 31 keeps all 18 conditions.

Table 31: WSP with free topology. For each trial the accuracy columns count exact outputs out of 16,384 outer inputs against correctly rounded MPFR values. Sine and square use both signs with magnitudes from 3 to 10, and self-power uses positive inputs. The last column gives the number of input-dependent operations in trial order. An invalid reply has no numerical prediction or RMSE.

		Exact outer predictions			Operation counts
Target	Operators	Trial 1	Trial 2	Trial 3	Trials 1, 2, 3
Square	Affine/ReLU	16,384	16,384	16,384	2543, 4081, 2833
Square	With product	16,384	16,384	16,384	1, 1, 1
Sine	Affine/ReLU	8,191	9,272	5,662	4067, 3959, 3790
Sine	With product	16,384	16,384	16,384	587, 526, 772
Positive self-power	Affine/ReLU	0	invalid	invalid	4026, invalid, invalid
Positive self-power	With product	16,384	16,384	16,380	2622, 3008, 2881

On these same outer sets the three affine/rectifier sine circuits have raw RMSE 1.47×10^{-16}, 1.03×10^{-16} and 1.07×10^{-16}. For affine/rectifier self-power, the sole valid circuit has RMSE 0.0166. With product gates, the third self-power circuit has RMSE 1.18×10^{-13}. All other evaluable rows and trials in Table 31 have RMSE zero. Each product-gate square reply uses a single binary64 multiplication, which computes the reference by definition.

Every affine/rectifier square proposal matched all 304 inputs specially chosen near rounding midpoints on the outer intervals and all 32,768 fresh ordinary outer inputs. Across 16,369 inputs sampled over a wider range of exponents, the three exact counts were 16,357, 16,294 and 16,364. Appendix Y proves the first circuit's result on both complete outer intervals. On 1,884 CORE-MATH sine inputs in those intervals the three affine/rectifier replies matched 1,026, 1,079 and 697, compared with 1,881, 1,884 and 1,883 with product gates. The product-gate self-power circuits missed one, one and two of the 896 near-midpoint probes below one defined in Appendix S. On the 8,192-input wide-exponent panel, the second and third product-gate self-power replies each produced nonfinite

outputs, so neither has finite raw RMSE there. The electronic results retain each trial, split, exact count and RMSE status. In this experiment Astra chose each topology. The same-graph BP comparisons in Appendix U concern the three separately engineered smooth circuits.

X What changes after one round of central feedback?

For each of the three scalar targets and two operator sets in Appendix W, we selected the first trial in advance using its replicate label, 9213. This gives six pairs of circuits. Each follow-up request included the original specification, the verbatim first reply, and a diagnostic computed on all 4,096 central inputs. For an executable circuit the diagnostic listed operation count, number of finite outputs, exact matches and matches at distance at most one ULP (including exact matches). It supplied the input, prediction and reference for only the three largest ULP errors. Nonfinite predictions received priority and the earlier input broke ties. If every case matched, the three supplied examples all showed equality. For a circuit that failed to compile, the prescribed diagnostic was the parser error and the first three central labels. Astra could return its previous answer. Each revision used one separate medium-effort Astra request with a 300-second timeout and the same 4,096-operation cap. The six requests contained only this central feedback.

Table 32: Effect of one WSP revision given central feedback. An arrow separates the first result from the revised one, and a single number denotes equality before and after. The operations column counts input-dependent operations. The other columns count exact matches to MPFR. Both circuits were evaluated on identical inputs. The existing outer sample was used for the first-response study. The fresh sample was drawn with an independent seed.

Target	Operators	Operations	Central (4,096)	Outer (16,384)	Fresh (32,768)
Square	Affine	2,543	4,096	16,384	32,768
Square	Product	1	4,096	16,384	32,768
Sine	Affine	4,067 → 3,878	2,055 → 3,185	8,191 → 11,650	16,126 → 23,157
Sine	Product	587	4,096	16,384	32,768
Self-power	Affine	4,026 → 4,049	1 → 797	0 → 1,088	2 → 2,265
Self-power	Product	2,622 → 3,358	4,096	16,384	32,768

All six revisions compiled. The recorded interaction for each has one model reply and no tool action. Three replies were byte-for-byte identical to the originals. Requests took 5.8–139.6 seconds. For affine/rectifier sine, 9,608 fresh outer inputs became exact while 2,577 formerly exact predictions changed to a neighboring or more distant float. Across those same 32,768 inputs, raw RMSE fell from 1.47×10^{-16} to 5.07×10^{-17}. For affine/rectifier self-power it fell from 0.0163 to 6.32×10^{-6}, with maximum ULP distance falling from 324,964 to 187. The other four pairs matched every fresh outer output and had RMSE zero before and after repair. The electronic table records all four evaluated splits, including a fresh central set. A single feedback round measures these six specific revisions, conditional on the first reply and the three cases shown.

Y A first-response ReLU square correct on the entire outer shell

We prove the following all-input statement for a first response in the free-topology WSP experiment. We analyzed the frozen reply `3354251e2` (replicate 9213, 2,543 expanded input-dependent operations). The graph multiplies a wire only by a fixed coefficient. A loop implements integer multiplication using affine operations and ReLUs. Let RN denote binary64 rounding to nearest, ties to even. Under the separate-operation contract of the circuit language in Appendix W, this reply returns $\mathrm{RN}(x^2)$ for *every* binary64 x with $3 \le |x| \le 10$.

Normalization. Initialize a nonnegative remainder r to $r_0 = |x|/16 \in [3/16, 5/8]$, exactly representable in this range. In decreasing order of the integer index j, a flag f_j records whether to rescale r. The flags at thresholds $a = 2^{-j}$, for $j = 512, 256, 128, 64, 32, 16, 8, 4, 2, 1$, test $r < a$ by

$$f_j = \mathrm{ReLU}\big(1 - 2^{j+53}\,\mathrm{ReLU}(r - (a - 2^{-j-53}))\big)\,.$$

The stored threshold in the inner subtraction is the predecessor of a. If $r < a$ then r is at most that predecessor, giving $f_j = 1$. At $r = a$ the subtraction is exactly 2^{-j-53}. For every representable $r \geq a$, monotonic rounding gives a difference at least that large, hence $f_j = 0$. Even the largest gain has finite product on this shell. With intermediate $h = \mathrm{ReLU}(r - f_j)$, the update $r \leftarrow h + (r - h)2^j$ therefore preserves r for a zero flag and scales r by 2^j for a unit flag, since $0 \leq r \leq 1$. The flags through $j = 4$ vanish. Those at $j = 2, 1$ normalize r_0 into $r \in [1/2, 1)$ by factors four, two, or one. Thus $m = 2^{53}r$ is an exactly represented integer in $[2^{52}, 2^{53})$.

Exact integer square. Use the integer base $B = 2^{27}$. A limb is a digit in this base. The first loop reads the 26 highest-order binary digits of m, where higher order means greater place value. It obtains the integer quotient $a = \lfloor m/B \rfloor$ and remainder $b = m - aB$. For an integer remainder u and power of two p, the bit test $\mathrm{ReLU}(1 - \mathrm{ReLU}(p - u))$ returns one if $u \geq p$ and zero otherwise. The next loop reads all 53 bits of m from most to least significant. Let v denote the integer represented by the bits already read. Let P denote their product with m, and H, M and L its high, middle and low limbs. After each prefix, the stored values satisfy

$$P = vm = HB^2 + MB + L, \qquad 0 \leq M, L < B.$$

Initially all limbs are zero. For bit t the updated product should be $2P + tm$. The two lower limbs add tb and ta respectively to the doubled old digits, with carry from the lower limb. The actual selectors $\mathrm{ReLU}(b - (1 - t)B)$ and $\mathrm{ReLU}(a - (1 - t)B)$ produce those summands exactly. Let z denote either provisional digit sum before carrying. Both are integers in $[0, 3B)$. The carry into the next digit is the quotient $\lfloor z/B \rfloor$. The circuit computes it by adding the two comparison bits at B and $2B$, subtracts the resulting multiple of B, and adds the carry to the next digit. The high digit is updated as $2H + \mathrm{carry}$. Each prefix product satisfies $P \leq m^2 < 2^{106}$, so $H < 2^{52}$. The digit and bit operations stay within the binary64 exact integer range. After the last iteration $P = m^2$.

One final rounding. The normalized real square is $r^2 = 2^{-52}(H + M/B + L/B^2)$. Define the sticky bit s to be one if any bit of the discarded low limb L is nonzero, and zero otherwise. Let v_{sq} denote the rounded normalized square computed by the circuit,

$$v_{\mathrm{sq}} = 2^{-52}\,\mathrm{RN}\left(H + \frac{M + s/2}{B}\right).$$

Here $M + s/2$ and division by B are exact. The integer H lies in $[2^{50}, 2^{52})$. Throughout the possible range of P/B^2, binary64 rounding midpoints lie on the $1/B$ lattice (the intervals between successive powers of two in this range have spacings $1/4$ and $1/2$). If $L = 0$, the unrounded argument is exact. If $L > 0$, both $H + M/B + L/B^2$ and $H + (M + 1/2)/B$ lie strictly inside the same interval between adjacent $1/B$ lattice sites. No rounding midpoint separates them. Thus replacing the low limb preserves the side of every midpoint, and $L = 0$ preserves an exact tie and its even-neighbor decision. The final addition therefore produces the correctly rounded value of P/B^2. The circuit undoes each active normalization by an exact division by four or sixteen, then multiplies by 256 to undo the initial division of the input by sixteen. On the shell all these rescalings by powers of two remain normal, so rounding commutes with them and the result is $\mathrm{RN}(x^2)$.

The shell proof allows a concrete failure closer to zero. At $x =$ `0x1.0000000000001p-512` the same frozen circuit differs from the reference by one subnormal spacing. A separate audit against once-rounded exact rationals reproduced 104/104 normalization-boundary cases, 2,466/2,466 limb-boundary cases, 2,222/2,222 exact rounding ties, and 1,192/1,192 additional near-midpoint cases in the shell. All four finite panels had raw RMSE zero. The argument above supplies the all-shell guarantee.

Z A separately engineered square circuit for every binary64 input in $[-10, 10]$

The first-response WSP square circuit in Appendix Y fails on some central inputs with normal outputs. For the scalar input $x = (1 + 2^{-52})2^{-508}$ it returns 2^{-1016}, two output ULPs below the correctly rounded square. Its intermediate value represents $x^2/256$ before a final power-of-two rescaling. We retain the first 2,492 scalar operations of that proposal and replace its output logic. The resulting graph has 3,352 operations (949 ReLUs), compared with 2,543 operations (732 ReLUs) in

the first reply. Both graphs use the same operation-by-operation binary64 evaluator. These numbers give construction costs.

Claim. Let x denote any binary64 input in $[-10, 10]$. Use the binary64 format of the Institute of Electrical and Electronics Engineers (IEEE) standard, round each graph operation separately to nearest with ties to even, and retain subnormal results (gradual underflow). Write RN for this rounding. The engineered WSP graph returns the bit pattern of $\mathrm{RN}(x^2)$ for every such x, including both signed zeros.

Normalization. Initialize a running nonnegative remainder r to $r_0 = \mathrm{RN}(|x|/16)$. At each integer $k = 512, 256, \ldots, 1$ in decreasing powers of two, define a flag f_k to be one if the current $r < 2^{-k}$ and zero otherwise. The shared prefix multiplies r by 2^k exactly when the flag is set. For $q = 2^{-k}$ the preceding float is $q-\delta$ with $\delta = 2^{-k-53}$. The implemented flag is $\mathrm{ReLU}(1-\delta^{-1}\,\mathrm{ReLU}(r-(q-\delta)))$. For $r \le q - \delta$ the inner ReLU is zero. For $r \ge q$, monotone rounding gives an inner difference at least δ, so the flag is zero. All intermediates are finite since $r < 1$ and $k \le 512$. The conditional update uses $h = \mathrm{ReLU}(r - f_k)$ and $r \leftarrow h + (r-h)2^k$, with $h = r$ or 0 exactly. The strict threshold keeps $0 \le r < 1$ throughout. Define the total rescaling exponent $S = \sum_k k f_k$. This integer also selects the output branch below. We obtain $r = r_0 2^S$. For $r_0 > 0$ the descending flags perform the binary expansion of the integer shift taking r_0 into $[1/2, 1)$, capped at 1023. For $r_0 = 0$, all flags are set. If $S \le 541$, the normalization reaches $[1/2, 1)$ and the initial division by 16 is exact. Indeed, inexact division could only produce $r_0 \le 2^{-1022}$, which requires a shift of at least 1021. Define the significand integer $m = 2^{53} r$ and its exact square $P = m^2$. In this case m lies in $[2^{52}, 2^{53})$ and

$$|x| = m2^{-49-S}, \qquad x^2 = P2^{-98-2S}, \qquad P = m^2.$$

Even if the shift saturates, m is an integer below 2^{53} because r_0 is a multiple of 2^{-1074} and $1023 + 53 \ge 1074$.

Normal outputs, $S \le 514$. Let H, M, L be the high, middle and low digits of P in base 2^{27}, as computed by the integer-product loop in the preceding proof. Thus $P = H2^{54} + M2^{27} + L$, with $H < 2^{52}$ and $0 \le M, L < 2^{27}$. The shared prefix uses the nonzero flag of L to return $\mathrm{RN}(r^2)$. Indeed, all rounding midpoints for $P/2^{54} \in [2^{50}, 2^{52})$ are on the 2^{-27} grid. Replacing $L/2^{54}$ by 2^{-28} when $L > 0$ preserves the side of every midpoint, and $L = 0$ preserves ties. The final 2^{-52} scaling of this normalized result is exact. We first multiply it by 256, then apply the inverse scale 2^{-2k} for each active flag. Every intermediate in this branch is normal, since the exact final square is at least 2^{-1022}, and each active inverse scaling decreases the positive value. Rounding commutes with these power-of-two scalings down to the normal endpoint. For the running result v the circuit computes a selector intermediate $h = \mathrm{ReLU}(v - 256 f_k)$, valid because $0 \le v \le 256$.

Subnormal outputs and the transition to zero. The positive subnormal output levels are successive integer multiples of 2^{-1074}. For $515 \le S \le 541$, let $j = 2(S - 515) \in \{0, 2, \ldots, 52\}$. Measured in these units, the exact result is $P/2^{54+j}$. Its integer part is $\lfloor H/2^j \rfloor$. The guard bit is the first bit below the retained integer unit. Numbering bits from zero at the lowest position, this is bit $j-1$ of H when $j > 0$. For $j = 0$ this bit is bit 26 of M. If a nonzero number is an integer divided by a power of two, its square cannot lie exactly halfway between two such output levels. A halfway value is an odd multiple of 2^{-1075}, whose exponent after removing powers of two is odd, whereas the corresponding exponent of a square is even. Consequently adding the first discarded bit to the integer part rounds exactly. The graph extracts successive bits of H by integer thresholding and subtraction, selects the appropriate prefix and bit from S, and scales the resulting integer (at most 2^{52}) by 2^{-1074} exactly. This includes a possible carry to the smallest normal. For $S \ge 542$, exact initial division gives $|x| = 16r2^{-S} < 2^{-538}$. Inexact initial division is confined to $|x| < 2^{-1018}$. Both cases round to positive zero. The final addition selects the normal branch for $S \le 514$, the subnormal calculation for $515 \le S \le 541$, and zero otherwise, giving positive zero for both zero inputs.

Checks. A rational oracle matched all 79,106 distinct bit patterns tested, including exponent-stratified inputs, neighbors of every input power of two in range, subnormal rounding transitions, and the counterexample to the first reply. An independently written integer-square-root stress test matched 45,468 inputs bracketing subnormal half-steps and a second exponent/boundary suite

matched all 40,779 inputs. All three panels had raw RMSE zero for the engineered circuit. On the same 79,106 inputs the first reply matched 78,026 and had RMSE 9.15×10^{-324}. We computed the final RMSE scaling in higher precision so this small positive error survives rounding. Inputs can occur in more than one suite. The archived generator and full circuit distinguish this engineered result from the frozen first reply.

Research disclosures

Broader impacts. Small networks whose outputs can be checked against a mathematical reference could make numerical mistakes easier to detect before deployment. A direct weight edit can also change outputs far beyond the examples used to test it. The measured differences between exact matches and small numerical errors illustrate why applications that depend on exact arithmetic need tests for their own ranges and data types.

Model assistance. Astra, a hosted large language model, is credited as an AI research collaborator in the byline. It generated the experimental weight arrays under the prompt conditions in Section 3 and the architecture proposals in Section 5. A coding agent also helped implement the experiments, replay the numerical results, review drafts and revise the manuscript. Francois Chaubard is the human corresponding author and retains responsibility for the final manuscript and release.

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